Chefaa

By

Abdullah Ismail Shaaban Ali
Dalia Yasser Mohamed Hamed
Eman Medhat Salem Elsayed
Haneen Shaban Abd-Elfattah Elkhouly
Mona Mohamed Ahmed Dewidar
Sondos Ahmed Shawkey Sherif

Supervised by

Dr.Arabi Keshk

Department of ……….
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Summary

Healthcare services often suffer from fragmented systems, making it difficult for patients to access doctors, laboratories, and pharmacies through a unified platform. Patients frequently face challenges in booking appointments, obtaining medical consultations, accessing laboratory services, and purchasing medications efficiently. On the other hand, healthcare providers often lack integrated tools to manage appointments, services, payments, and communication with patients.

To address these challenges, we developed **Chefaa**, an AI-powered online healthcare platform that connects four main stakeholders: **Patients, Doctors, Pharmacies, and Medical Laboratories**, under the supervision of a secure administration system. To ensure trust and reliability, doctors, pharmacists, and laboratory owners must submit professional licenses and verification documents before gaining access to the platform.

The platform enables doctors to create and manage multiple clinics, organize schedules without conflicts, monitor appointments, manage patient attendance and payments, issue digital prescriptions, and track their financial performance through intelligent analytics dashboards. Pharmacies can manage medicine inventories, receive and track orders, provide delivery services, and monitor profits through AI-generated insights. Laboratories can publish their available services and securely deliver medical test results to patients through the application.

For patients, the platform offers a seamless healthcare experience by allowing them to search for healthcare providers, book, cancel, or reschedule appointments, order medications online, and make secure electronic payments. In addition, patients can benefit from AI-powered services, including a medical chatbot for healthcare guidance and a diagnostic assistant capable of analyzing uploaded medical reports and radiology images, providing simplified explanations and recommending suitable medical specialists when necessary.

The platform also integrates advanced Artificial Intelligence features such as daily briefings, business analytics, financial insights, and intelligent healthcare assistance for all

stakeholders. By combining healthcare services, digital payments, appointment management, pharmacy operations, laboratory services, and AI technologies into a single ecosystem, Shefaa aims to improve healthcare accessibility, efficiency, and user experience while reducing administrative complexity for both patients and healthcare providers.

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Chapter 1: Introduction

1.1 Introduction

The rapid growth of digital technologies has transformed many sectors, including healthcare. Despite these advancements, many healthcare services still suffer from fragmented systems that make it difficult for patients to access medical care efficiently. Patients often face challenges in finding suitable doctors, booking appointments, obtaining laboratory services, purchasing medications, and managing medical records. Healthcare providers also struggle with appointment scheduling, patient management, communication, and business analytics.

To address these issues, we developed **Chefaa**, an AI-powered healthcare platform that integrates patients, doctors, pharmacies, laboratories, and administrators into a unified digital ecosystem. The platform streamlines healthcare processes, improves accessibility, enhances communication, and provides intelligent services through Artificial Intelligence technologies.

1.2 Problem Definition

The healthcare sector faces several challenges that affect both patients and healthcare providers:

1.2.1 Difficulty Accessing Healthcare Services

Patients often spend significant time searching for suitable doctors, pharmacies, and laboratories. The lack of centralized healthcare platforms makes access to services inefficient and inconvenient.

1.2.2 Inefficient Appointment Management

Traditional appointment systems frequently lead to scheduling conflicts, missed appointments, and poor clinic organization, affecting both patients and doctors.

1.2.3 Limited Communication Between Stakeholders

Doctors, patients, pharmacies, and laboratories usually operate independently, creating communication gaps and delays in delivering healthcare services.

1.2.4 Lack of Intelligent Healthcare Assistance

Many patients require quick medical guidance, interpretation of medical reports, and recommendations for appropriate specialists, which are not always readily available.

1.2.5 Limited Business and Performance Analytics

Healthcare providers often lack tools that help them analyze appointments, revenues, service performance, and operational efficiency.

1.3 Objectives

The main objectives of the proposed system are:

- Develop an integrated healthcare platform connecting patients, doctors, pharmacies, and laboratories.
- Facilitate online appointment booking, cancellation, and rescheduling.
- Enable doctors to manage clinics, schedules, prescriptions, and patient records efficiently.
- Provide pharmacies with inventory management and order-tracking capabilities.
- Allow laboratories to manage services and securely deliver test results.
- Implement secure online payment services.
- Integrate Artificial Intelligence features for healthcare assistance, analytics, and daily summaries.
- Improve healthcare accessibility, efficiency, and user experience.
- Ensure security and trust through verification and authentication mechanisms.

1.4 Tools Used to Build and Develop the System

1.4.1 Frontend Tools & Technologies

- Visual Studio Code
- React.js
- Redux Toolkit
- React Router DOM
- Axios
- HTML5
- CSS3
- JavaScript (ES6+)
- Bootstrap / Custom CSS
- Chart.js
- SweetAlert2
- Socket.io Client

1.4.2 Backend Tools & Technologies

- Visual Studio Code
- Node.js
- Express.js
- MongoDB
- Mongoose
- JSON Web Token (JWT)
- Multer
- Socket.io
- Node-Cron
- Cloudinary
- OpenAI API
- google-auth-library
- helmet

- moment
- morgan
- nodemailer
- passport
- passport-google-oauth20
- uuid
- validator
- winston

1.4.3 Artificial Intelligence Technologies

- OpenAI GPT Models
- Medical Report Analysis
- AI Medical Chatbot
- Daily Brief Generation
- Revenue and Performance Analytics

1.4.4 UI/UX Design Tools

- Figma

1.4.5 Mobile Application Development Tools

Chapter 2: Previous Work

2.1 Introduction

Many digital healthcare platforms have been developed to improve access to medical services, appointment booking, online consultations, pharmacy services, and laboratory management. These systems aim to reduce waiting times, enhance communication between patients and healthcare providers, and simplify healthcare management. However, most existing solutions focus on a limited set of services and rarely provide a fully integrated healthcare ecosystem supported by Artificial Intelligence.

2.2 Comparative Study: Similar Applications

The following applications were studied as references during the development of Chefaa.

1. Vezeeta
   - ➢ Features:
     - Doctor search and appointment booking.
     - Online consultations.
     - Patient reviews and ratings.
     - Pharmacy services in some regions.
   - ➢ Limitations:
     - Limited integration between doctors, pharmacies, and laboratories.
     - Limited AI-powered healthcare assistance.
     - No advanced clinic-management analytics for providers.
2. Altibbi
   - ➢ Features:
     - Medical consultations.
     - Health articles and medical information.
     - AI-based symptom checker.

- Limitations:
  - Focuses mainly on consultations and health information.
  - Does not provide a complete clinic, pharmacy, and laboratory management system.
  - Limited operational tools for healthcare providers.

3. Sehaty

- Features:
  - Appointment booking.
  - Access to medical records.
  - Government healthcare services.
- Limitations:
  - Primarily designed for government healthcare services.
  - Limited pharmacy inventory and order-management capabilities.
  - Limited AI-driven business analytics.

2.3 Comparison Between Existing Systems and Shefaa

Feature	Vezeeta	Altibbi	Sehaty	Shefaa
Doctor Search	✓	✓	✓	✓
Online Appointment Booking	✓	**Limited**	✓	✓
Appointment Cancellation & Rescheduling	**Limited**	✗	✓	✓
Multiple Clinic Management	✗	✗	✗	✓
Clinic Schedule Organization	✗	✗	✗	✓

Patient Attendance Tracking	✗	✗	✗	✓
Digital Prescriptions	**Limited**	✓	✗	✓
Pharmacy Integration	**Limited**	✗	**Limited**	✓
Pharmacy Inventory Management	✗	✗	✗	✓
Medicine Ordering System	**Limited**	✗	✗	✓
Order Tracking	✗	✗	✗	✓
Delivery & Pickup Options	**Limited**	✗	✗	✓
Laboratory Services	✗	✗	**Limited**	✓
Upload & Receive Lab Results	✗	✗	**Limited**	✓
Secure Online Payment	**Limited**	**Limited**	✓	✓
Real-Time Notifications	**Limited**	✗	✓	✓
AI Medical Chatbot	✗	**Limited**	✗	✓
AI Healthcare Guidance	**Limited**	✓	✗	✓
AI Analysis of Medical Reports	✗	✗	✗	✓
AI Analysis of Radiology Images	✗	✗	✗	✓
Doctor Daily Briefing	✗	✗	✗	✓
Pharmacy Daily Briefing	✗	✗	✗	✓
Revenue Analytics	✗	✗	✗	✓
AI Business Insights	✗	✗	✗	✓

Platform Commission Tracking	✗	✗	✗	✓
Real-Time Chat System	✗	**Limited**	✗	✓
Provider Verification System	**Limited**	**Limited**	✓	✓
Integrated Healthcare Ecosystem	**Partial**	**Partial**	**Partial**	✓

2.4 Discussion

The comparative study demonstrates that existing healthcare platforms typically focus on a specific aspect of healthcare services. Vezeeta mainly concentrates on doctor discovery and appointment booking, Altibbi focuses on medical consultations and health information, while Sehaty provides access to governmental healthcare services. However, none of these systems combines doctors, pharmacies, laboratories, patients, payment services, business analytics, and Artificial Intelligence within a single integrated platform.

Shefaa distinguishes itself by providing a comprehensive healthcare ecosystem that enables healthcare providers to manage their operations efficiently while offering patients seamless access to healthcare services. Furthermore, the integration of Artificial Intelligence for medical assistance, report analysis, daily briefings, and business analytics represents a significant advancement beyond the capabilities offered by existing solutions.

Chapter 3: System Analysis

3.1 Introduction

System analysis is a crucial phase in software development that focuses on understanding the requirements, functionalities, and interactions within the proposed system. This chapter presents the analysis of the Shefaa platform, including stakeholders, functional requirements, non-functional requirements, and system use cases.

3.2 System Overview

Shefaa is an AI-powered healthcare platform designed to connect patients, doctors, pharmacies, laboratories, and administrators within a unified ecosystem. The platform aims to simplify healthcare access, appointment management, medicine ordering, laboratory services, online payments, and healthcare analytics.

The system consists of five main actors:

- Administrator
- Doctor
- Pharmacy
- Laboratory
- Patient

Each actor has specific responsibilities and permissions that contribute to the overall operation of the platform.

3.3 Stakeholders Analysis

3.3.1 Administrator

The administrator is responsible for managing the entire platform and ensuring the authenticity of healthcare providers.

Responsibilities:

- Verify doctors, pharmacies, and laboratories.

- Approve or reject registration requests.
- Manage platform users.
- Monitor system activities.
- Manage platform commissions and revenues.

3.3.2 Doctor

Responsibilities:

- Create and manage clinics.
- Organize schedules and appointments.
- Manage patient visits.
- Issue prescriptions.
- View analytics and revenues.
- Access AI-powered daily briefings.

3.3.3 Pharmacy

Responsibilities:

- Manage medicine inventory.
- Receive and process orders.
- Track deliveries.
- View financial reports and analytics.

3.3.4 Laboratory

Responsibilities:

- Manage laboratory services.
- Receive test requests.
- Upload and send test results.
- Manage laboratory information.

3.3.5 Patient

Responsibilities:

- Search for doctors and healthcare services.
- Book, cancel, and reschedule appointments.

- Order medicines online.
- View prescriptions and laboratory results.
- Use AI healthcare services.

3.4 Functional Requirements

Functional requirements describe the services, operations, and behaviors that the Shefaa platform must provide to its users. The system supports five main actors: Patients, Doctors, Pharmacies, Laboratories, and Administrators.

3.4.1 Authentication and Authorization

The system shall provide a secure authentication mechanism that allows users to register and log into the platform according to their roles. Patients can register directly, while doctors, pharmacies, and laboratories must submit professional verification documents during registration. These documents are reviewed by the administrator before account activation.

The system shall support:

- User registration and login.
- Role-based access control.
- Password recovery and reset functionality.
- Secure session management using JWT authentication.
- Account verification for healthcare providers.

3.4.2 Profile Management

The platform shall allow all users to create and maintain their profiles. Each user type has a customized profile according to their role within the system.

The system shall allow users to:

- View profile information.
- Update personal details.
- Change passwords.
- Upload profile images and supporting documents.

- Manage account settings.

Doctors can additionally provide information such as specialization, qualifications, clinic information, consultation fees, and available schedules. Pharmacies and laboratories can maintain their business information, services, and contact details.

3.4.3 Notification Management

The platform shall provide a notification system to keep users informed about important events and activities.

The system shall allow users to:

- Receive real-time notifications.
- View notification history.
- Mark notifications as read.
- Configure notification preferences.

Medicine Time

Notifications may include appointment confirmations, prescription updates, laboratory results, payment confirmations, order status updates, and administrative announcements.

3.4.4 Appointment Management

The appointment management module enables patients and doctors to efficiently organize healthcare visits.

Patients shall be able to:

- Search for doctors.
- View doctor schedules and available time slots.
- Book appointments online.
- Cancel appointments.
- Reschedule appointments.
- View appointment history.

Doctors shall be able to:

- View all appointments.

- Manage clinic schedules.
- Mark appointments as completed.
- Track attendance status.
- Identify missed appointments.
- Block patients who repeatedly fail to attend appointments.

The system shall automatically prevent scheduling conflicts and overlapping appointments.

3.4.5 Prescription Management

The prescription module enables doctors to generate and manage digital prescriptions while allowing patients to access their treatment information.

Doctors shall be able to:

- Create prescriptions.
- Update existing prescriptions.
- Review prescription history.
- Add medications, dosage instructions, and medical notes.
- Request laboratory tests.

Patients shall be able to:

- View prescriptions.
- Access previous prescriptions.
- Share prescriptions with pharmacies when ordering medications.

3.4.6 Clinic Management

The clinic management module enables doctors to organize and maintain multiple clinics from a centralized dashboard.

Doctors shall be able to:

- Create clinics.
- Update clinic information.
- Delete clinics.

- Manage clinic addresses.
- Configure working days and operating hours.
- Define appointment durations.
- View clinic schedules and statistics.

Patients shall be able to:

- View clinic information.
- View available appointment slots.
- Select preferred clinic locations.

3.4.7 Pharmacy Management

The pharmacy module provides a complete medicine ordering and inventory management system.

Patients shall be able to:

- Search for pharmacies.
- Search for medicines.
- View pharmacy profiles.
- View medicine information.
- Create medicine orders.
- Track order status.
- Confirm order receipt.
- Submit reviews and ratings.
- Pay online.

Pharmacies shall be able to:

- Receive customer orders.
- Process and update order status.
- Manage deliveries.
- Monitor inventory levels.
- Track revenue and sales performance.

3.4.8 Laboratory Management

The laboratory module facilitates communication between laboratories and patients.

Laboratories shall be able to:

- Manage laboratory profiles.
- Add and update services.
- Activate or deactivate services.
- Receive laboratory requests.
- Upload test results.
- Manage patient records related to laboratory services.

Patients shall be able to:

- Request laboratory services.
- View laboratory information.
- Receive and download laboratory reports.

3.4.9 Catalog and Inventory Management

The platform shall provide inventory and catalog management functionality for pharmacies and laboratories.

Pharmacies shall be able to:

- Add medicines.
- Edit medicine information.
- Update stock quantities.
- Restock inventory.
- Remove unavailable products.

Laboratories shall be able to:

- Add services.
- Modify service details.
- Activate or deactivate services.

3.4.10 Billing and Financial Management

The platform shall provide financial management capabilities for users and administrators.

Users shall be able to:

- View billing summaries.
- Access payment history.
- Make secure online payments.

Administrators shall be able to:

- Generate billing reports.
- Manage financial transactions.
- Track platform commissions.
- View revenue analytics.
- Monitor payment activities.

3.4.11 Artificial Intelligence Services

The system shall integrate Artificial Intelligence technologies to improve healthcare accessibility and decision support.

The AI module shall provide:

- Medical chatbot assistance for patients.
- Healthcare guidance and recommendations.
- AI-generated daily briefings for doctors and pharmacies.
- Financial analysis and performance insights.
- Pharmacy stock analysis and recommendations.
- Medical report interpretation.
- Radiology image analysis.
- Specialist recommendations based on uploaded medical documents.

The AI services are designed to support healthcare decision-making and should not replace professional medical consultation.

3.4.12 Administration Module

The administration module provides complete control over the platform.

Administrators shall be able to:

- Verify doctors, pharmacies, and laboratories.
- Approve or reject registration requests.
- Block or delete accounts.
- Manage roles and permissions.
- Monitor system logs.
- Configure platform settings.
- View system-wide analytics.
- Manage backup and restoration operations.
- Monitor platform health and performance.

The administrator is responsible for maintaining system security, reliability, and compliance with platform policies.

3.5 Non-Functional Requirements

Security

- ➢ The system shall use JWT-based authentication.
- ➢ Passwords shall be securely encrypted.
- ➢ Role-based authorization shall be enforced.
- ➢ Payment transactions shall be processed securely.
- ➢ Only verified healthcare providers shall access professional features.

Performance

- ➢ System response time should be minimal during normal operation.
- ➢ The platform shall support concurrent users efficiently.
- ➢ Real-time updates shall be delivered using Socket.io.

Reliability

- The system shall maintain high availability.
- Data loss shall be minimized through backup mechanisms.
- Critical transactions shall be logged and tracked.

Scalability

- The system shall support future growth in users, clinics, pharmacies, and laboratories.
- Additional AI services and healthcare modules can be integrated without major redesign.

Usability

- The user interface shall be intuitive and easy to navigate.
- The platform shall support responsive web and mobile access.
- Healthcare services shall be accessible with minimal user effort.

Maintainability

- The system architecture shall support future updates and enhancements.
- Modules shall be designed independently to simplify maintenance.

Availability

- The platform shall be available 24/7 except during scheduled maintenance periods.

Compatibility

- The system shall function correctly on modern web browsers.
- The platform shall support both desktop and mobile devices.

Data Integrity

- Medical, financial, and appointment records shall remain accurate and consistent.
- Unauthorized modifications to critical data shall be prevented.

3.6 **System Modeling**

We will represent our system using the following diagrams:

• Use case diagram.

• Class Diagram.

• Sequence diagram.

• Data flow diagram (DFD).

• Entity relationship diagram (ERD).

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3.6.1 Use Case Specifications

1. UC-01: User Login

Item	Description
Use Case ID	UC-01
Use Case Name	User Login
Primary Actor	Patient, Doctor, Pharmacy, Laboratory, Admin
Description	Allows users to access the platform securely.
Preconditions	User account exists and is active.
Main Flow	1. Enter email and password.2. System validates credentials.3. System identifies user role.4. User is redirected to dashboard.
Alternative Flow	Invalid credentials or inactive account.
Postconditions	User gains access to authorized functions.

2. UC-02: Book Appointment

Item	Description
Use Case ID	UC-02
Use Case Name	Book Appointment
Primary Actor	Patient
Description	Allows patients to reserve appointments with doctors.
Preconditions	Patient is logged in and slot is available.
Main Flow	1. Search doctor.2. Select clinic.3. Choose available slot.4. Confirm booking.5. System creates appointment.
Alternative Flow	Selected slot becomes unavailable.
Postconditions	Appointment is successfully scheduled.

3. UC-03: Cancel / Reschedule Appointment

Item	Description
Use Case ID	UC-03
Use Case Name	Cancel or Reschedule Appointment
Primary Actor	Patient
Description	Allows patients to modify existing appointments.
Preconditions	Appointment exists.
Main Flow	1. Open appointment list.2. Select appointment.3. Choose cancel or reschedule.4. Confirm action.
Alternative Flow	Appointment already completed.
Postconditions	Appointment status is updated.

4. UC-04: Create Clinic

Item	Description
Use Case ID	UC-04
Use Case Name	Create Clinic
Primary Actor	Doctor
Description	Allows doctors to create and manage clinics.
Preconditions	Doctor account is approved.
Main Flow	1. Enter clinic information.2. Define working days.3. Define available slots.4. Save clinic.
Alternative Flow	Invalid clinic data.
Postconditions	Clinic becomes visible to patients.

5. UC-05: Manage Appointments

Item	Description
Use Case ID	UC-05
Use Case Name	Manage Appointments
Primary Actor	Doctor
Description	Allows doctors to manage patient appointments.
Preconditions	Doctor is logged in.
Main Flow	1. View appointments.2. Mark patient arrived.3. Mark appointment completed.4. Mark payment status.
Alternative Flow	Patient does not attend appointment.
Postconditions	Appointment status is updated.

6. UC-06: Create Prescription

Item	Description
Use Case ID	UC-06
Use Case Name	Create Prescription
Primary Actor	Doctor
Description	Allows doctors to issue electronic prescriptions.
Preconditions	Patient appointment exists.
Main Flow	1. Open patient record.2. Add medicines.3. Add dosage instructions.4. Save prescription.
Alternative Flow	Missing prescription information.

Postconditions	Prescription becomes available to patient.

7. UC-07: Create Medicine Order

Item	Description
Use Case ID	UC-07
Use Case Name	Create Medicine Order
Primary Actor	Patient
Description	Allows patients to order medicines from pharmacies.
Preconditions	Medicine is available.
Main Flow	1. Search medicine.2. Add to cart.3. Select delivery method.4. Confirm order.
Alternative Flow	Medicine out of stock.
Postconditions	Order is created successfully.

8. UC-08: Online Payment

Item	Description
Use Case ID	UC-08
Use Case Name	Online Payment
Primary Actor	Patient
Description	Allows users to pay for appointments and medicine orders online.
Preconditions	Payment request exists.
Main Flow	1. Select payment method.2. Enter payment details.3. Complete transaction.4. Receive confirmation.
Alternative Flow	Payment fails.
Postconditions	Payment status becomes Paid.

UC-09: Manage Inventory

Item	Description
Use Case ID	UC-09
Use Case Name	Manage Inventory
Primary Actor	Pharmacy
Description	Allows pharmacies to manage medicine stock.
Preconditions	Pharmacy account is approved.
Main Flow	1. Add medicine.2. Update medicine information.3. Restock inventory.4. Delete medicine if necessary.
Alternative Flow	Invalid medicine information.
Postconditions	Inventory database is updated.

9. UC-10: Process Pharmacy Order

Item	Description
Use Case ID	UC-10
Use Case Name	Process Pharmacy Order
Primary Actor	Pharmacy
Description	Allows pharmacies to process medicine orders.
Preconditions	Order exists.
Main Flow	1. View order.2. Accept order.3. Prepare medicines.4. Mark order ready.5. Complete delivery.
Alternative Flow	Order cancelled.
Postconditions	Order status becomes completed.

10. UC-11: Upload Lab Result

Item	Description
Use Case ID	UC-11
Use Case Name	Upload Lab Result
Primary Actor	Laboratory
Description	Allows laboratories to upload test results.
Preconditions	Lab request exists.
Main Flow	1. Open request.2. Upload result file.3. Save result.4. Notify patient.
Alternative Flow	Invalid file upload.
Postconditions	Result becomes available to patient.

11. UC-12: Analyze Medical Report using AI

Item	Description
Use Case ID	UC-12
Use Case Name	Analyze Medical Report
Primary Actor	Patient
Description	Allows patients to upload reports and receive AI analysis.
Preconditions	Patient is logged in.
Main Flow	1. Upload report.2. AI processes file.3. Generate explanation.4. Recommend specialist if needed.
Alternative Flow	Unsupported file format.
Postconditions	Analysis results are displayed.

12. UC-13: Verify Healthcare Provider

Item	Description
Use Case ID	UC-13
Use Case Name	Verify Healthcare Provider
Primary Actor	Admin
Description	Allows administrators to verify doctors, pharmacies, and laboratories.
Preconditions	Registration request exists.
Main Flow	1. Review submitted documents.2. Validate credentials.3. Approve or reject request.4. Notify applicant.
Alternative Flow	Missing or invalid documents.
Postconditions	Account status is updated.

13. UC-14: Manage Users

Item	Description
Use Case ID	UC-14
Use Case Name	Manage Users
Primary Actor	Admin
Description	Allows administrators to manage platform users.
Preconditions	Administrator logged in.
Main Flow	1. View users.2. Search users.3. Activate, deactivate, or block accounts.4. Save changes.
Alternative Flow	User not found.
Postconditions	User status is updated.

14. UC-15: Generate Billing & Reports

Item	Description
Use Case ID	UC-15
Use Case Name	Generate Billing Reports
Primary Actor	Admin
Description	Allows administrators to monitor revenues and transactions.
Preconditions	Financial data exists.
Main Flow	1. Access billing module.2. Generate report.3. View revenue statistics.4. Export report if needed.
Alternative Flow	No available data.
Postconditions	Financial reports are generated successfully.

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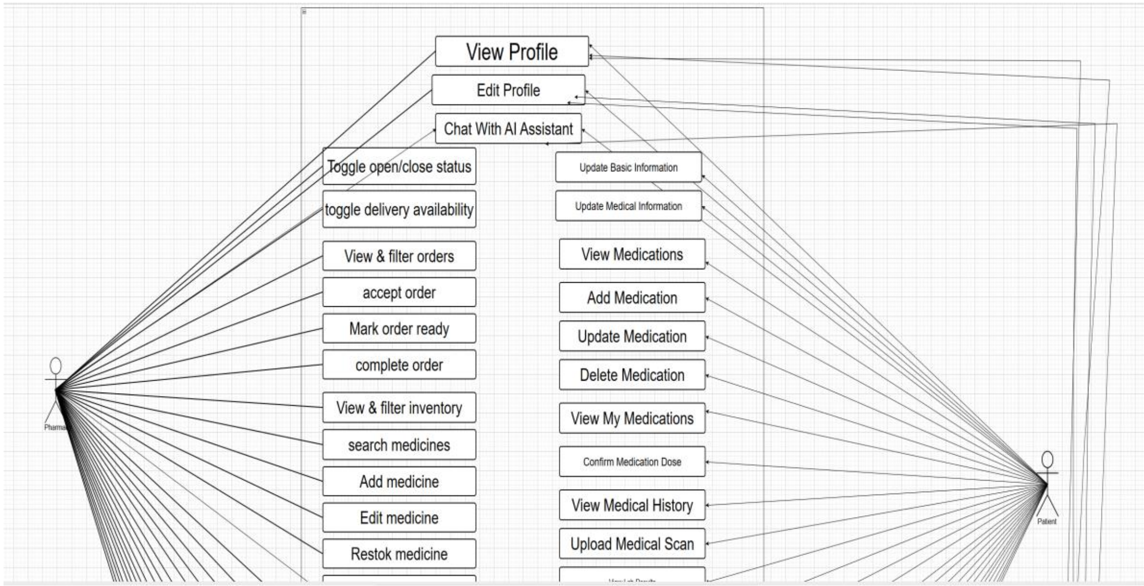
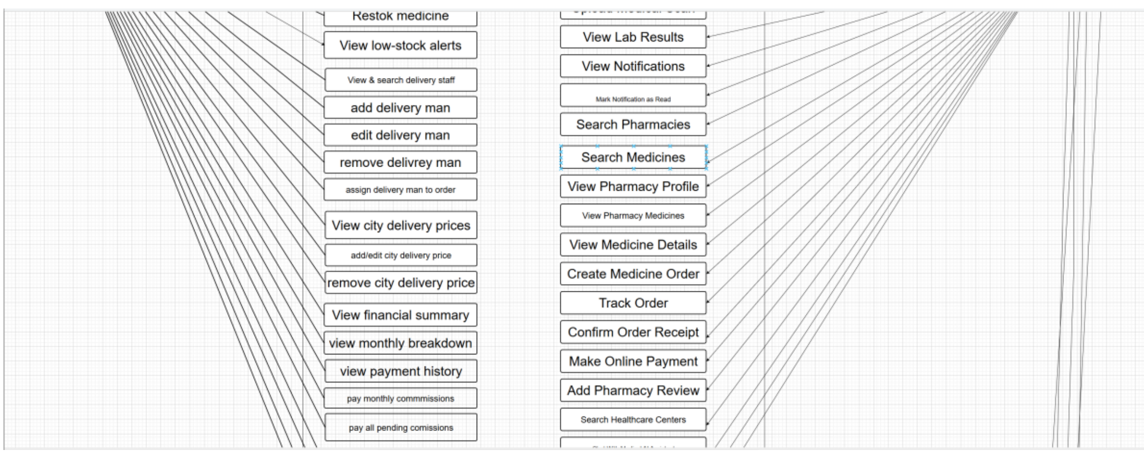
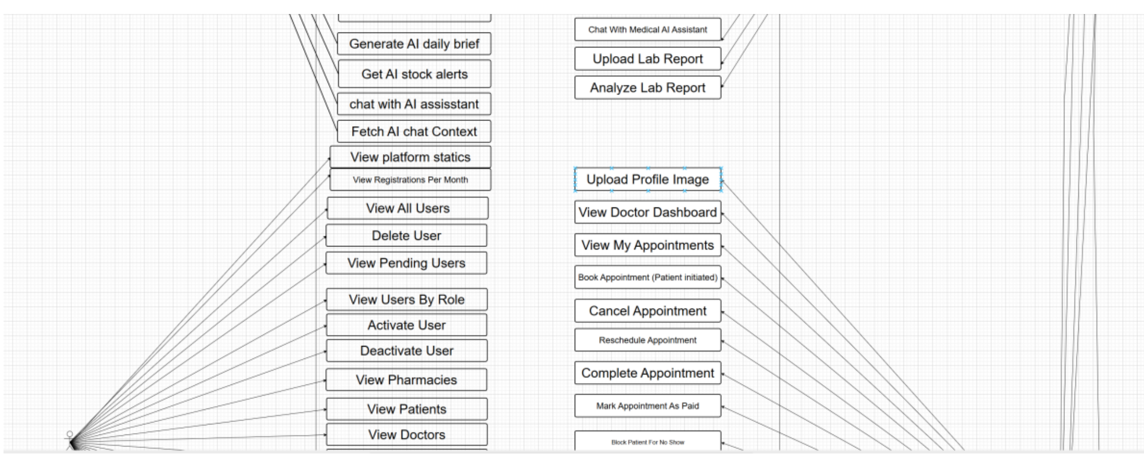
3.6.2 Use Case Diagram







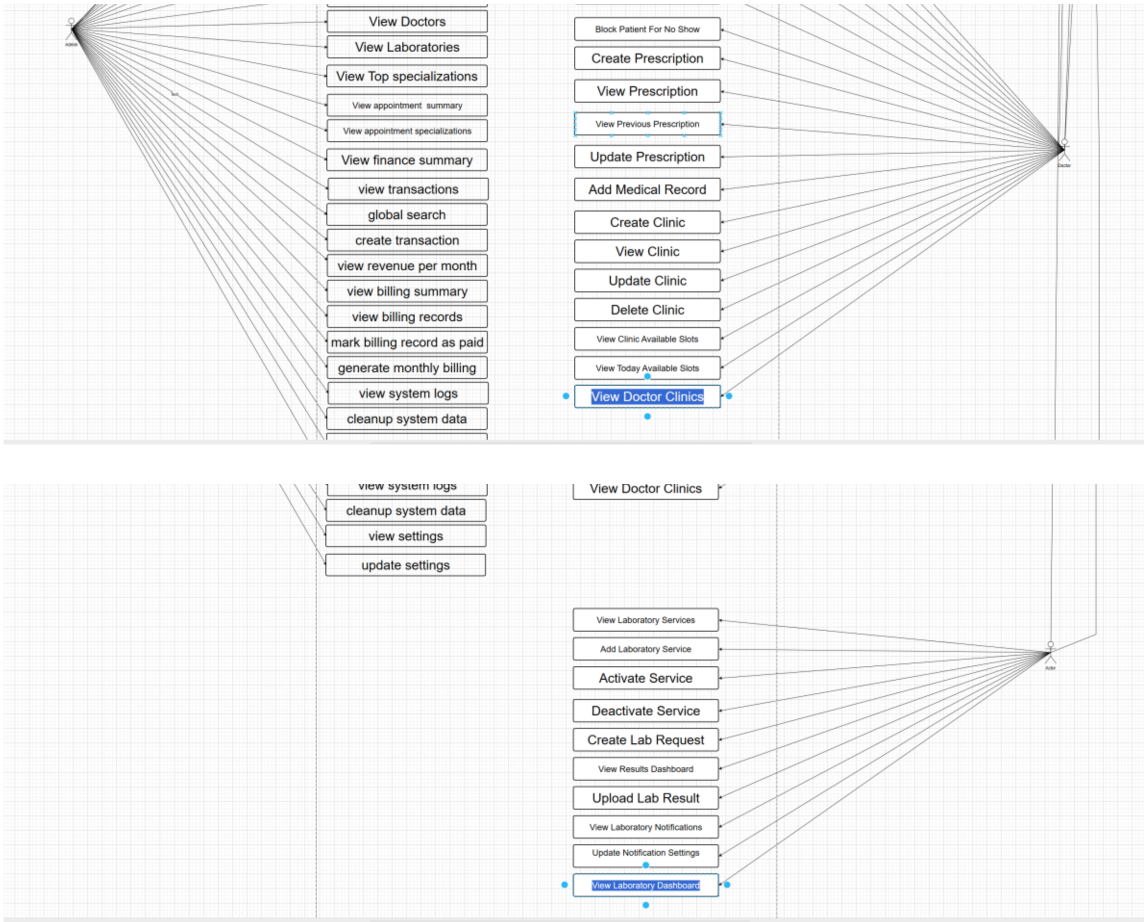


Figure 3.1: Use Case Diagram of the Chefaa Healthcare Platform

3.6.2 Class Diagram

The following class diagram illustrates the main classes of the Chefaa Healthcare Platform and the relationships between them. The diagram was derived directly from the MongoDB collections and Mongoose schemas implemented in the system.

Classes

• User Class represents all registered users in the platform, including patients, doctors, pharmacies, laboratories, and administrators. It contains authentication, authorization, and account management information.

• Patient Class represents patients who use the platform to book appointments, order medications, upload medical reports, and access healthcare services.

• Doctor Class represents healthcare providers who manage clinics, appointments, prescriptions, and patient medical records.

• Clinic Class represents medical clinics associated with doctors and contains scheduling, pricing, and location information.

• Appointment Class manages appointment reservations between patients and doctors, including scheduling and payment information.

• Prescription Class represents electronic prescriptions created by doctors and may include medications, laboratory tests, imaging requests, and follow-up recommendations.

• MedicalRecord Class stores patient medical history, diagnoses, treatments, prescriptions, and attachments.

• Lab Class represents laboratories and radiology centers registered in the system.

• Service Class represents laboratory and radiology services offered by laboratories.

• LabRequest Class manages laboratory requests submitted by patients and stores uploaded test results.

• Pharmacy Class represents pharmacies that provide medication ordering and delivery services.

• MedicineStock Class manages pharmacy inventory, medication availability, pricing, and stock quantities.

• Order Class represents medication orders created by patients and processed by pharmacies.

• DeliveryMan Class represents delivery personnel responsible for delivering pharmacy orders.

• Review Class stores ratings and feedback submitted by patients regarding doctors and pharmacies.

• Transaction Class manages financial transactions, online payments, commissions, and platform fees.

• MonthlyPayment Class stores pharmacy commission settlements and monthly payment records.

• BillingRecord Class tracks financial obligations, revenues, and billing information for platform entities.

• Notification Class manages notifications and alerts sent to users across the platform.

• RefreshToken Class handles user session management and authentication tokens.

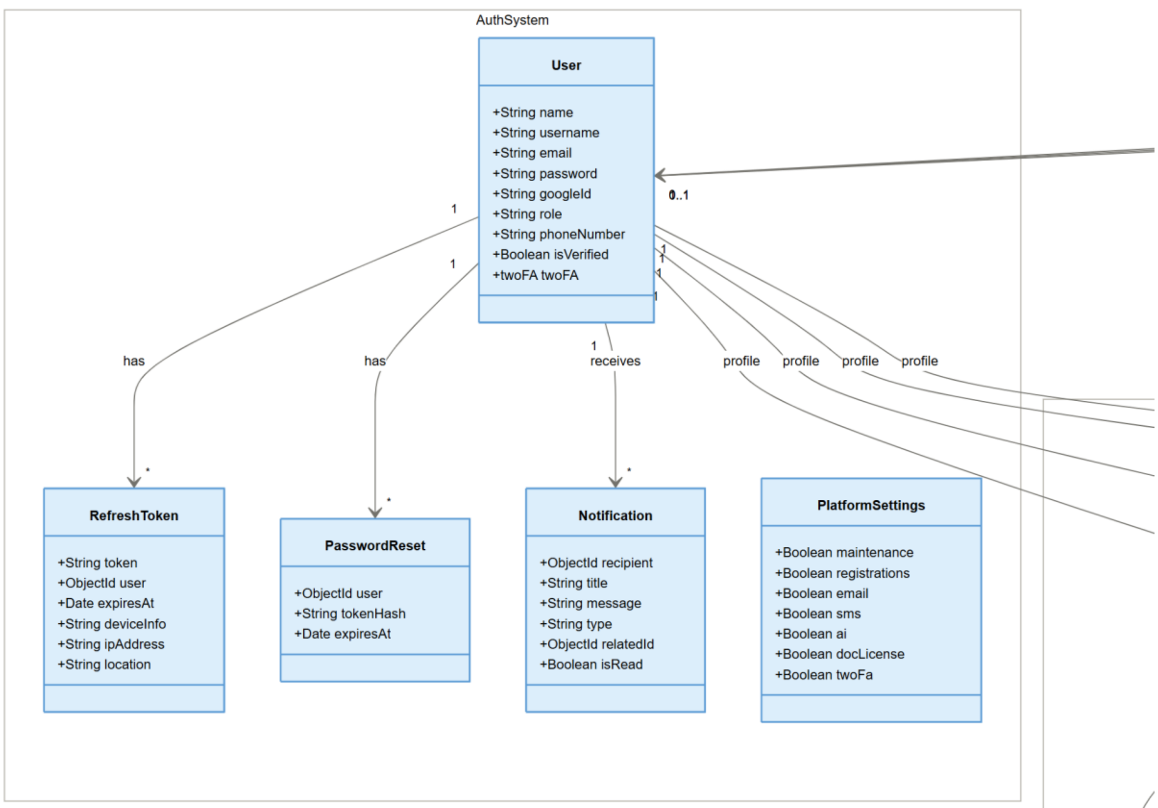
• PasswordReset Class manages password recovery requests and reset tokens.

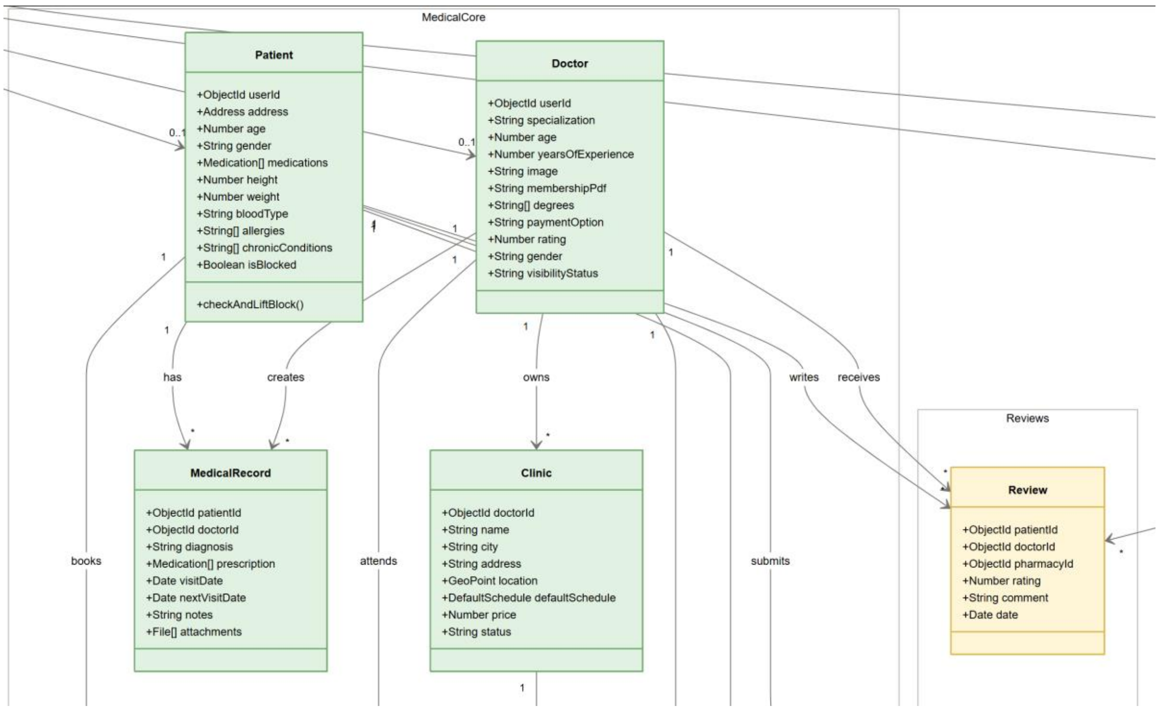
• PlatformSettings Class stores global platform configurations such as AI services, registration controls, notifications, and maintenance settings.

Relationships

The relationships between classes reflect the healthcare workflow implemented within the platform. A user may have a patient, doctor, pharmacy, or laboratory profile. Doctors manage clinics and appointments, while patients can book appointments and receive prescriptions. Prescriptions may generate medication orders through pharmacies, while laboratories process service requests and upload test results. Financial transactions, billing records, notifications, and authentication services support the overall operation of the healthcare ecosystem.

The following figure presents the complete Class Diagram of the Chefaa Healthcare Platform.




MedicalCore
Patient
+ObjectId userId
+Address address
+Number age
+String gender
+Medication[] medications
+Number height
+Number weight
+String bloodType
+String[] allergies
+String[] chronicConditions
+Boolean isBlocked
+checkAndLiftBlock()
Doctor
+ObjectId userId
+String specialization
+Number age
+Number yearsOfExperience
+String image
+String membershipPdf
+String[] degrees
+String paymentOption
+Number rating
+String gender
+String visibilityStatus
has
creates
owns
writes
receives
books
attends
submits
MedicalRecord
+ObjectId patientId
+ObjectId doctorId
+String diagnosis
+Medication[] prescription
+Date visitDate
+Date nextVisitDate
+String notes
+File[] attachments
Clinic
+ObjectId doctorId
+String name
+String city
+String address
+GeoPoint location
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+String status
Reviews
Review
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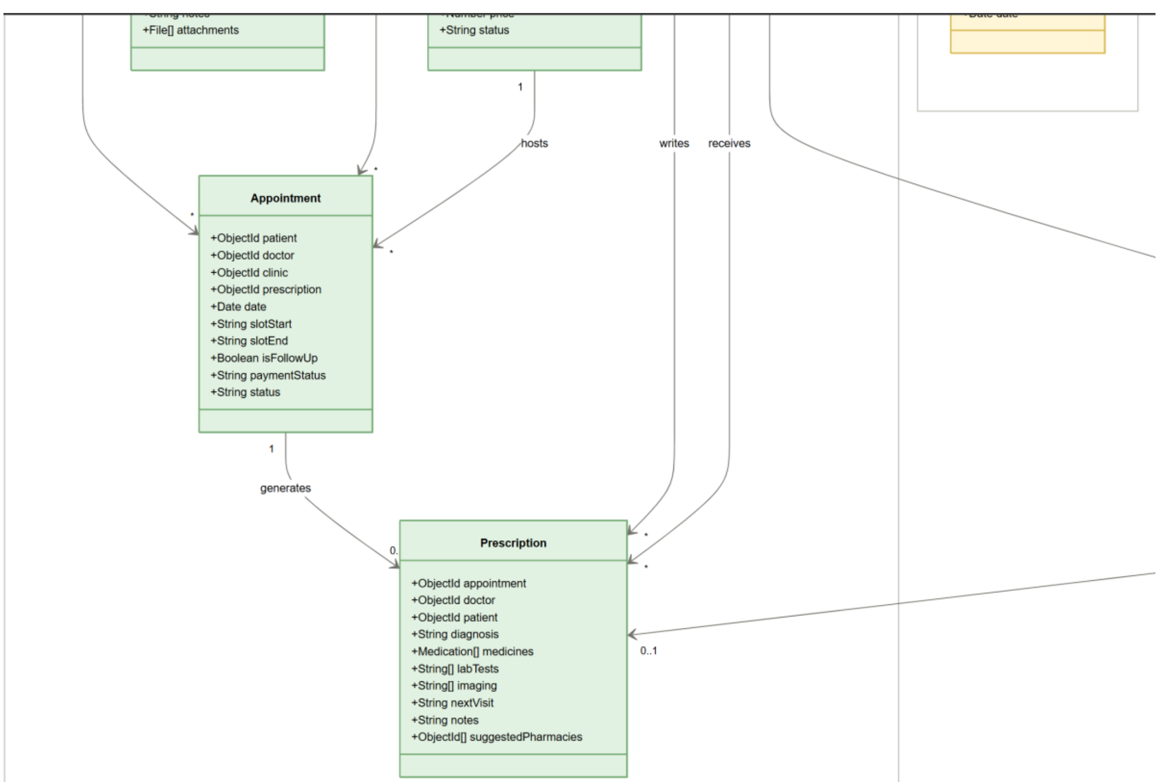
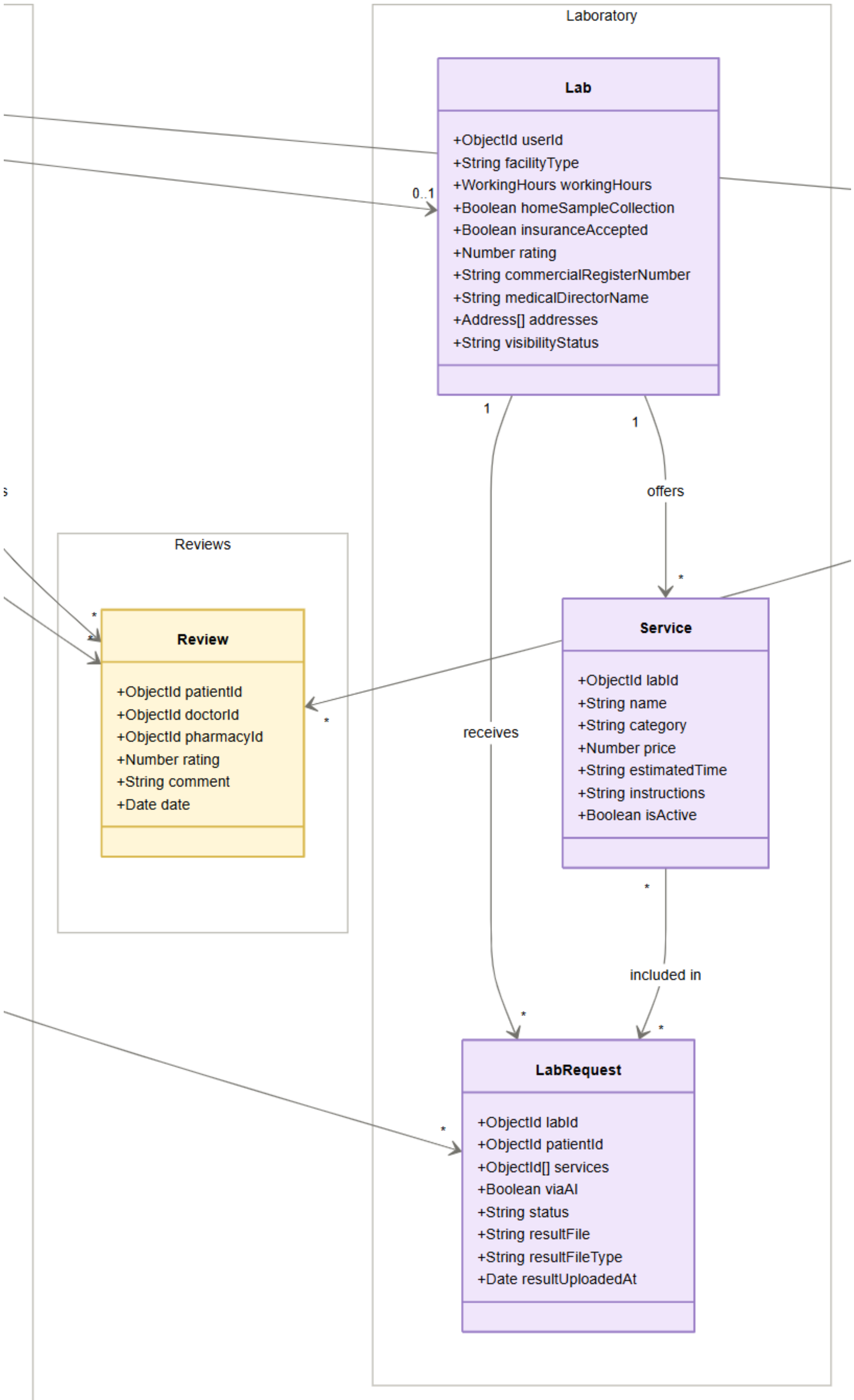
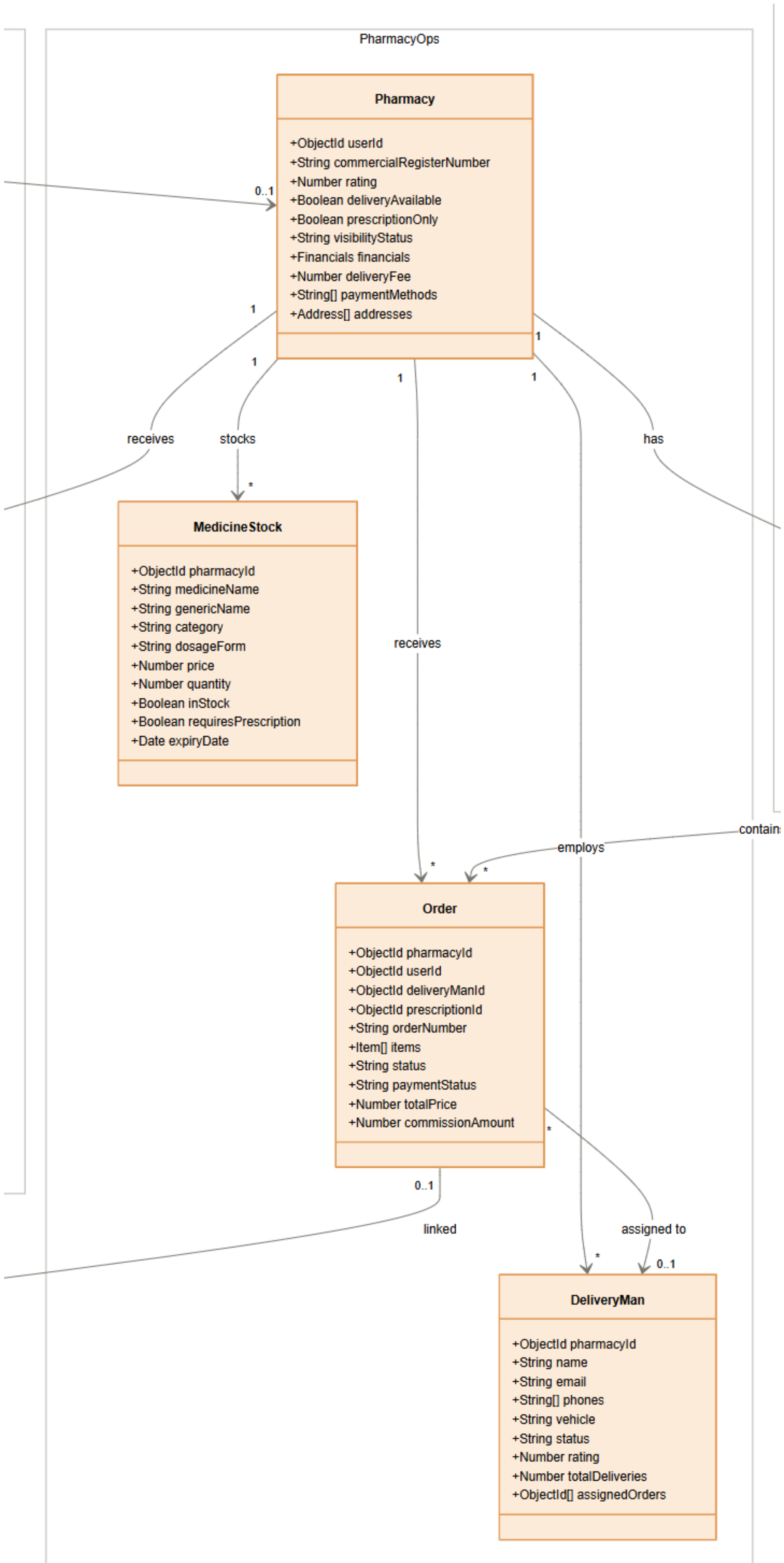
+ObjectId doctorId
+ObjectId pharmacyId
+Number rating
+String comment
+Date date


+File[] attachments
+String status
hosts
writes
receives
Appointment
+ObjectId patient
+ObjectId doctor
+ObjectId clinic
+ObjectId prescription
+Date date
+String slotStart
+String slotEnd
+Boolean isFollowUp
+String paymentStatus
+String status
generates
Prescription
+ObjectId appointment
+ObjectId doctor
+ObjectId patient
+String diagnosis
+Medication[] medicines
+String[] labTests
+String[] imaging
+String nextVisit
+String notes
+ObjectId[] suggestedPharmacies


Laboratory
Lab
+ObjectId userId
+String facilityType
+WorkingHours workingHours
+Boolean homeSampleCollection
+Boolean insuranceAccepted
+Number rating
+String commercialRegisterNumber
+String medicalDirectorName
+Address[] addresses
+String visibilityStatus
0..1
1
1
offers
*
Reviews
Review
+ObjectId patientId
+ObjectId doctorId
+ObjectId pharmacyId
+Number rating
+String comment
+Date date
*
*
Service
+ObjectId labId
+String name
+String category
+Number price
+String estimatedTime
+String instructions
+Boolean isActive
receives
*
included in
*
*
LabRequest
+ObjectId labId
+ObjectId patientId
+ObjectId[] services
+Boolean viaAI
+String status
+String resultFile
+String resultFileType
+Date resultUploadedAt
*


PharmacyOps
Pharmacy
+ObjectId userId
+String commercialRegisterNumber
+Number rating
+Boolean deliveryAvailable
+Boolean prescriptionOnly
+String visibilityStatus
+Financials financials
+Number deliveryFee
+String[] paymentMethods
+Address[] addresses
0..1
1
1
1
1
1
receives
stocks
has
*
MedicineStock
+ObjectId pharmacyId
+String medicineName
+String genericName
+String category
+String dosageForm
+Number price
+Number quantity
+Boolean inStock
+Boolean requiresPrescription
+Date expiryDate
receives
employs
contain
*
*
Order
+ObjectId pharmacyId
+ObjectId userId
+ObjectId deliveryManId
+ObjectId prescriptionId
+String orderNumber
+Item[] items
+String status
+String paymentStatus
+Number totalPrice
+Number commissionAmount
*
0..1
linked
assigned to
*
0..1
DeliveryMan
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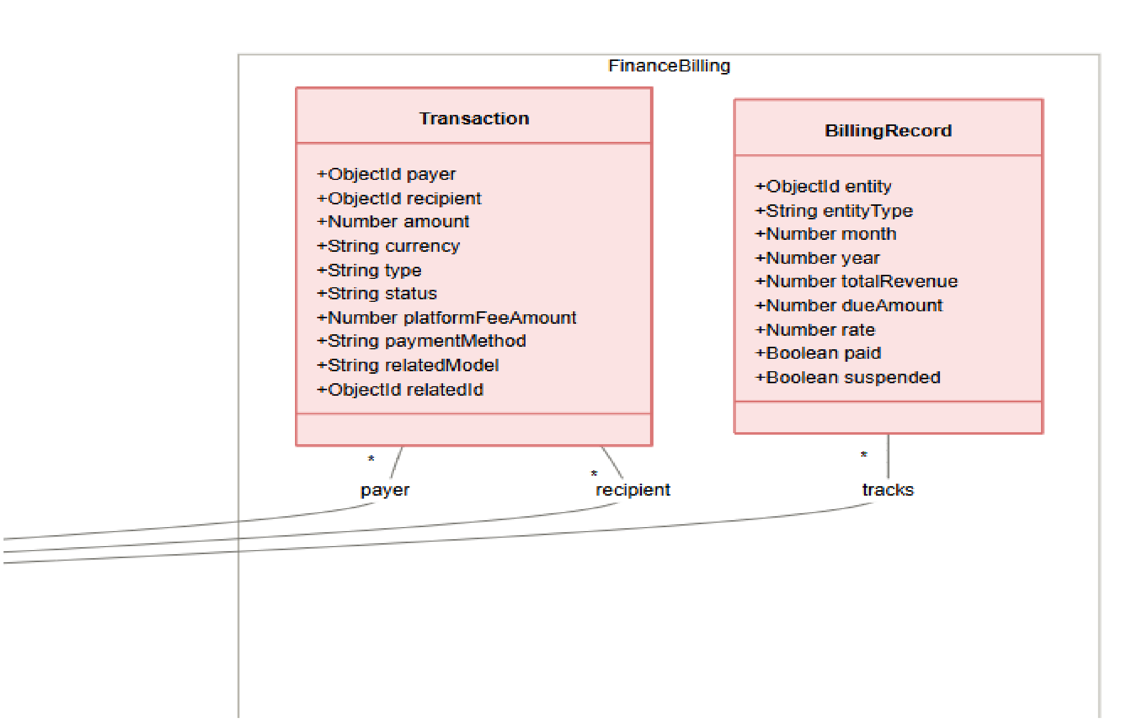
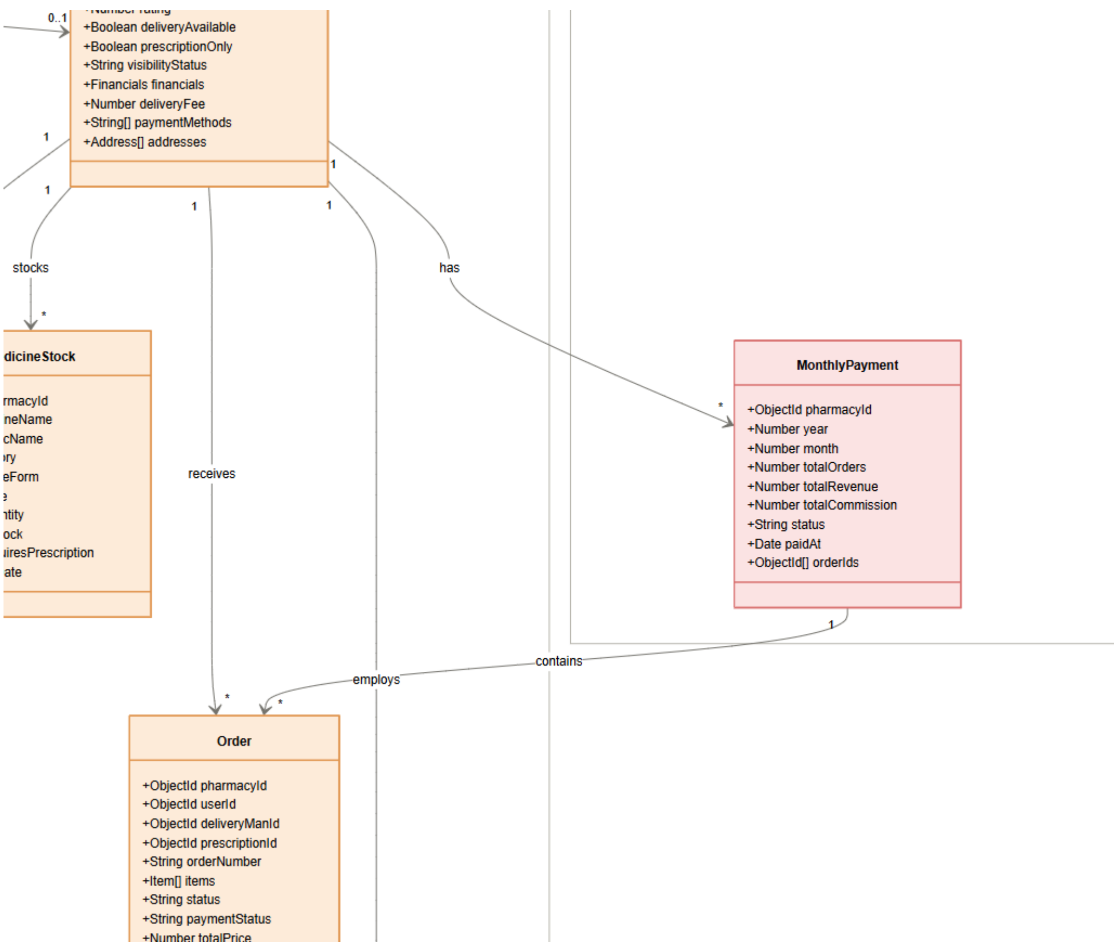
+String name
+String email
+String[] phones
+String vehicle
+String status
+Number rating
+Number totalDeliveries
+ObjectId[] assignedOrders


FinanceBilling
Transaction
+ObjectId payer
+ObjectId recipient
+Number amount
+String currency
+String type
+String status
+Number platformFeeAmount
+String paymentMethod
+String relatedModel
+ObjectId relatedId
BillingRecord
+ObjectId entity
+String entityType
+Number month
+Number year
+Number totalRevenue
+Number dueAmount
+Number rate
+Boolean paid
+Boolean suspended
*
payer
*
recipient
*
tracks


0..1
+Boolean deliveryAvailable
+Boolean prescriptionOnly
+String visibilityStatus
+Financials financials
+Number deliveryFee
+String[] paymentMethods
+Address[] addresses
1
1
1
1
1
stocks
*
has
receives
MonthlyPayment
+ObjectId pharmacyId
+Number year
+Number month
+Number totalOrders
+Number totalRevenue
+Number totalCommission
+String status
+Date paidAt
+ObjectId[] orderIds
*
1
contains
employs
*
*
Order
+ObjectId pharmacyId
+ObjectId userId
+ObjectId deliveryManId
+ObjectId prescriptionId
+String orderNumber
+Item[] items
+String status
+String paymentStatus

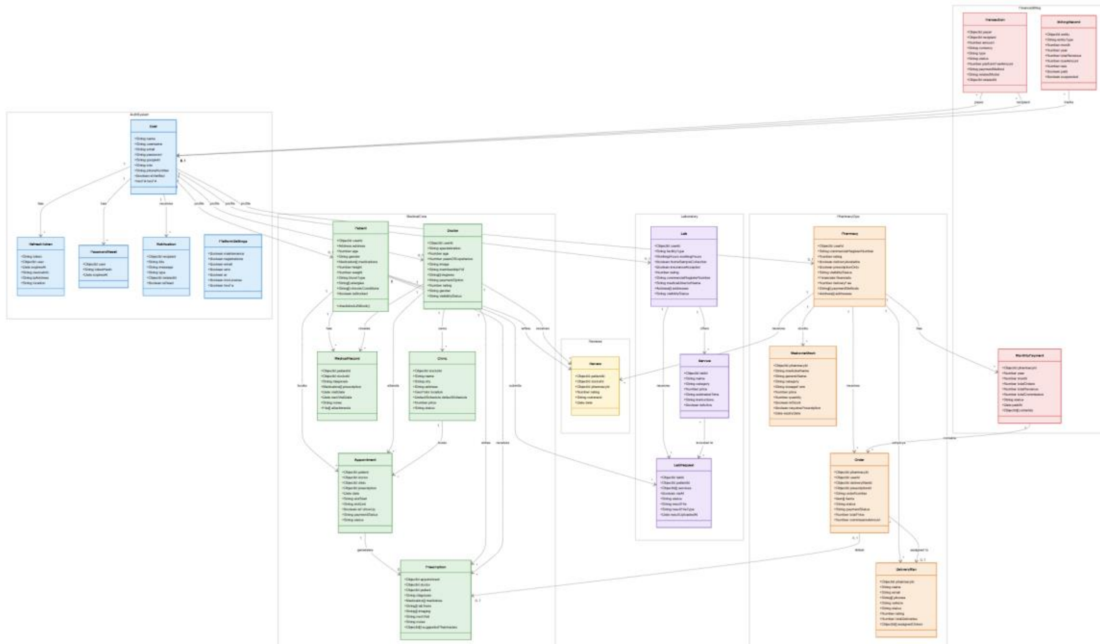


Figure 3.2: Class Diagram of the Chefaa Healthcare Platform

3.6.3 Sequence Diagram

Doctor Registration and Admin Approval Sequence Diagram

The following sequence diagram illustrates the interactions between a doctor, the system, and the administrator during the doctor registration and verification process in the Chefaa Healthcare Platform. The diagram demonstrates how doctors create accounts, submit professional verification documents, undergo administrative review, and receive approval or rejection notifications.

Participants:

- **Doctor:** The healthcare professional who registers and submits verification documents.
- **System:** Handles account creation, document storage, verification requests, and notifications.
- **Admin:** Reviews submitted documents and decides whether to approve or reject the doctor's application.
- **Notification:** Sends approval or rejection messages to doctors.

Flow of Interactions:

1. Doctor Registration

- Doctor → System: Submits registration information including personal details, specialization, and account credentials.
- System → Doctor: Validates registration data and creates a doctor account with an "Unverified" status.

2. Upload Verification Documents

- Doctor → System: Uploads professional verification documents (medical license, membership certificate, or professional credentials).

• System → System: Stores uploaded documents securely in the database.

• System → Admin: Generates a verification request and notifies the administrator.

3. Review Verification Request

• Admin → System: Requests the list of pending doctor verification applications.

• System → Admin: Returns submitted doctor information and uploaded documents.

• Admin → System: Reviews the provided credentials and supporting documents.

4. Verification Decision

Alt [Documents Approved]

• Admin → System: Approves the doctor's verification request.

• System → System: Updates the doctor's account status from "Unverified" to "Verified".

• System → Notification: Generates an approval notification.

• Notification → Doctor: Sends account approval confirmation.

5. Rejection Flow

Alt [Documents Rejected]

• Admin → System: Rejects the verification request and optionally provides a rejection reason.

• System → System: Records the rejection decision.

• System → Notification: Generates a rejection notification.

• Notification → Doctor: Sends rejection message with the reason for rejection.

6. Access Granted

• Doctor → System: Attempts to access doctor services.

• System → Doctor: Grants access only if the account status is verified.

7. Send Notifications

• System → Notification: Creates status update notifications.

• Notification → Doctor: Sends verification status updates and account activation information.

This sequence diagram ensures that all doctors undergo a secure verification process before being allowed to provide healthcare services through the platform. The workflow improves platform credibility, protects patients from unverified practitioners, and maintains compliance with healthcare service requirements.

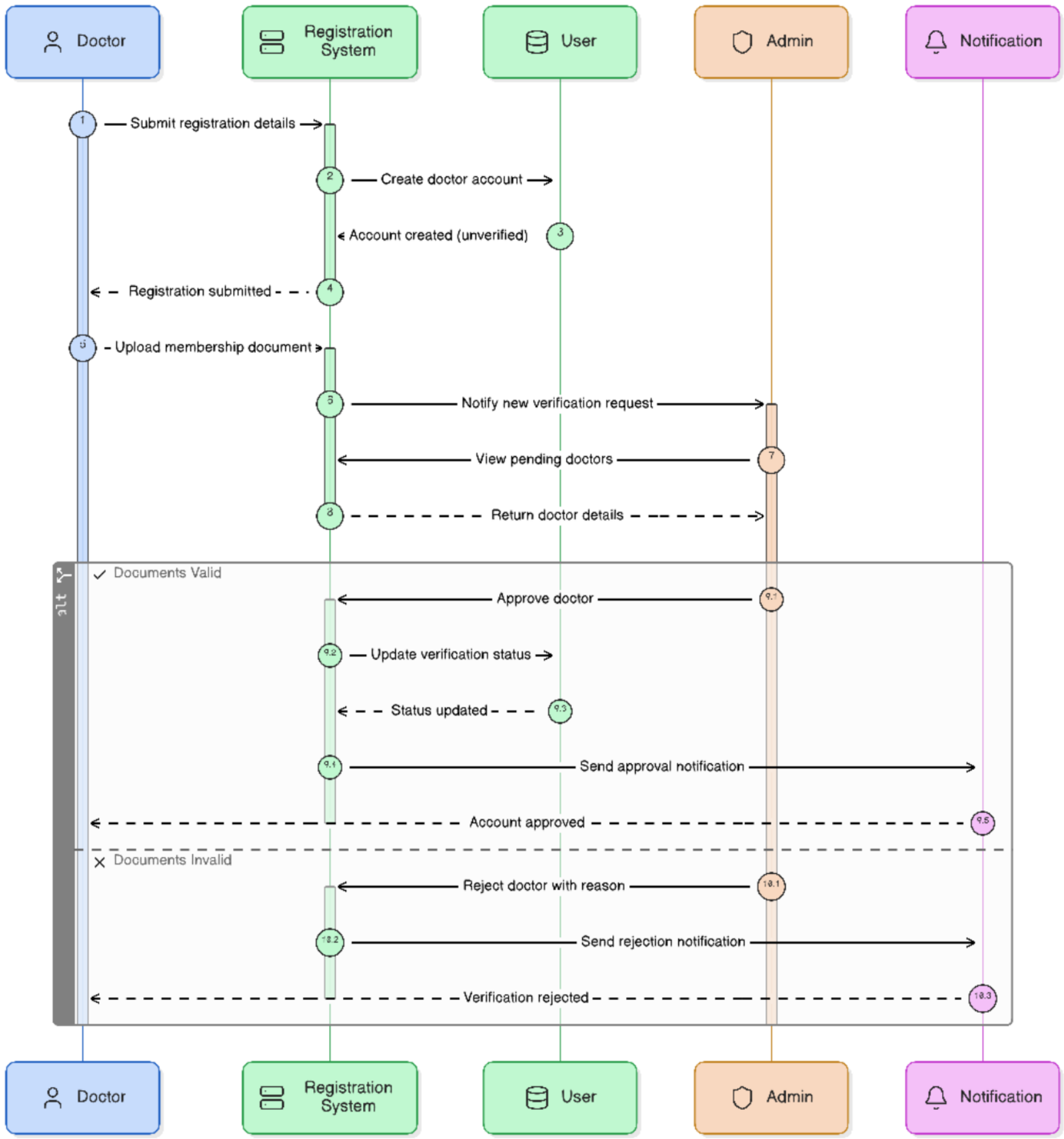


Figure 3.6.1: Doctor Registration and Admin Approval Sequence Diagram

15. Create Prescription Sequence Diagram

The following sequence diagram illustrates the workflow of a doctor creating and issuing an electronic prescription within the Chefaa Healthcare Platform during a patient appointment. It demonstrates how medical data is retrieved, how prescriptions are generated, and how patients are notified once a prescription is issued.

Participants:

• **Doctor:** The healthcare provider responsible for diagnosing the patient and issuing the prescription.

• **System:** Manages appointment data, patient medical records, prescription creation, and data storage.

• **Appointment System:** Provides access to appointment details and links prescriptions to specific visits.

• **Patient Record:** Stores and retrieves patient medical history and clinical information.

• **Notification Service:** Sends alerts to patients when a new prescription is created.

Flow of Interactions:

1. Access Appointment

• Doctor → Appointment System: Opens a specific patient appointment.

• Appointment System → Doctor: Returns appointment details including patient information and visit context.

2. Retrieve Patient Medical Data

• Doctor → System: Requests patient medical history.

• System → Patient Record: Fetches stored medical data including previous diagnoses, allergies, and treatments.

• Patient Record → System: Returns complete patient medical information.

• System → Doctor: Displays patient medical data for clinical evaluation.

3. Create Prescription

• Doctor → System: Initiates prescription creation.

• Doctor → System: Adds diagnosis, medications, dosage instructions, laboratory tests, imaging requests, and medical notes.

• System → System: Validates and stores prescription data.

4. Link Prescription to Appointment

• System → Appointment System: Attaches prescription to the corresponding appointment record.

• Appointment System → System: Confirms successful linking of prescription and visit.

• System → Doctor: Confirms that the prescription has been saved successfully.

5. Send Patient Notification

• System → Notification Service: Triggers prescription notification event.

• Notification Service → Patient: Sends alert informing the patient that a new prescription has been issued and is available for viewing.

6. Finalization

• Doctor → System: Completes the consultation session.

• System → Appointment System: Updates appointment status to "Completed" with prescription attached.

This sequence ensures that prescriptions are accurately generated, securely stored, and properly associated with patient appointments. It also guarantees immediate communication with patients, improving continuity of care and overall healthcare efficiency.

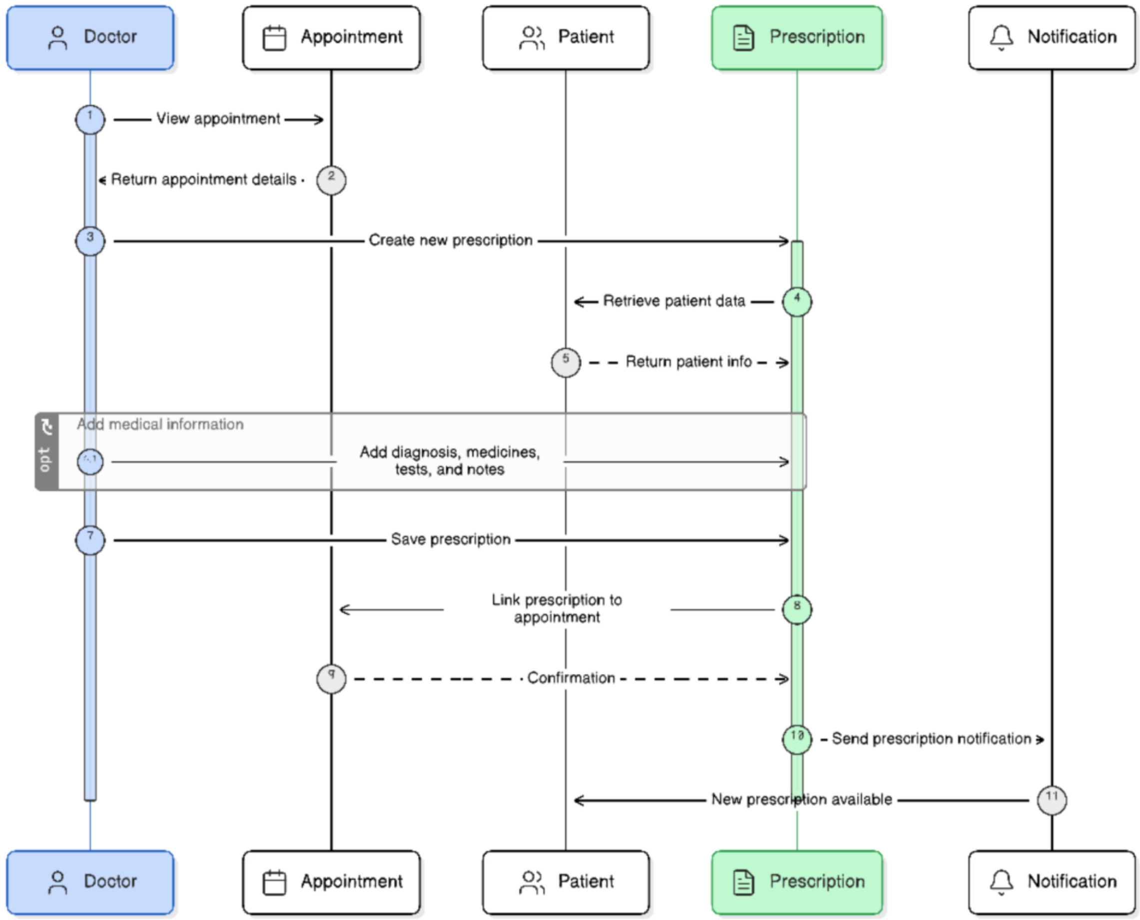


Figure 3.6.2: Create Prescription Sequence Diagram

16. Pharmacy Order Process Sequence Diagram

The following sequence diagram illustrates the complete lifecycle of a medicine ordering process within the Pharmacy module of the Healthcare Platform. It covers all stages starting from product search, order creation, payment processing, pharmacy handling, and final delivery to the patient.

Participants:

• Patient: The user who searches for medicines and places an order.

• Pharmacy System: Handles medicine search, stock validation, order processing, and status management.

• Stock Database: Stores real-time information about medicine availability and inventory levels.

• Payment Gateway: Processes online payments when prepayment is selected.

• Pharmacy: Responsible for accepting, preparing, and packaging the order.

• Delivery Personnel: Handles delivery of the order to the patient.

• Notification Service: Sends updates regarding order status changes.

Flow of Interactions:

1. Search for Medicines

• Patient → Pharmacy System: Searches for required medicines.

• Pharmacy System → Stock Database: Queries medicine availability and stock status.

• Stock Database → Pharmacy System: Returns available medicines and quantities.

• Pharmacy System → Patient: Displays available products.

2. Create Order

• Patient → Pharmacy System: Selects medicines and creates an order request.

• Pharmacy System → Stock Database: Validates stock availability for selected items.

• Stock Database → Pharmacy System: Confirms or rejects availability.

• Pharmacy System → Patient: Displays order summary and confirmation details.

3**. Payment Process (Alternative Flow)**

Alt [Online Payment]

• Patient → Payment Gateway: Initiates online payment.

• Payment Gateway → Pharmacy System: Confirms successful transaction.

• Pharmacy System → Patient: Updates order status as "Paid" and confirmed.

Alt [Cash on Delivery]

• Pharmacy System → Patient: Marks order as "Pending Payment" to be collected upon delivery.

4. **Order Confirmation**

• Pharmacy System → Pharmacy: Sends confirmed order details for preparation.

• Pharmacy → Pharmacy System: Accepts the order and updates status to "Preparing."

5. **Delivery Assignment**

• Pharmacy → Delivery Personnel: Assigns order for delivery.

• Delivery Personnel → Pharmacy System: Updates status to "On the Way."

• Delivery Personnel → Patient: Delivers the order.

6. **Order Completion**

• Patient → Pharmacy System: Confirms receipt of order.

• Pharmacy System → Stock Database: Updates inventory after order fulfillment.

• Pharmacy System → Notification Service: Sends order completion notification.

• Notification Service → Patient: Confirms successful delivery.

This sequence ensures a complete and structured workflow for pharmacy operations, including inventory validation, payment handling, order tracking, and final delivery, improving efficiency and reliability of the healthcare platform.

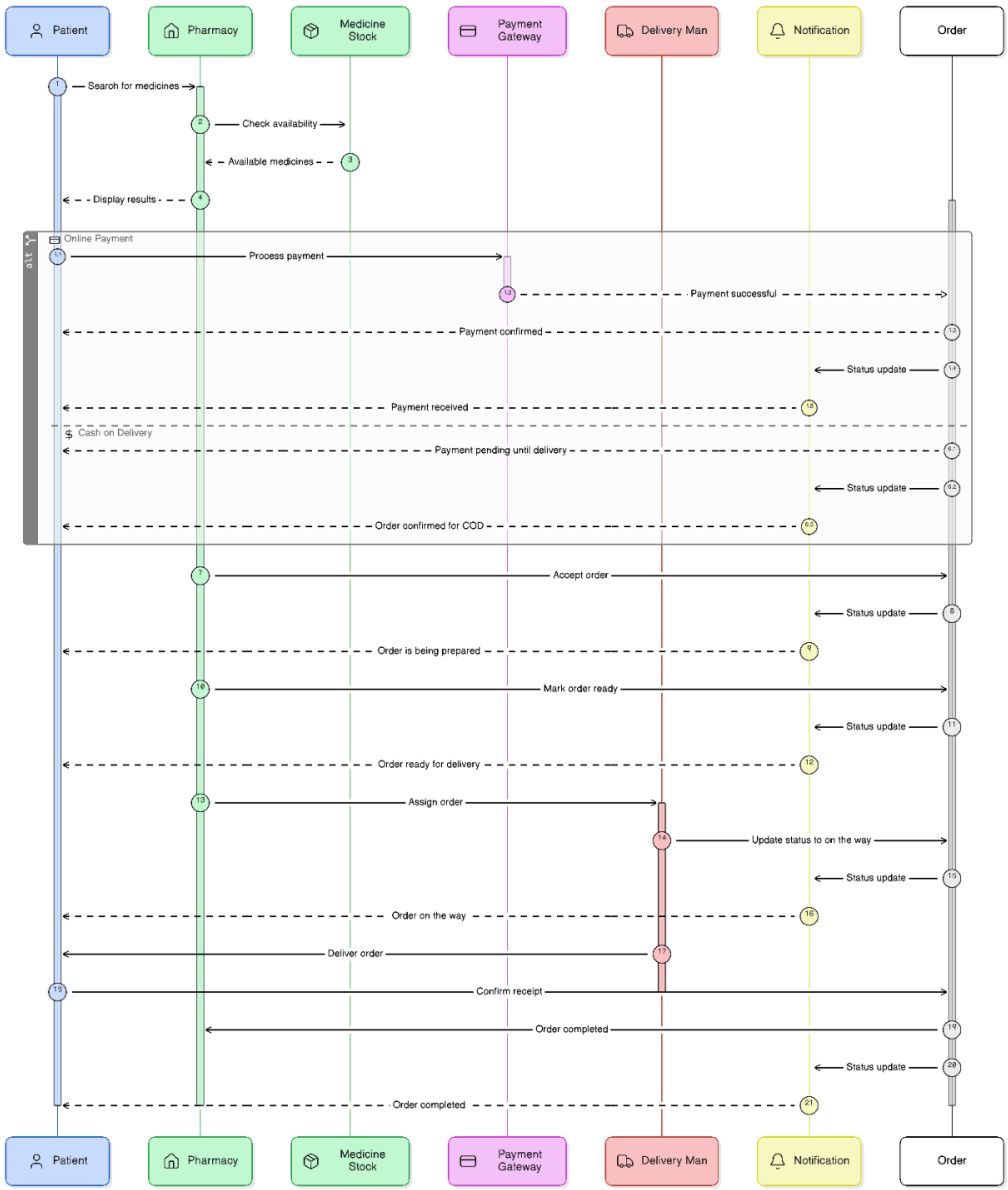


Figure 3.6.3: Pharmacy Order Process Sequence Diagram

17. Lab Request and Upload Result Sequence Diagram

The following sequence diagram describes the end-to-end workflow of laboratory services within the Healthcare Platform. It illustrates how patients request lab tests, how laboratories process these requests, and how results are uploaded and delivered back to patients through the system.

Participants:

• **Patient:** The user who requests laboratory tests and views the final results.

• **Lab System:** Manages lab services, request handling, and result storage.

• **Lab Service Database:** Stores available laboratory tests and their details.

• **Laboratory:** Responsible for accepting requests, performing tests, and uploading results.

• **Notification Service:** Sends alerts to patients when results are available.

Flow of Interactions:

1. Browse Laboratory Services

• Patient → Lab System: Requests to view available laboratory services.

• Lab System → Lab Service Database: Retrieves list of available tests and services.

• Lab Service Database → Lab System: Returns available laboratory services.

• Lab System → Patient: Displays available lab tests.

2. Submit Lab Request

• Patient → Lab System: Selects a test and submits a lab request.

• Lab System → Laboratory: Forwards the lab request for processing.

• Laboratory → Lab System: Accepts and confirms the request.

• Lab System → Patient: Confirms that the request has been successfully submitted.

3. Test Processing

• Laboratory → Laboratory: Performs the required medical test.

• Laboratory → Lab System: Updates test status to "In Progress" during execution.

4. Upload Lab Results

• Laboratory → Lab System: Uploads completed test results and attachments.

• Lab System → Lab System: Stores results securely and links them to the patient record.

5. Notify Patient

• Lab System → Notification Service: Triggers result availability notification.

• Notification Service → Patient: Sends alert informing the patient that lab results are ready for viewing.

6. View Results

• Patient → Lab System: Requests to view uploaded lab results.

• Lab System → Patient: Displays complete test results and reports.

This sequence ensures a structured laboratory workflow, enabling smooth communication between patients and laboratories while maintaining accurate tracking, secure data storage, and timely result delivery.

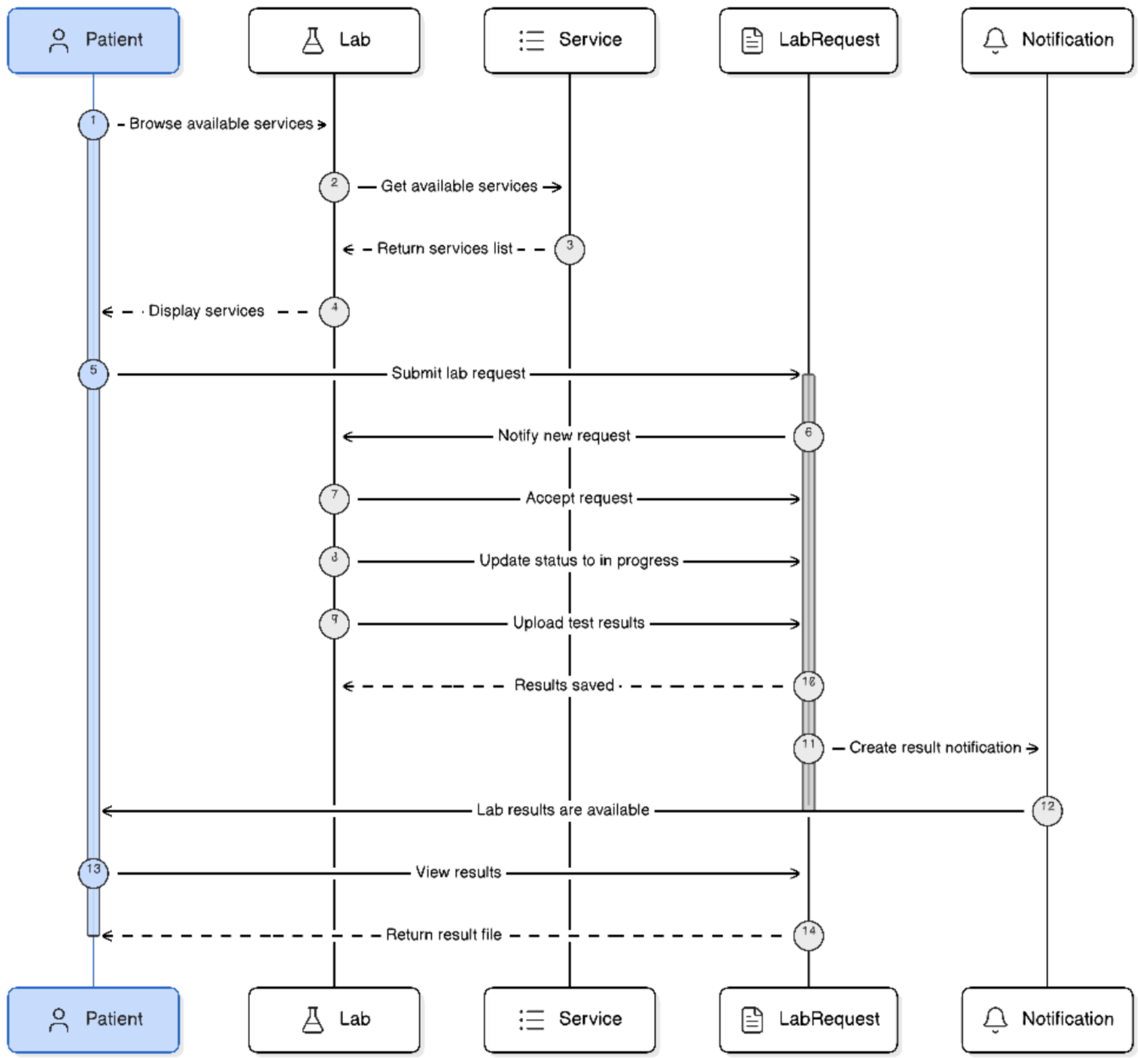


Figure 3.6.4: Lab Request and Upload Result Sequence Diagram

AI Medical Report Analysis Sequence Diagram

The following sequence diagram describes the workflow of the AI-powered medical report analysis feature within the Healthcare Platform. It illustrates how patient-uploaded medical reports are validated, processed using AI models, analyzed, and finally presented to the user along with recommendations.

Participants:

• **Patient:** The user who uploads medical reports for AI-based analysis and receives results.

• **AI Analysis System:** Handles file validation, data extraction, AI processing, and result generation.

• **File Validator:** Ensures uploaded medical files meet format and quality requirements before processing.

• **AI Engine:** Performs intelligent analysis of extracted medical data using trained AI models.

• **Recommendation Engine:** Generates interpretations, insights, and suggestions for relevant medical specialists.

• **Database:** Stores uploaded reports and generated analysis results.

• **Notification Service:** Notifies the patient when analysis is completed.

Flow of Interactions:

1. Upload Medical Report

• Patient → AI Analysis System: Uploads medical report or diagnostic file.

• AI Analysis System → File Validator: Sends file for validation.

• File Validator → AI Analysis System: Confirms whether the file is valid or rejected.

2. Data Extraction and Processing

• AI Analysis System → AI Engine: Sends validated report for processing.

• AI Engine → AI Engine: Extracts relevant medical data and performs AI-based interpretation.

• AI Engine → AI Analysis System: Returns analyzed medical insights.

3. Generate Recommendations

• AI Analysis System → Recommendation Engine: Sends AI analysis results.

• Recommendation Engine → AI Analysis System: Generates possible interpretations and recommended specialist consultations.

• AI Analysis System → Patient: Displays analysis results and suggested medical guidance.

4. Store Results

• AI Analysis System → Database: Stores uploaded reports and generated analysis outputs.

• Database → AI Analysis System: Confirms successful storage.

5. Send Notification

• AI Analysis System → Notification Service: Triggers completion notification.

• Notification Service → Patient: Sends alert that medical report analysis is completed and available.

6. Access Results

• Patient → AI Analysis System: Requests to view previous analysis reports.

• AI Analysis System → Patient: Retrieves and displays stored analysis history.

This sequence demonstrates the integration of artificial intelligence within the healthcare platform to enhance patient support by providing automated medical report interpretation, improving accessibility to insights, and assisting in early medical guidance.

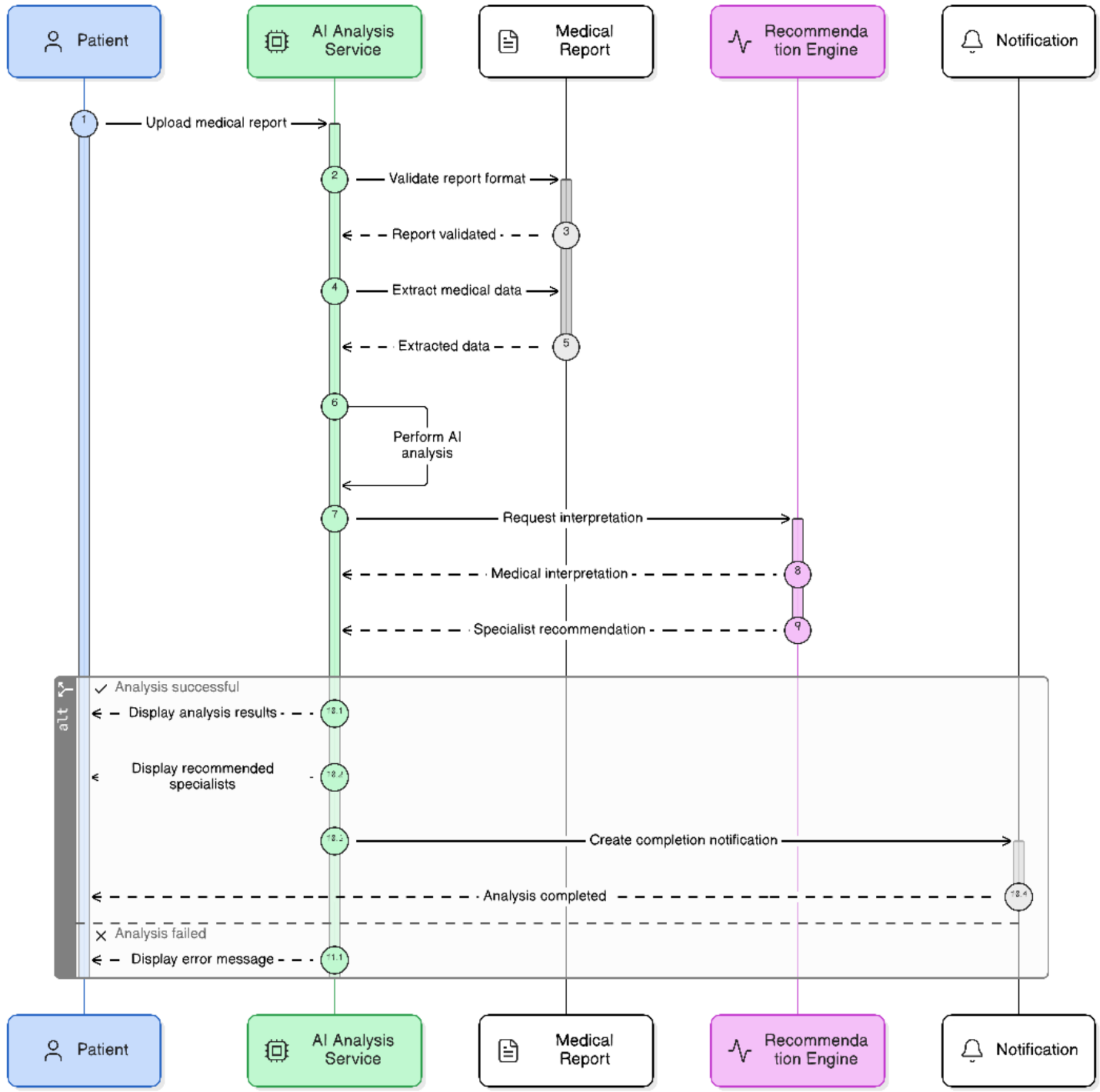


Figure 3.6.5:AI Medical Report Analysis Sequence Diagram

3.6.4 Data Flow Diagram

Data Flow Diagram (DFD)

A Data Flow Diagram (DFD) is a graphical representation used to illustrate how data moves throughout a system. It shows the flow of information between external entities, system processes, and data stores. The DFD helps in understanding how the Chefaa Healthcare Platform processes, stores, and exchanges data among different stakeholders, including patients, doctors, pharmacies, laboratories, administrators, and external services.

The Chefaa platform integrates multiple healthcare services within a single ecosystem, enabling appointment management, prescription handling, laboratory services, pharmacy orders, online payments, and AI-powered healthcare assistance. The DFD provides a clear visualization of these interactions and demonstrates how information flows across the system components.

1. Level 0 Data Flow Diagram (Context Diagram)

The Level 0 Data Flow Diagram (DFD), also known as the Context Diagram, presents a high-level view of the Chefaa Healthcare Platform. At this level, the entire platform is represented as a single process that interacts with external entities through data flows. The purpose of this diagram is to define the system boundaries and illustrate how information enters and leaves the platform without showing internal processing details.

Main Process

0.0 Chefaa Healthcare Platform

The Chefaa Healthcare Platform serves as an integrated healthcare ecosystem that connects patients, doctors, pharmacies, laboratories, administrators, and external services. The platform facilitates healthcare service delivery by managing appointments, prescriptions, laboratory requests, medication orders, online payments, notifications, and AI-powered healthcare assistance.

External Entities and Their Roles

1. Patient

The Patient is the primary user of the platform who utilizes healthcare services provided through the system.

Data Flow from Patient to System:

• Appointment booking requests.

• Medication orders.

• Medical reports and laboratory documents.

• Payment information.

• Search and consultation requests.

Data Flow from System to Patient:

• Appointment confirmations.

• Prescription information.

• Laboratory results.

• Order status updates.

• Notifications and reminders.

---

2. Doctor

Doctors provide healthcare consultations and medical services through the platform.

Data Flow from Doctor to System:

• Profile information.

• Clinic schedules.

• Prescriptions.

• Appointment updates.

Data Flow from System to Doctor:

• Appointment requests.

• Patient information.

• Revenue analytics.

• Notifications and reminders.

---

3. Pharmacy

Pharmacies manage medication inventories and process patient orders.

Data Flow from Pharmacy to System:

• Inventory information.

• Medicine availability.

• Order processing updates.

Data Flow from System to Pharmacy:

• Medication orders.

• Prescription details.

• Delivery requests.

• Financial reports.

---

4. Laboratory

Laboratories provide diagnostic and radiology services through the platform.

Data Flow from Laboratory to System:

• Laboratory reports.

• Diagnostic results.

• Service information.

Data Flow from System to Laboratory:

• Laboratory requests.

• Patient information.

• Test requirements.

---

5. Administrator

The Administrator is responsible for monitoring and managing the entire platform.

Data Flow from Administrator to System:

• User management operations.

• Configuration settings.

• Approval and verification decisions.

Data Flow from System to Administrator:

• Analytics reports.

• System monitoring data.

• Financial reports.

---

6. Payment Gateway

The Payment Gateway is an external service responsible for processing online payments.

Data Flow from System to Payment Gateway:

• Payment requests.

• Transaction details.

Data Flow from Payment Gateway to System:

• Payment confirmations.

• Transaction status updates.

---

7. AI Services

AI Services provide intelligent healthcare support and data analysis.

Data Flow from System to AI Services:

• Medical reports.

• Laboratory results.

• User questions and requests.

Data Flow from AI Services to System:

• Analysis results.

• Medical recommendations.

• AI chatbot responses.

---

8. SMS / Push Notification Service

The Notification Service is responsible for delivering alerts, reminders, and updates to users.

Data Flow from System to Notification Service:

• Appointment reminders.

• Prescription notifications.

• Order updates.

Data Flow from Notification Service to System:

• Delivery status confirmations.

---

9. Email / Cloud Storage Service

The Email and Cloud Storage Service handles document storage, backups, and email communications.

Data Flow from System to Email / Cloud Storage:

• Medical reports.

• Prescriptions.

• User documents.

• Backup files.

Data Flow from Email / Cloud Storage to System:

• Stored documents.

• Backup data.

• Email delivery confirmations.

The following figure illustrates the Context Diagram (Level 0 DFD) of the Shefaa Healthcare Platform and highlights the interactions between the platform and its external entities.

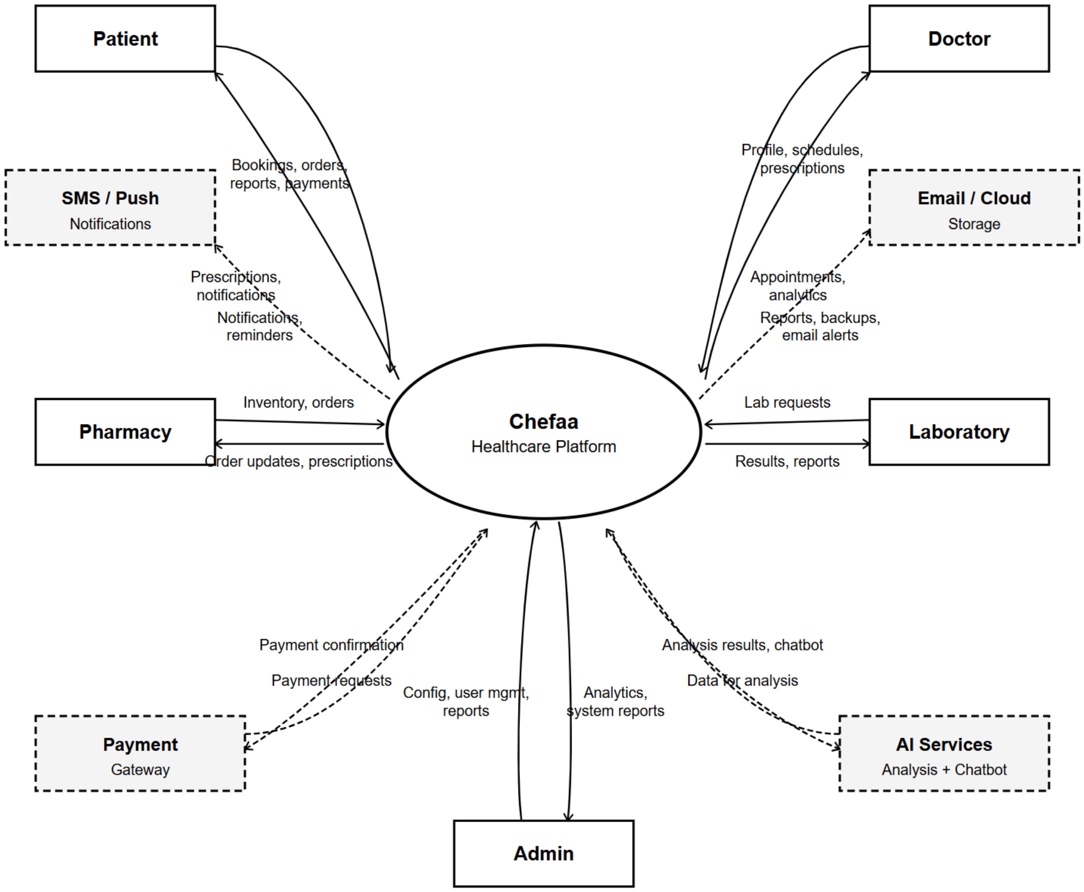


Figure 3.3: Class Diagram of the Chefaa Healthcare Platform

2. Level 1 Data Flow Diagram

The Level 1 Data Flow Diagram (DFD) provides a detailed decomposition of the Healthcare Management System into its major functional processes. Unlike the Context Diagram (Level 0), which represents the entire system as a single process, the Level 1 DFD illustrates the main subsystems, their interactions with external entities, and the data stores used to support system operations.

The system is divided into eight major processes:

P1 – Registration & Verification

This process handles user registration and login activities. Patients can create accounts and authenticate their identities. User information is stored and retrieved from the Users Database (DS1) to ensure secure access to the system.

P2 – Appointment Management

This process enables patients to search for doctors and book appointments. Doctors can manage their available schedules and appointment slots. Appointment information is maintained in the Appointments Database (DS2) to support scheduling and appointment tracking.

P3 – Prescription Management

Doctors create and manage electronic prescriptions for patients through this process. Prescription details are stored in the Prescriptions Database (DS3), while relevant medical information is recorded in the Medical Records Database (DS4). Patients can access their prescriptions through the system.

P4 – Pharmacy Orders

This process allows patients to order prescribed medications from participating pharmacies. Pharmacies receive and process medicine orders, update order statuses, and manage inventory information. Order records are stored in the Orders Database (DS5), while medicine availability is maintained in the Medicine Inventory Database (DS6).

P5 – Lab Result Management

Doctors can submit laboratory test requests, and laboratories can upload test results once analyses are completed. Laboratory requests are stored in the Lab Requests Database (DS7), while notifications related to test results are maintained in the Notifications Database (DS8).

P6 – AI Medical Analysis

This process provides AI-powered medical insights and recommendations. Patients can submit queries or request analysis of medical information. The system communicates with the external AI Analysis Service, which processes the request and returns analytical results to support healthcare decision-making.

P7 – Payment Processing

The Payment Processing module manages financial transactions related to appointments, laboratory services, and pharmacy orders. Payment requests are sent to the Payment Gateway, and transaction confirmations are recorded in the Payments Database (DS9).

P8 – Admin Management

This process supports administrative operations, including reviewing pending requests, approving actions, monitoring system activities, and maintaining audit records. Administrative data is stored and retrieved from the Admin/Audit Log Database (DS10).

The Level 1 DFD demonstrates how the Healthcare Management System integrates multiple healthcare services into a single platform. It illustrates the flow of information between patients, doctors, pharmacies, laboratories, administrators, external services, and internal databases, ensuring efficient management of healthcare operations and patient care.


Patient
Register / login
Book appt.
Medicine order
P1
Registration & Verification
store/read
DS1 Users DB
Prescription
Appt. details
P2
Appointment Management
store/read
DS2 Appointments DB
Patient list
Doctor
Manage slots
Create Rx
Rx confirmed
P3
Prescription Management
store
store
DS3 Prescriptions DB
DS4 Medical Records DB
AI insights
Order status
Order assigned
P4
Pharmacy Orders
read/update
DS5 Orders DB
DS6 Medicine Inventory DB
Pharmacy
Process order
Lab results
P5
Lab Result Management
store
notify
DS7 Lab Requests DB
DS8 Notifications DB
Lab requests
Chatbot query
Laboratory
Upload results
P6
AI Medical Analysis
request analysis
results
AI Analysis
Service
P7
Payment Processing
payment req.
confirmation
store
DS9 Payments DB
Payment
Gateway
Admin
Approval decisions
Pending requests
P8
Admin Management
store/read
DS10 Admin / Audit Log DB

Figure 3.4: Level 1 Data Flow Diagram of the Chefaa Healthcare Platform

Level 2 Data Flow Diagram (P2 – Appointment Management)

The Level 2 Data Flow Diagram (DFD) provides a detailed breakdown of the Appointment Management process. It illustrates how patients search for doctors, book appointments, how doctors manage appointment schedules, how notifications are generated, and how payment information is recorded. The diagram also shows the interaction between external entities, system processes, and data stores involved in the appointment lifecycle.

The Appointment Management process consists of five main sub-processes:

P2.1 Search Doctors

In this process, the patient submits a search query to find suitable doctors based on criteria such as specialization, location, or availability. The system retrieves the relevant appointment and availability information from the Appointments Database (DS2) and presents the available options. Once the patient selects a suitable time slot, the information is passed to the appointment booking process.

P2.2 Book Appointment

After selecting a doctor and an available slot, the patient submits a booking request. The system creates the appointment record and stores the associated clinic information in the Clinics Database (DS-C). A confirmation message is then sent to the patient, and the booked appointment is forwarded for further management.

P2.3 Manage Appointment

This process allows doctors to manage their appointments by updating schedules, canceling appointments, or rescheduling them when necessary. The system maintains and updates doctor-related information in the Doctors Database (DS-D). Any modifications to the appointment status are recorded and forwarded to the notification process.

P2.4 Track & Notify

Once an appointment status changes, the system generates and manages notifications for the relevant users. Notification records are stored in the Notifications Database (DS8). This process ensures that patients and doctors are informed about confirmations, cancellations, rescheduling requests, and other appointment updates. If payment is required, the process triggers the payment recording stage.

P2.5 Record Payment

In this process, payment information is exchanged with the Payment Gateway to complete the transaction. The system stores payment records in the Payments Database (DS9) and receives transaction confirmation from the payment service. This ensures that appointment-related payments are securely recorded and verified.

Overall, this Level 2 DFD demonstrates the complete appointment management workflow, beginning with doctor search and appointment booking, followed by appointment management, notification handling, and payment recording. The process ensures efficient coordination between patients, doctors, clinics, and payment services while maintaining accurate and up-to-date records throughout the system.

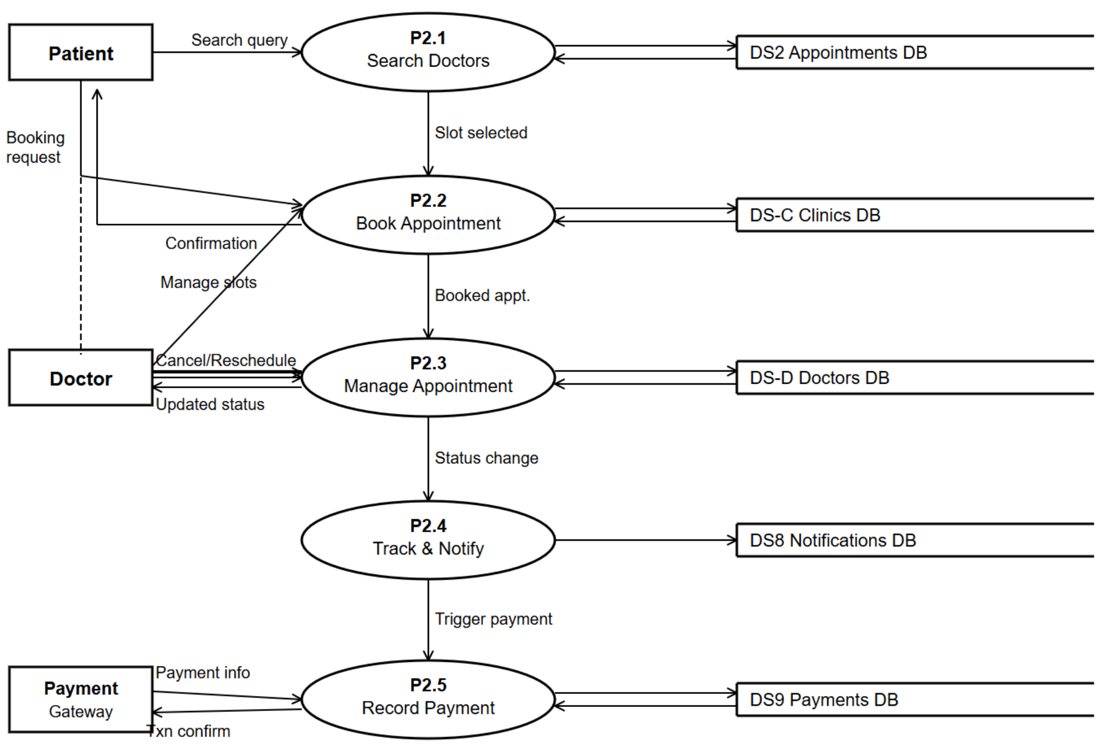


Figure 3.5: Level 2 Data Flow Diagram Appointment Management of the Chefaa Healthcare Platform

Level 2 Data Flow Diagram (P4 – Pharmacy Order Management)

The Level 2 Data Flow Diagram (DFD) provides a more detailed view of the Pharmacy Order Management process. It breaks down the main process into several sub-processes that describe how a patient searches for medicines, places an order, how the pharmacy processes and prepares the order, how delivery is managed, and finally how the payment is completed. The diagram also illustrates the interaction between external entities, internal processes, and data stores used throughout the order lifecycle.

The Pharmacy Order Management process consists of five main sub-processes:

P4.1 Browse Medicines

In this process, the patient searches for available medicines by submitting a search query. The system retrieves medicine information from the Medicine Inventory database (DS6) and displays the available products. Once the patient selects the desired medicine, the information is forwarded to the order creation process.

P4.2 Create Order

After selecting the required medicines, the patient places an order. The system creates a new order record and stores the order details in the Orders Database (DS5). An order confirmation is then provided to the patient, and the order is forwarded for processing.

P4.3 Process Order

The pharmacy receives the order details and verifies medicine availability and order information. The pharmacy prepares and fulfills the order, while the system updates the notification records in the Notifications Database (DS8). Once the order is ready, it is assigned for delivery.

P4.4 Manage Delivery

This process manages the delivery operation. Delivery information is stored and retrieved from the Delivery Database. The pharmacy can track the delivery status, and the system monitors the order until successful delivery is completed.

P4.5 Process Payment

After the delivery process is completed, payment information is exchanged with the Payment Gateway. The system records payment details in the Payments Database (DS9) and receives a transaction confirmation, ensuring that the payment has been successfully processed.

Overall, this Level 2 DFD demonstrates the complete workflow of pharmacy order management, from medicine browsing and order placement to delivery tracking and payment processing, ensuring efficient coordination between patients, pharmacy staff, delivery services, and payment systems.

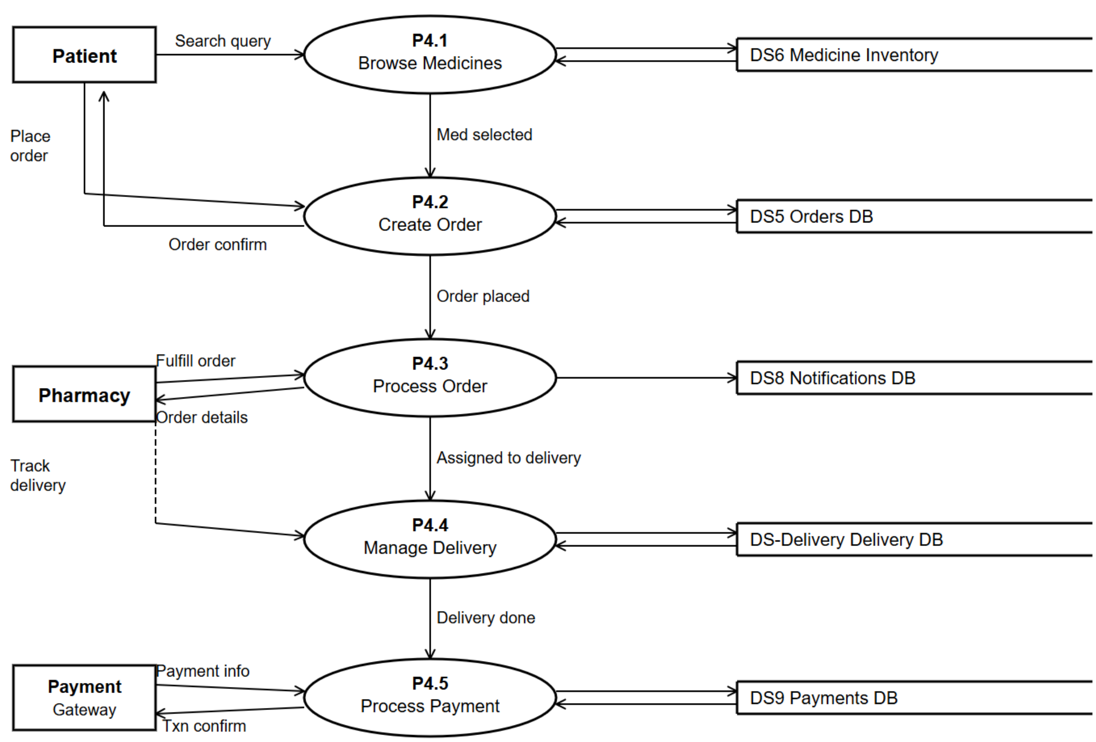


Figure 3.4: Level 2 Data Flow Diagram Pharmacy Order Management of the Chefaa Healthcare Platform.

Level 2 Data Flow Diagram (P5 – Laboratory Services Management)

The Level 2 Data Flow Diagram (DFD) provides a detailed representation of the Laboratory Services Management process. It illustrates how laboratory test requests are received from doctors, how analyses are conducted by laboratory staff, how test results are uploaded and stored, how patients are notified, and how medical records are updated. The diagram also shows the interaction between external entities, internal processes, and data stores involved throughout the laboratory workflow.

The Laboratory Services Management process consists of five main sub-processes:

P5.1 Receive Lab Request

In this process, the doctor submits a laboratory test request for a patient. The system records the request details and stores them in the Lab Requests Database (DS7). Once the request is successfully received, it is forwarded to the laboratory analysis process for further action.

P5.2 Conduct Analysis

After receiving the laboratory request, the laboratory performs the required medical tests and analyses. Sample results are processed and associated with the corresponding laboratory services stored in the Lab Services Database (DS-Lab Services). Once the analysis is completed, the generated results are prepared for uploading.

P5.3 Upload Results

In this process, laboratory staff upload the completed test results and supporting documents. The files are stored in cloud storage, and the system receives the generated file URLs for future access. Notification records are also created and stored in the Notifications Database (DS8) to inform relevant users that results are available.

P5.4 Notify & Deliver

Once the laboratory results have been uploaded, the system notifies the patient that the results are ready. Patients can access and view their laboratory reports through the system. At the same time, the results are linked to the patient's information stored in the Medical Records Database (DS4), ensuring secure and organized access to medical data.

P5.5 Update Medical Record

After successful delivery and notification, the patient's medical record is updated with the newly generated laboratory results. This ensures that the patient's health history remains complete and up-to-date, allowing healthcare providers to access accurate information for future consultations and treatments.

Overall, this Level 2 DFD demonstrates the complete laboratory services workflow, starting from the doctor's laboratory request and ending with the integration of laboratory results into the patient's medical record. The process ensures efficient communication between doctors, laboratory staff, patients, cloud storage services, and healthcare databases while maintaining accurate and accessible medical information.

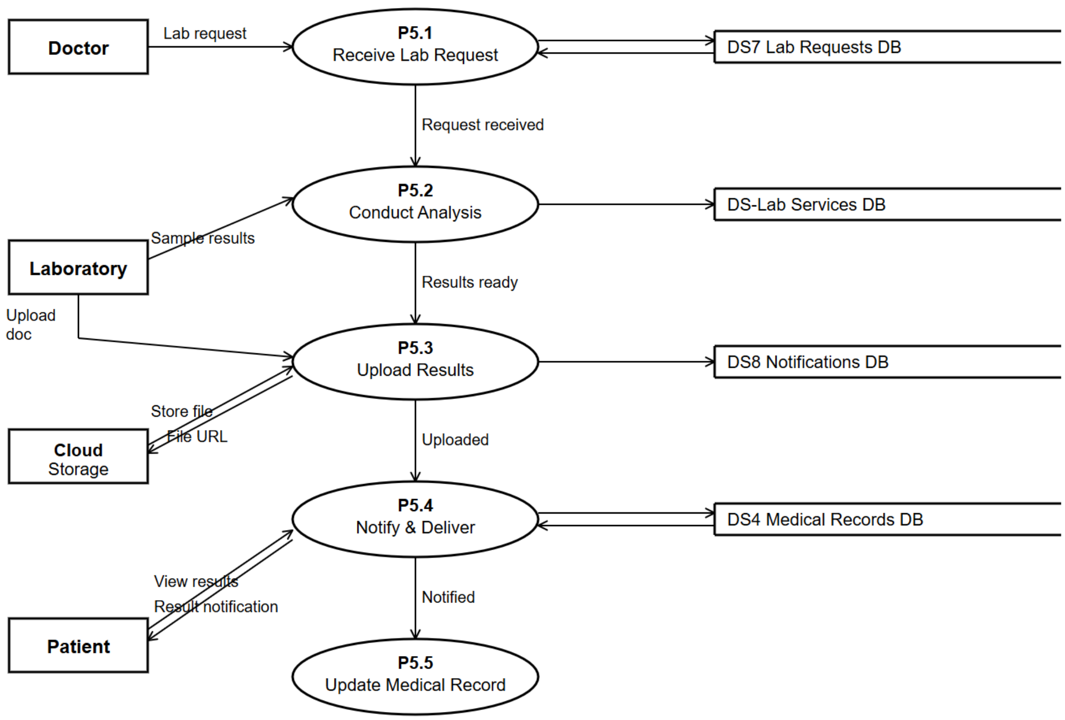


Figure 3.4: Level 2 Data Flow Diagram Laboratory Services Management of the Chefaa Healthcare Platform.

Level 2 Data Flow Diagram (P6 – AI-Based Medical Report Analysis & Specialist Recommendation)

The Level 2 Data Flow Diagram (DFD) provides a detailed representation of the AI-Based Medical Report Analysis and Specialist Recommendation process. It illustrates how patients upload medical reports, how reports are validated and processed by the AI engine, how analytical insights are generated and stored, how suitable specialists are recommended, and how patients are notified of the final results. The diagram also shows the interaction between external entities, internal processes, cloud storage services, AI services, and healthcare databases involved throughout the workflow.

The AI-Based Medical Report Analysis and Specialist Recommendation process consists of six main sub-processes:

P6.1UploadReport

In this process, the patient uploads a medical report, scan, or laboratory document through the system. The uploaded file is stored and linked to the patient's information in the Medical Records Database (DS4). Once the file is successfully received, the system confirms the upload and forwards the report to the validation process for further processing.

P6.2Validate&Store

After receiving the uploaded report, the system validates the file format, completeness, and required metadata. Valid reports are stored and prepared for AI processing. Relevant report information is also recorded in the AI Insights Database (DS-AI Insights) to support future analysis and tracking. Once validation is completed, the report is forwarded to the AI analysis process.

P6.3SendtoAIEngine

In this process, the validated report data is sent to the AI Analysis Service for automated examination. Supporting files may be stored in cloud storage, and document URLs are generated for secure access. The AI service receives the report data and performs advanced analysis to identify medical findings, patterns, and potential health concerns. The generated analysis results are then returned to the system.

P6.4ReceiveAIAnalysis

Once the AI engine completes its processing, the system receives the generated analytical insights and recommendations. The process retrieves relevant specialist information from the Doctors Database (DS-Doctors) and matches the patient's medical condition with the most appropriate healthcare professionals. The resulting insights and doctor matches are prepared for recommendation delivery.

P6.5RecommendSpecialist

Based on the AI-generated analysis and matching results, the system recommends the most suitable medical specialist for the patient. The recommendation process may interact with the AI Chatbot Service to provide additional explanations, answer patient inquiries, and present personalized guidance regarding the recommended specialist and medical findings.

P6.6NotifyPatient

After the specialist recommendation is generated, the system notifies the patient that the analysis results and recommendations are available. Notification records are stored in the Notifications Database (DS8), ensuring that patients can securely access their analysis reports, AI insights, and specialist recommendations through the platform.

Overall, this Level 2 DFD demonstrates the complete AI-powered medical report analysis workflow, starting from the patient's report upload and ending with specialist recommendation and patient notification. The process ensures efficient integration between patients, AI services, cloud storage, chatbot services, healthcare databases, and notification systems while providing accurate medical insights and personalized healthcare recommendations.

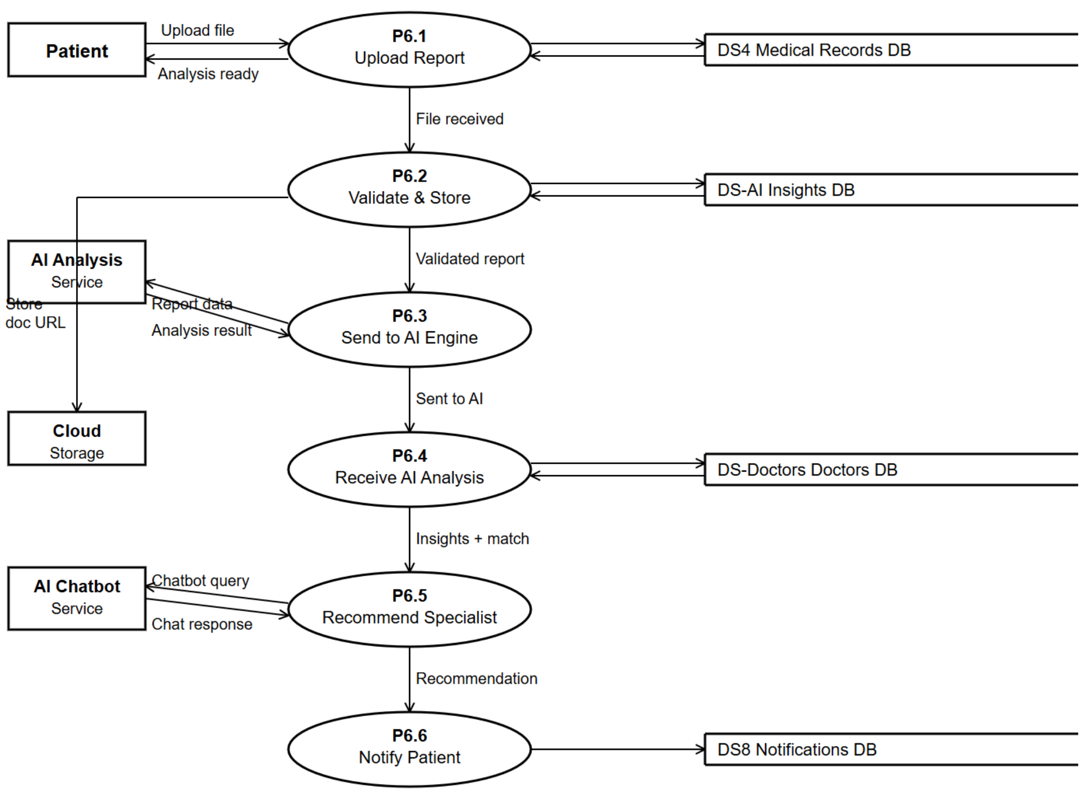


Figure 3.4: Level 2 Data Flow Diagram AI-Based Medical Report Analysis & Specialist Recommendation of the Chefaa Healthcare Platform.

3.6.5 ERD (Entity Relationship diagram)

The Entity-Relationship Diagram (ERD) is a visual representation of the system's data model. It defines the entities used in the Healthcare Management System, their attributes, and the relationships between them. This diagram helps design the underlying database schema and ensures proper data integrity, consistency, and organization across all system modules.

1. Entity Identification

Based on the system's main functionalities, the following entities have been identified:

•User
•Doctor
•Patient
•Clinic
•Appointment
•MedicalRecord
•MedicalRecordAttachment
•MedicalRecordPrescription
•Review
•ClinicScheduledDay
•ClinicScheduleBreak
•PatientMedication
•PatientMedicationAdherence
•Lab
•LabRequest
•LabService
•LabAddress
•Transaction

•Pharmacy

•PharmacyAddress

•PharmacyWorkingHour

•PharmacyCityDeliveryPrice

•DeliveryMan

•Order

•OrderItem

•OrderStatusHistory

•MedicineStock

•Prescription

•PrescriptionFile

•PrescriptionMedicine

•Notification

•RefreshToken

•PasswordReset

•MonthlyPayment

•BillingRecord

• Platform Setting

2. Relationship Identification

Relationships between the entities are identified based on system operations and user interactions:

User – Doctor (One-to-One)

Each doctor account is associated with one user account, and each doctor profile belongs to one user.

User – Patient (One-to-One)

Each patient profile is linked to one user account.

User – Lab (One-to-One)

Each laboratory account is associated with one user account.

User – Pharmacy (One-to-One)

Each pharmacy account is linked to one user account.

User – Notification (One-to-Many)

A user can receive multiple notifications, while each notification belongs to one user.

User – RefreshToken (One-to-Many)

A user may have multiple active authentication tokens for different sessions.

User – PasswordReset (One-to-Many)

A user can request multiple password reset operations over time.

Doctor – Clinic (One-to-Many)

A doctor can manage multiple clinics, while each clinic belongs to one doctor.

Doctor – Appointment (One-to-Many)

A doctor can have many appointments with patients.

Patient – Appointment (One-to-Many)

A patient can book multiple appointments.

Clinic – Appointment (One-to-Many)

A clinic can host multiple appointments.

Appointment – Medical Record (One-to-One)

Each appointment may generate a medical record.

Patient – Medical Record (One-to-Many)

A patient can have multiple medical records throughout their medical history.

Medical Record – Medical Record Attachment (One-to-Many)

A medical record may contain multiple attached files and reports.

Medical Record – Medical Record Prescription (One-to-Many)

A medical record can include multiple prescribed medications.

Patient – Review (One-to-Many)

A patient can submit multiple reviews.

Doctor – Review (One-to-Many)

A doctor can receive multiple reviews from patients.

Clinic – Clinic Scheduled Day (One-to-Many)

A clinic can have multiple scheduled working days.

Clinic Scheduled Day – Clinic Schedule Break (One-to-Many)

Each scheduled day may contain multiple break periods.

Patient – Patient Medication (One-to-Many)

A patient can have multiple medications assigned.

Patient Medication – Patient Medication Adherence (One-to-Many)

Medication adherence records track a patient's compliance with medication schedules.

Patient – Lab Request (One-to-Many)

A patient can submit multiple laboratory requests.

Lab – Lab Request (One-to-Many)

A laboratory can process multiple lab requests.

Lab – Service (One-to-Many)

A laboratory can offer multiple medical services.

Lab – Lab Address (One-to-Many)

A laboratory may have one or more registered addresses.

Lab Request – Service (Many-to-Many)

A lab request may include multiple services, and each service may be included in multiple requests.

Pharmacy – Pharmacy Address (One-to-Many)

A pharmacy can have multiple registered addresses.

Pharmacy – Pharmacy Working Hour (One-to-Many)

A pharmacy can define multiple working-hour records.

Pharmacy – Delivery Man (One-to-Many)

A pharmacy can manage multiple delivery personnel.

Pharmacy – Medicine Stock (One-to-Many)

A pharmacy maintains multiple medicine inventory records.

Pharmacy – Order (One-to-Many)

A pharmacy can receive and process multiple orders.

Delivery Man – Order (One-to-Many)

A delivery man can deliver multiple orders.

Order – Order Item (One-to-Many)

An order contains multiple order items.

Order – Order Status History (One-to-Many)

An order can have multiple status updates throughout its lifecycle.

Medicine Stock – Order Item (One-to-Many)

Each order item references a medicine from the pharmacy inventory.

Appointment – Prescription (One-to-One)

An appointment may generate a prescription.

Prescription – Prescription File (One-to-Many)

A prescription can contain multiple attached files or images.

Prescription – Prescription Medicine (One-to-Many)

A prescription can include multiple prescribed medicines.

Prescription – Order (One-to-Many)

A prescription may be used to create one or more pharmacy orders.

Monthly Payment – Billing Record (One-to-Many)

Each monthly payment can generate one or more billing records.

3. Entities and Their Attributes

User

•user_id(PrimaryKey)

•name

•email

•password

•role

• status

Doctor

•doctor_id(PrimaryKey)

•user_id(ForeignKeyreferencingUser)

•name

• status

Patient

•patient_id(PrimaryKey)

•user_id(ForeignKeyreferencingUser)

•name

• status

Clinic

•clinic_id(PrimaryKey)

•doctor_id(ForeignKeyreferencingDoctor)

•name

• status

Appointment

•appointment_id(PrimaryKey)

•patient_id(ForeignKeyreferencingPatient)

•doctor_id(ForeignKeyreferencingDoctor)

•clinic_id(ForeignKeyreferencingClinic)

• status

Medical Record

•medical_record_id(PrimaryKey)

•patient_id(ForeignKeyreferencingPatient)

•doctor_id(ForeignKeyreferencingDoctor)

• status

Review

•review_id(PrimaryKey)

• status

Lab

•lab_id(PrimaryKey)

•user_id(ForeignKeyreferencingUser)

•name

• status

Lab Request

•request_id(PrimaryKey)

•patient_id(ForeignKeyreferencingPatient)

•lab_id(ForeignKeyreferencingLab)

• status

Service

•service_id(PrimaryKey)

• lab_id (Foreign Key referencing Lab)

Pharmacy

•pharmacy_id(PrimaryKey)

•user_id(ForeignKeyreferencingUser)

•name

• status

Order

•order_id(PrimaryKey)

•pharmacy_id(ForeignKeyreferencingPharmacy)

•user_id(ForeignKeyreferencingUser)

•delivery_man_id(ForeignKeyreferencingDeliveryMan)

•prescription_id(ForeignKeyreferencingPrescription)

• status

Prescription

•prescription_id(PrimaryKey)

•appointment_id(ForeignKeyreferencingAppointment)

• status

Notification

•notification_id(PrimaryKey)

• user_id (Foreign Key referencing User)

Refresh Token

•token_id(PrimaryKey)

•user_id(ForeignKeyreferencingUser)

• status

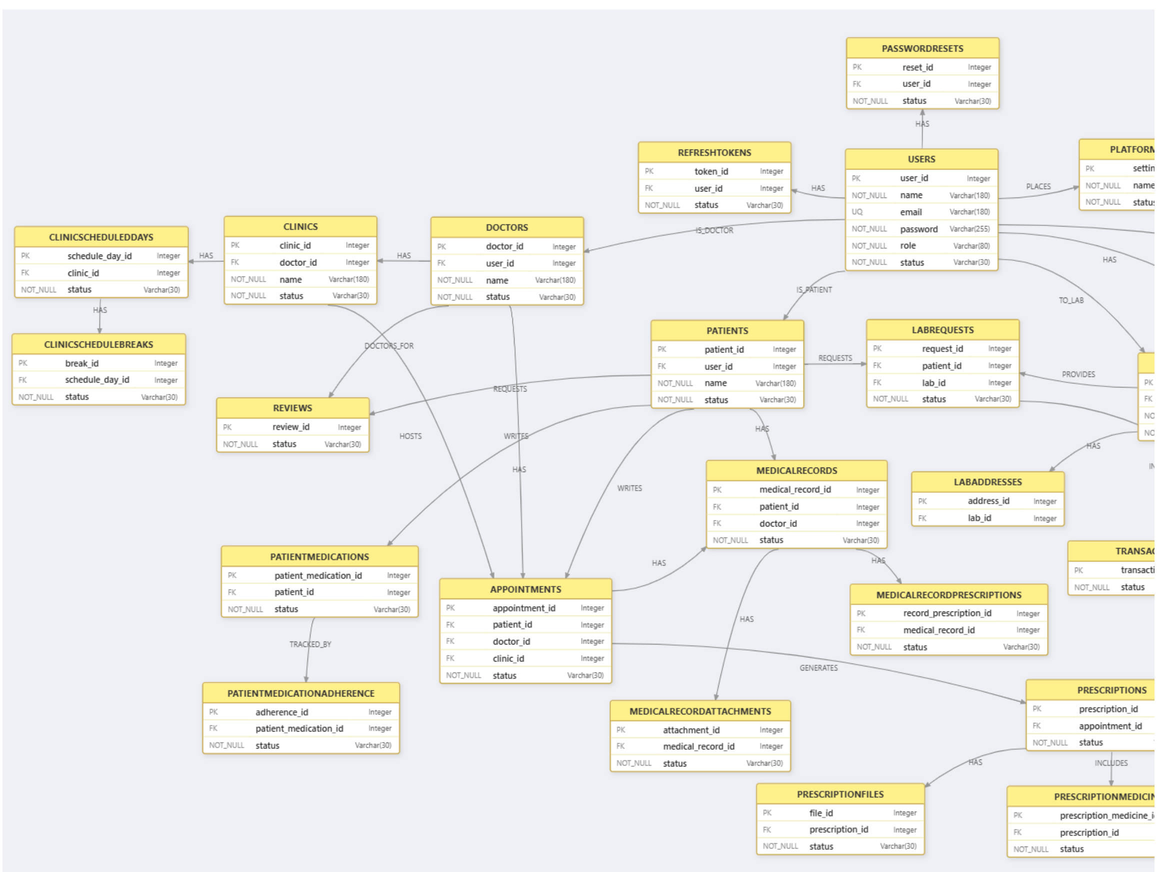
Password Reset

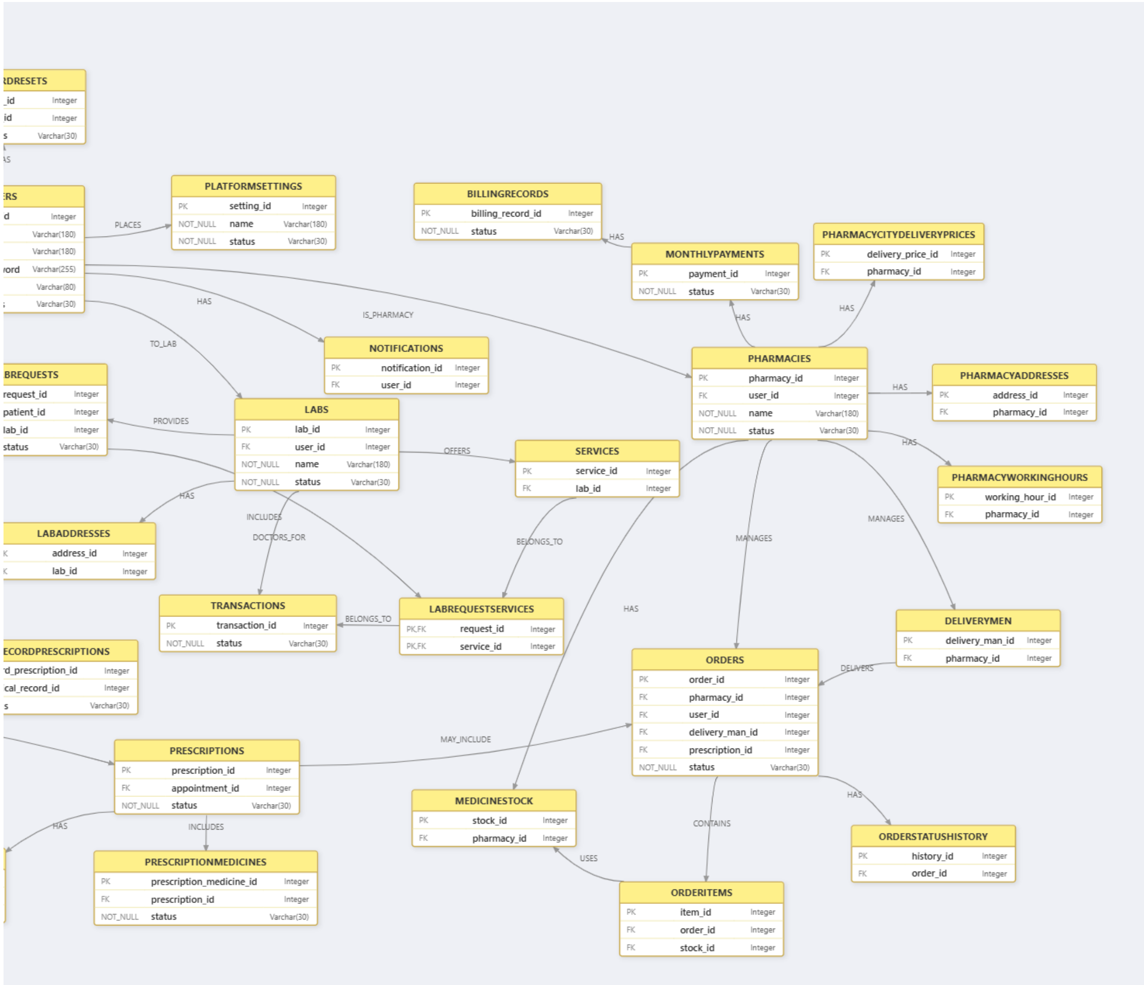
•reset_id(PrimaryKey)

•user_id(ForeignKeyreferencingUser)

• status

Additional entities such as MedicalRecordAttachment, PatientMedication, PharmacyAddress, MedicineStock, OrderItem, BillingRecord, and others follow the same structure using primary keys and foreign keys to maintain referential integrity throughout the system.





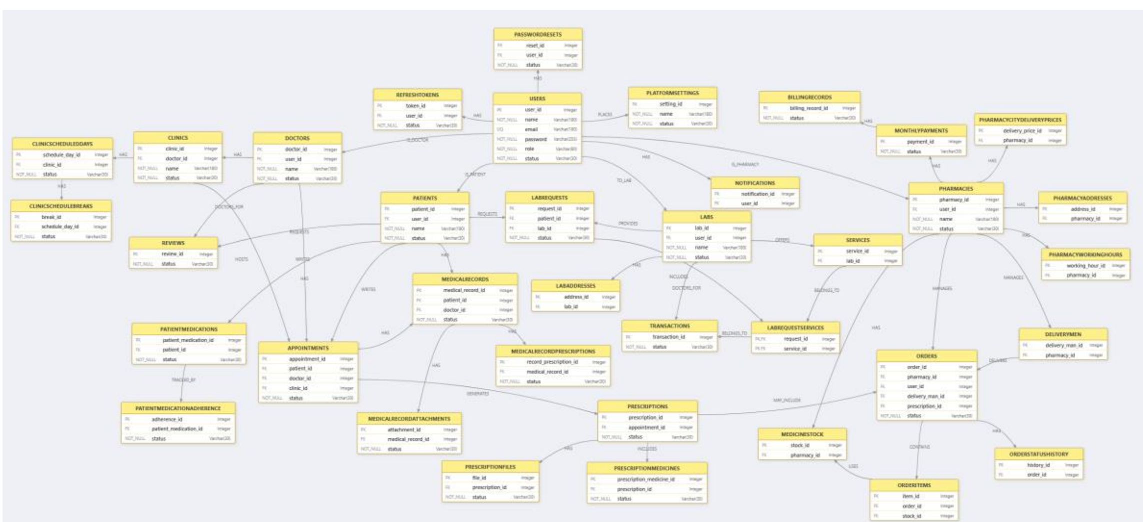


Figure 3.5: Chefaa ERD Diagram

Chapter 4: Business Model

Customer Segments

• Patients

Individuals seeking accessible healthcare services through a unified platform. They can book appointments, receive prescriptions, order medications, access laboratory services, and benefit from AI-powered medical assistance. The platform targets patients of all age groups who value convenience and remote healthcare access.

• Doctors and Healthcare Providers

General practitioners and specialists looking for efficient tools to manage clinics, schedules, appointments, electronic prescriptions, and financial analytics. The platform helps healthcare providers digitize their daily operations.

• Pharmacies

Community and retail pharmacies aiming to expand their reach through online medication ordering, inventory management, delivery services, and digital payment integration.

• Laboratories and Radiology Centers

Medical laboratories and diagnostic centers that provide testing and imaging services and require a secure channel to communicate results and interact with patients and physicians.

• Healthcare Organizations

Clinics and healthcare institutions interested in improving patient engagement, reducing administrative work, and integrating digital healthcare services.

Value Propositions

• Integrated Healthcare Platform

Chefaa provides a complete healthcare ecosystem that connects patients, doctors, pharmacies, laboratories, and administrators within a single platform, simplifying communication and healthcare management.

• Intelligent Appointment Management

Patients can search for doctors, view profiles, schedule appointments, and receive notifications, while doctors can organize clinics and manage appointment slots efficiently.

• Electronic Prescription System

Doctors can create digital prescriptions containing medications, diagnoses, laboratory tests, imaging requests, and follow-up recommendations, reducing paperwork and improving accessibility.

• AI Medical Analysis

Artificial Intelligence assists patients by analyzing laboratory reports and radiology images and providing explanations and recommendations for suitable specialists.

• AI Chatbot Assistant

An intelligent chatbot offers medical guidance, answers healthcare-related questions, and provides daily summaries and insights for doctors and pharmacies.

• Pharmacy Order Management

Patients can order medications online, choose delivery or pickup options, track orders, and complete payments electronically.

• Laboratory Result Management

Laboratories can upload test results directly to patients through the platform, enabling secure and efficient communication.

• Financial Analytics and Reporting

Healthcare providers receive AI-generated statistics, revenue reports, and performance analyses to support decision-making.

Channels

• Web Application

A responsive web platform developed using Next.js and React, allowing users to access healthcare services from desktop devices.

• Mobile Application

A cross-platform mobile application built using Flutter, providing users with healthcare services anywhere and anytime.

• Notifications and Emails

Push notifications and email services are used to inform users about appointments, prescriptions, laboratory results, and order status updates.

• Social Media Platforms

Marketing and awareness campaigns are conducted through social media channels to increase user engagement and promote healthcare services.

Customer Relationships

• AI-Powered Assistance

Patients and healthcare providers receive continuous support through intelligent chatbots and personalized recommendations.

• Customer Support Services

Users can contact the support team through chat, email, or phone for technical assistance and inquiries.

• Notification System

The system sends reminders for appointments, order updates, laboratory results, and payment confirmations.

• Feedback and Rating System

Patients can evaluate doctors, pharmacies, and laboratories, helping maintain service quality and improving user experience.

Key Resources

• Software Infrastructure

The platform consists of web and mobile applications supported by scalable backend servers and cloud databases.

• Artificial Intelligence Models

AI services are used for medical report analysis, chatbot functionality, and business analytics.

• Databases

MongoDB databases store users, appointments, prescriptions, orders, laboratory results, and financial records.

• Payment Gateway Integration

Secure payment services support online transactions for appointments and pharmacy orders.

• Development Team

Frontend, backend, mobile, AI engineers, and designers contribute to building and maintaining the platform.

Key Activities

• Software Development and Maintenance

Continuous development, testing, and updating of web and mobile applications.

• AI Model Development

Improving AI-based healthcare assistance, report analysis, and business intelligence systems.

• Database and Security Management

Maintaining secure storage and protecting medical records and user information.

• Customer Support and Monitoring

Providing user support and ensuring smooth platform operation.

• Healthcare Provider Verification

Validating doctors, pharmacies, and laboratories before granting access to the platform.

Key Partnerships

• Hospitals and Clinics

Collaboration with healthcare institutions to expand service coverage.

• Pharmacies and Laboratories

Partnerships with pharmacies and diagnostic centers to provide integrated healthcare services.

• Payment Service Providers

Integration with secure payment gateways to facilitate online transactions.

• Cloud Service Providers

Cloud infrastructure providers support storage, scalability, and system reliability.

• AI Service Providers

External AI technologies enhance medical analysis and intelligent assistance capabilities.

Cost Structure

• Development Costs

Expenses related to web application, mobile application, backend services, and AI model development.

• Cloud Hosting Costs

Database hosting, file storage, and server infrastructure expenses.

• Operational Costs

Customer support, maintenance, monitoring, and administrative operations.

• Marketing Costs

Digital marketing campaigns and promotional activities.

• Third-Party Service Costs

Payment gateways, email services, cloud storage, and AI APIs.

Revenue Streams

• Commission on Appointments

A percentage of consultation fees is collected from completed appointments.

• Pharmacy Order Commissions

The platform receives a commission from completed medication orders.

• Premium Subscription Services

Doctors, pharmacies, and laboratories can subscribe to advanced analytics and premium features.

• Advertising and Promotion Services

Healthcare providers can promote their services through featured listings and advertisements.

• Healthcare Analytics Services

Advanced AI-powered reports and business intelligence tools can be offered as premium services.

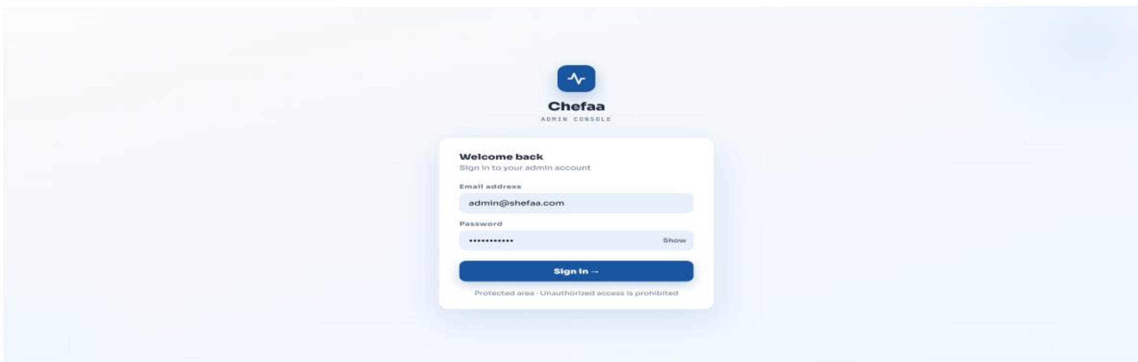
• Future Partnerships with Healthcare Organizations

Additional revenue can be generated through collaborations with hospitals, clinics, and insurance providers.

Chapter 5: System Design

<ADMIN>

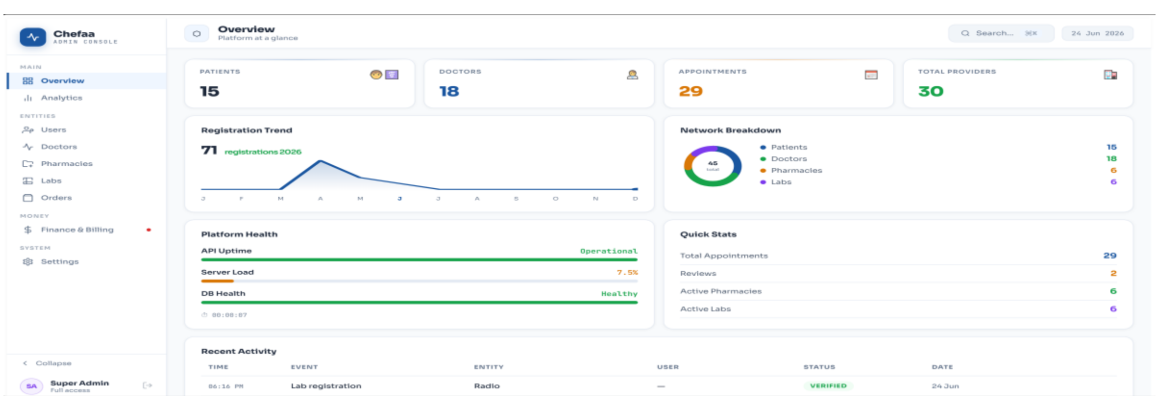
1. Admin Login Page



Description:

This screen represents the administrator login interface of the Chefaa platform. It provides a secure authentication mechanism that allows authorized administrators to access the system dashboard. The login process ensures that only verified users can manage platform operations, maintain security, and monitor healthcare service providers.

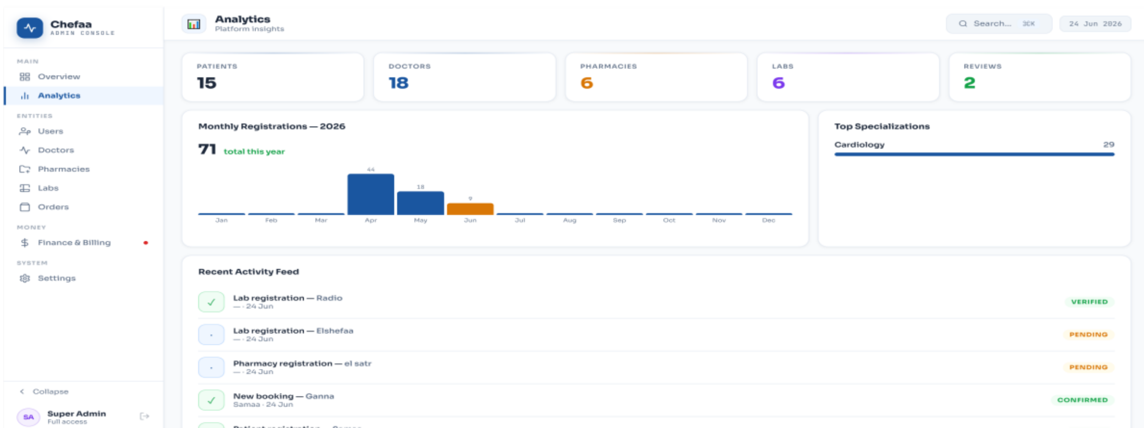
2. Admin Dashboard (Overview)



Description:

The Overview Dashboard serves as the central control panel for the administrator. It presents key platform statistics, including the number of patients, doctors, appointments, pharmacies, and laboratories. Interactive charts and activity logs provide real-time insights into system performance, helping administrators monitor platform growth and operational health efficiently.

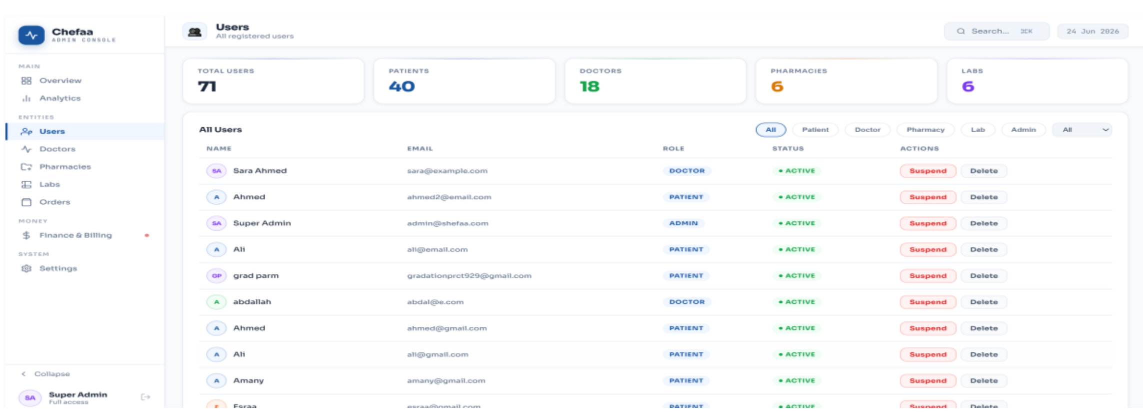
3. Analytics Page



Description:

The Analytics module provides comprehensive statistical reports and visual representations of platform activities. Administrators can track registration trends, user growth, provider engagement, and specialty distributions. This feature supports data-driven decision-making and helps evaluate overall system performance.

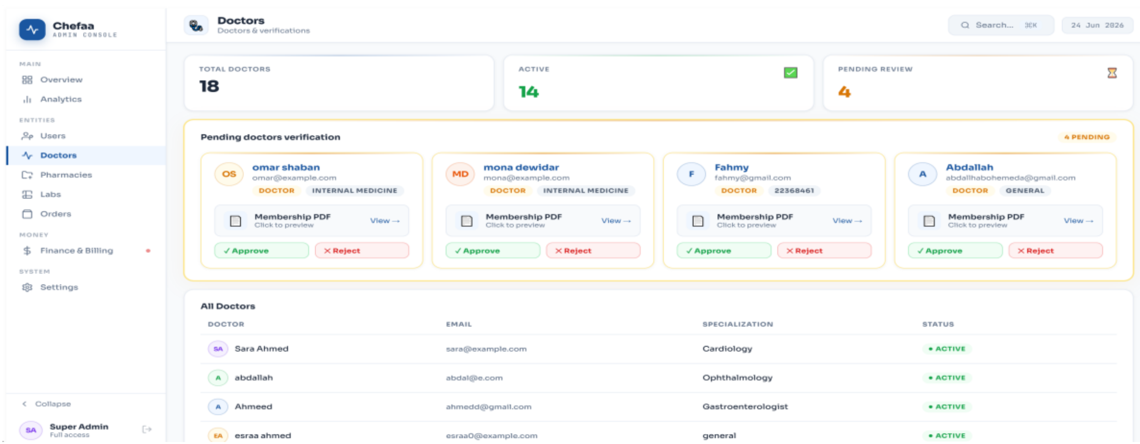
4. User Management Page



Description:

This page enables administrators to manage all registered users within the platform. User information, roles, and account statuses are displayed in an organized table. Administrators can suspend, activate, or review accounts to ensure compliance with platform policies and maintain a secure healthcare environment.

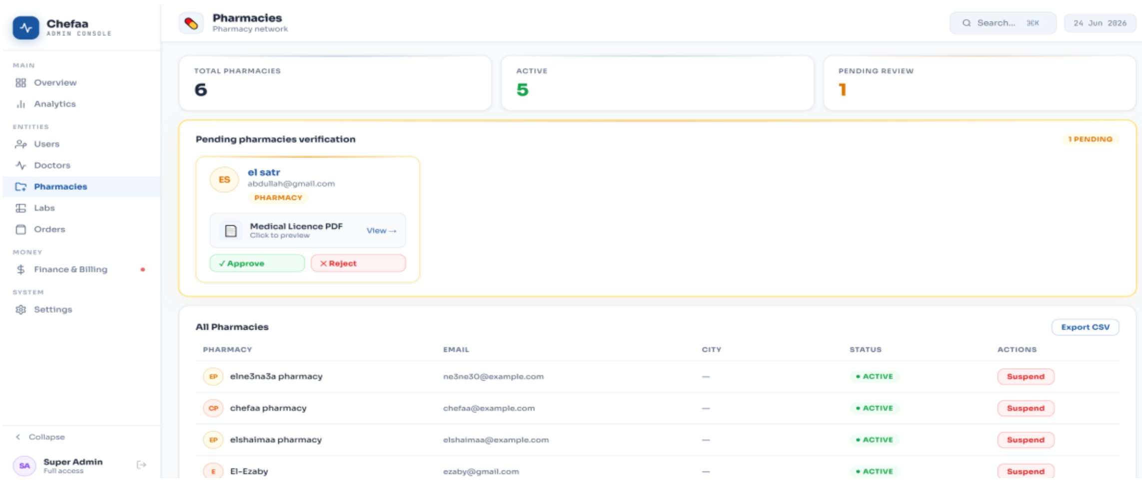
5. Doctor Verification and Management Page



Description:

The Doctor Management module allows administrators to review healthcare professionals' registration requests. Submitted medical licenses and verification documents are evaluated before approval. This process guarantees that only qualified and licensed doctors can provide medical services through the platform.

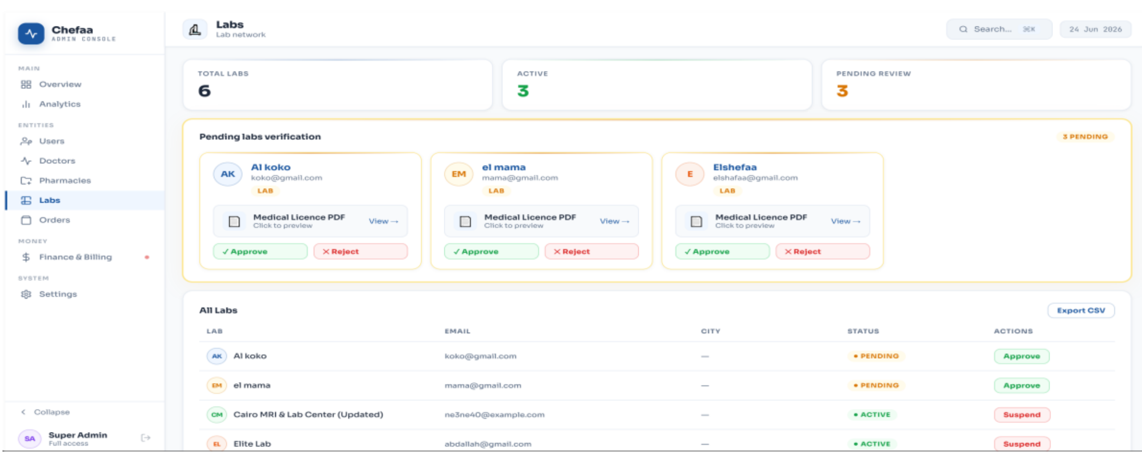
6. Pharmacy Verification and Management Page



Description:

This interface is dedicated to pharmacy management and verification. Administrators can review pharmacy licenses, approve registration requests, and monitor active pharmacies. The verification process ensures the authenticity and credibility of pharmaceutical service providers.

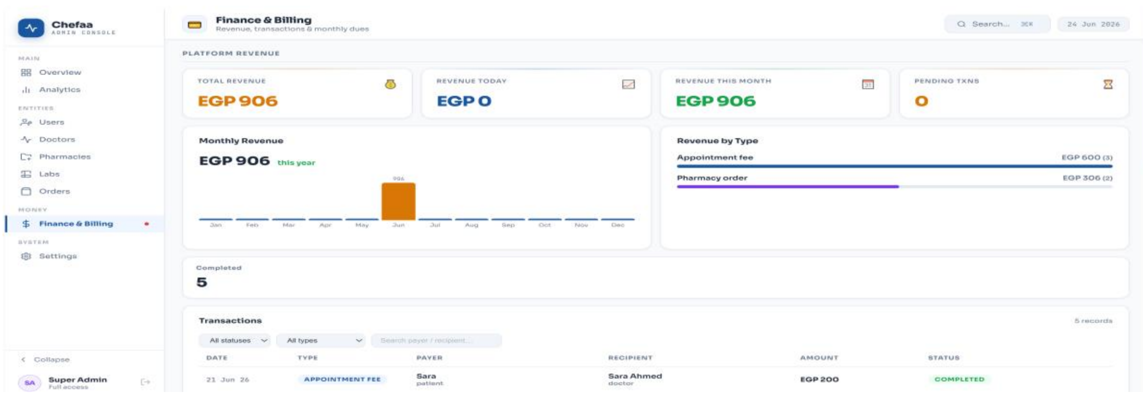
7. Laboratory Verification and Management Page



Description:

The Laboratory Management module enables administrators to verify laboratories before granting access to the platform. Uploaded licenses and accreditation documents are reviewed to ensure compliance with healthcare regulations and maintain service quality standards.

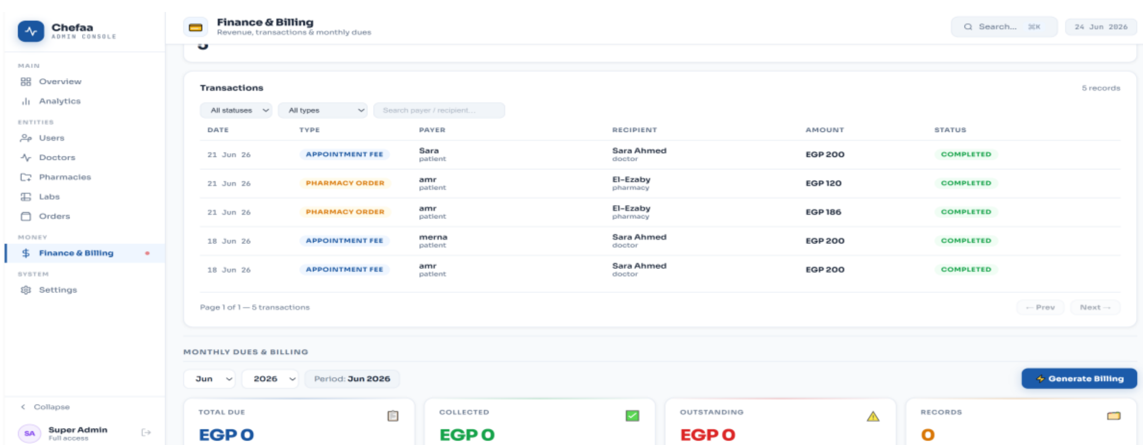
8. Finance and Billing Dashboard



Description:

The Finance and Billing Dashboard provides a detailed overview of platform revenues, completed transactions, and payment distributions. Administrators can monitor financial performance through charts and summaries, allowing effective management of platform earnings and billing activities

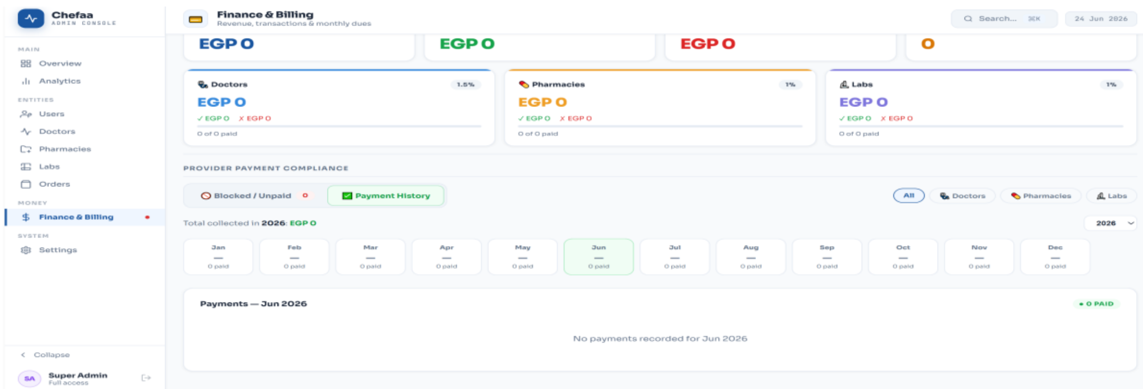
9. Transactions Management



Description:

This screen displays all financial transactions conducted through the platform, including appointment fees and pharmacy orders. The transaction history provides transparency, allowing administrators to track payments, monitor transaction statuses, and maintain financial accountability.

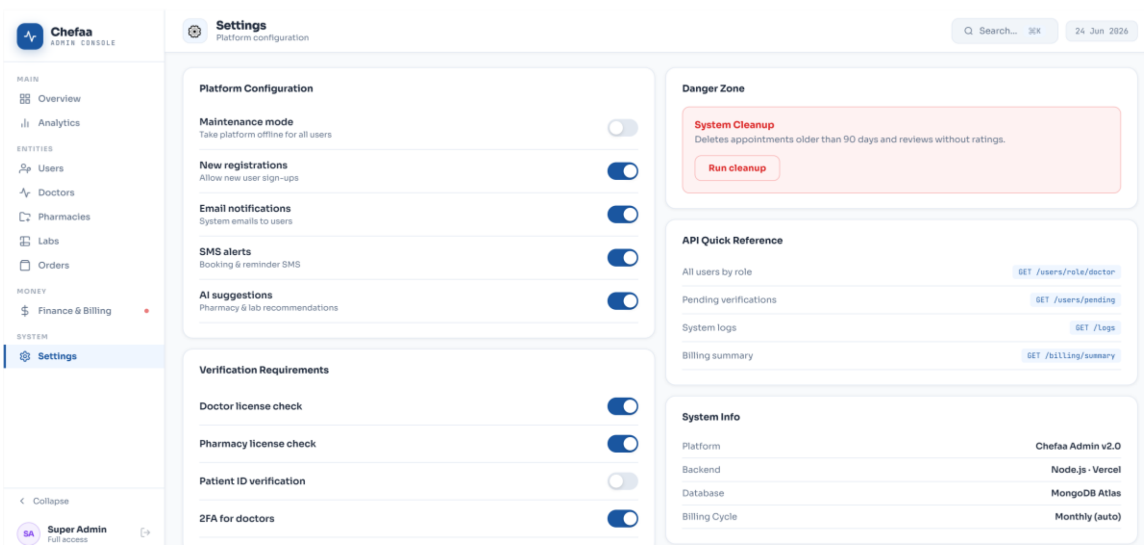
10. Monthly Dues and Provider Compliance& Payment History Monitoring



Description:

This feature provides a historical view of provider payments and billing records. Administrators can review monthly payment statuses, analyze payment trends, and ensure that all healthcare providers fulfill their financial obligations to the platform.

12. System Settings Page

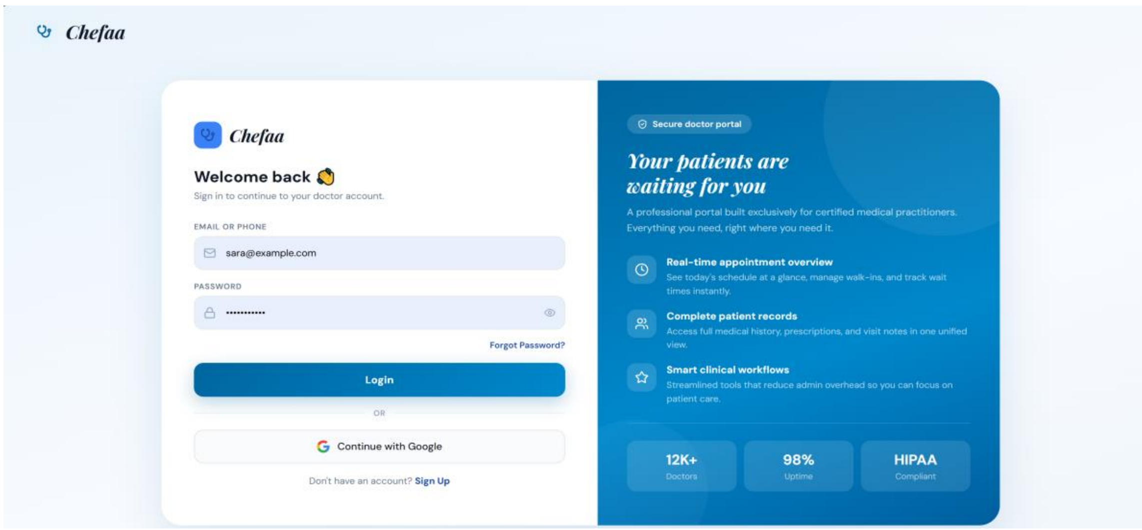


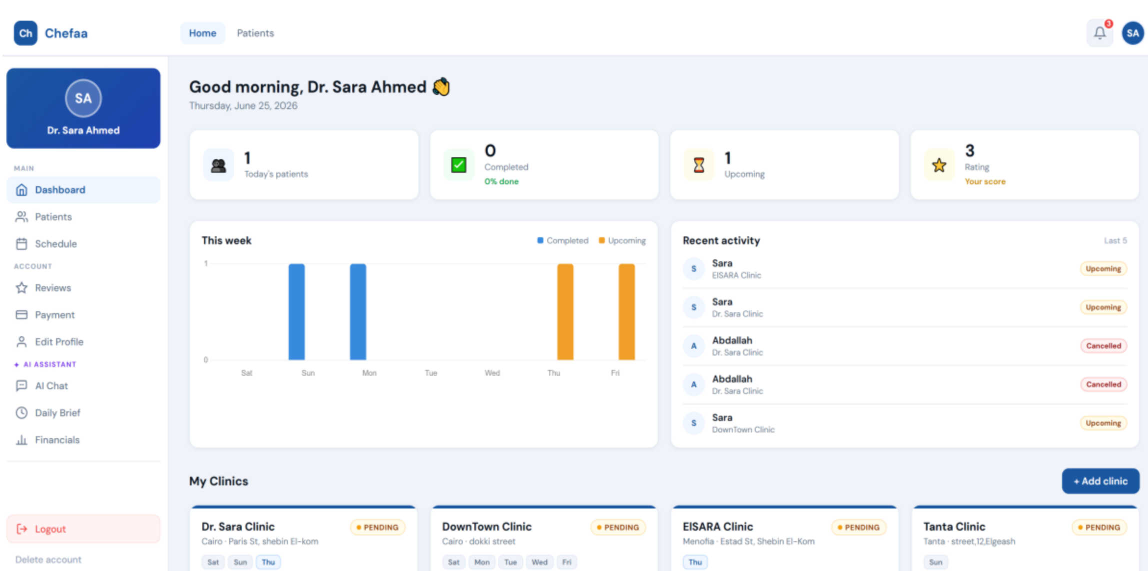
Description:

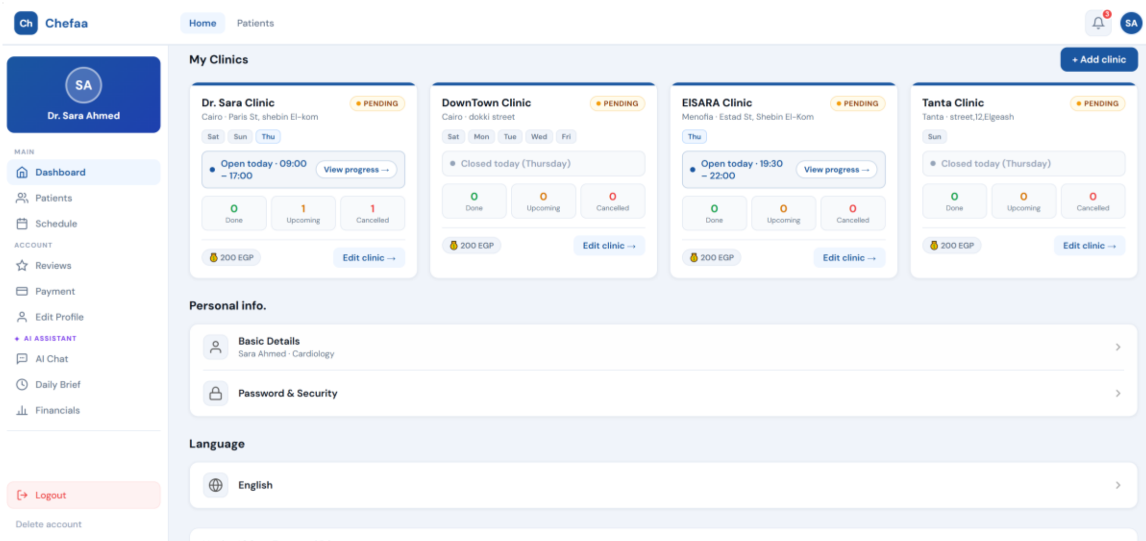
The Settings module allows administrators to configure platform-wide preferences and operational rules. Features include maintenance mode, registration controls, notification settings, verification requirements, and security configurations. This module ensures flexibility and efficient management of the entire healthcare ecosystem.

<DOCTOR>

Doctor Authentication & Dashboard



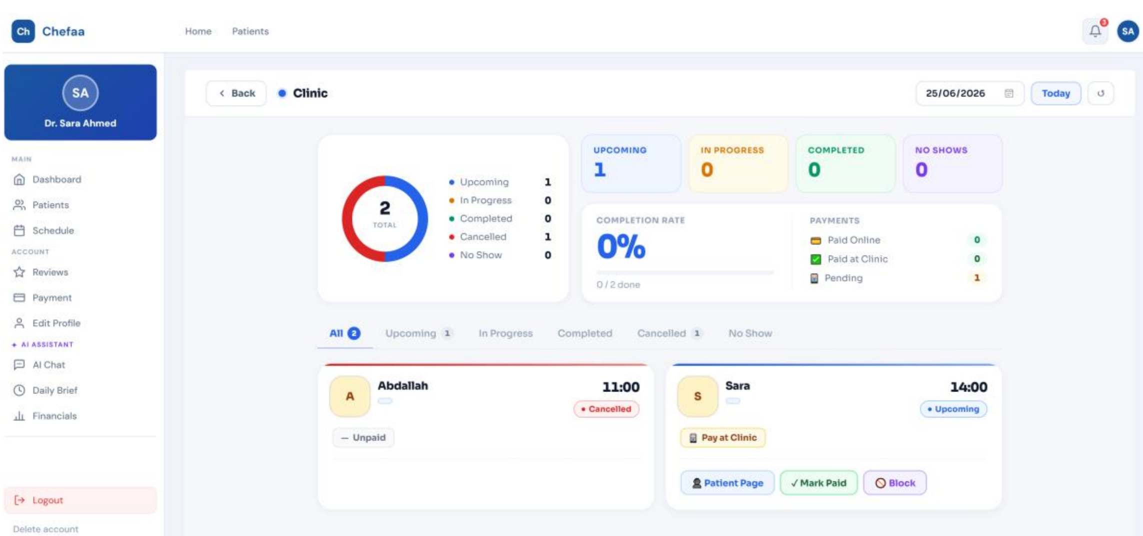


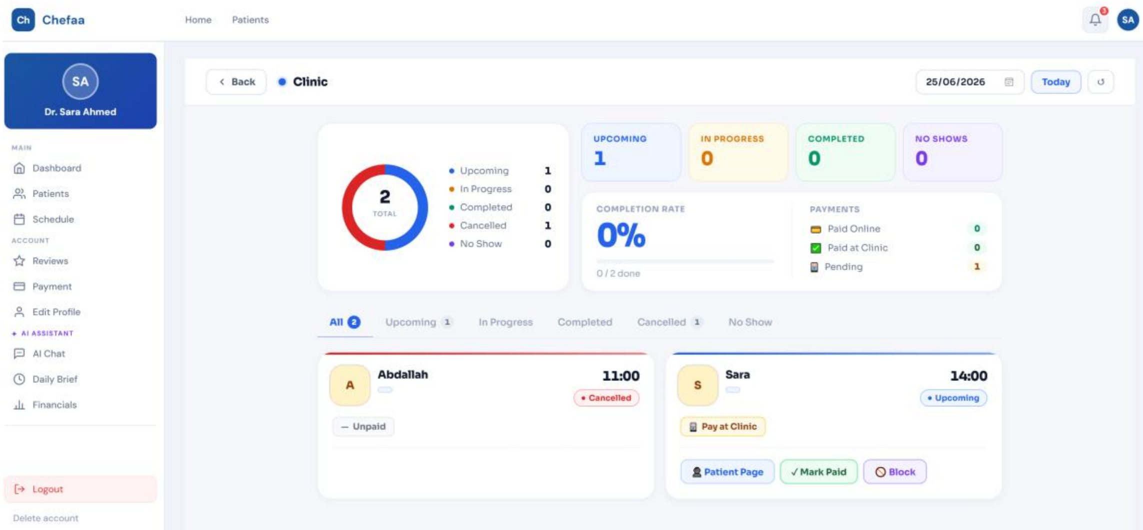


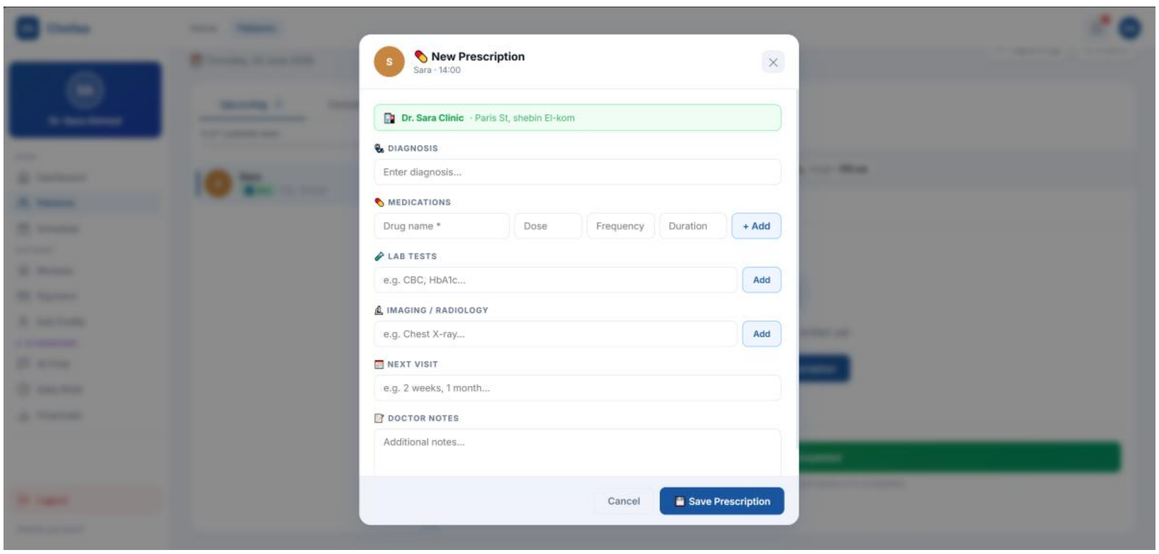
Description:

These screenshots demonstrate the doctor's authentication process and main dashboard. After logging in securely, doctors can access an overview of their clinics, appointments, patient statistics, upcoming schedules, and recent activities. The dashboard serves as a centralized workspace that provides quick access to essential clinical and administrative information.

Clinic & Patient Management



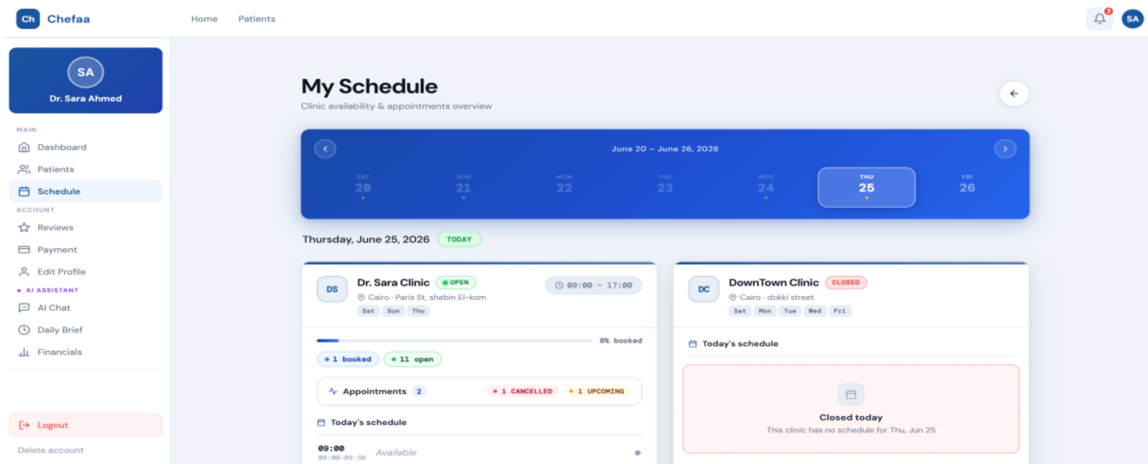


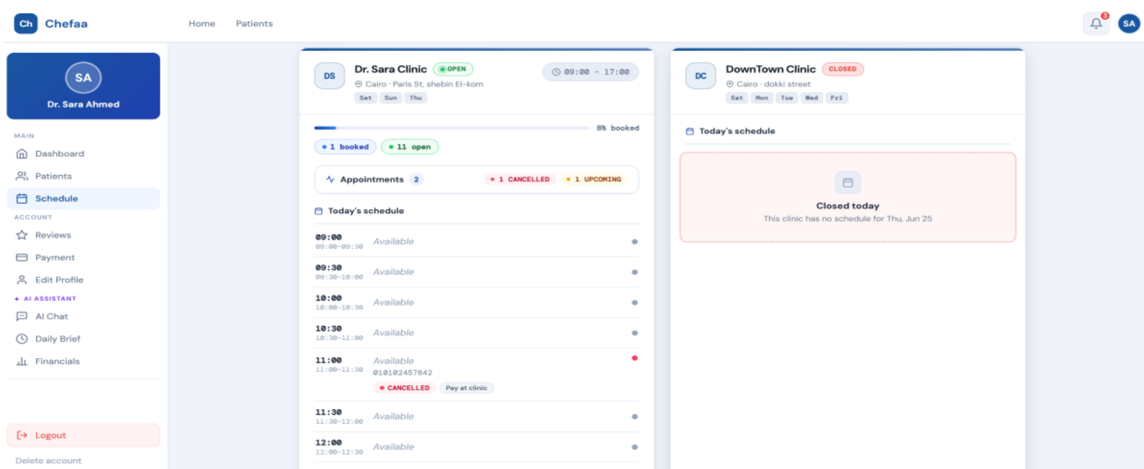


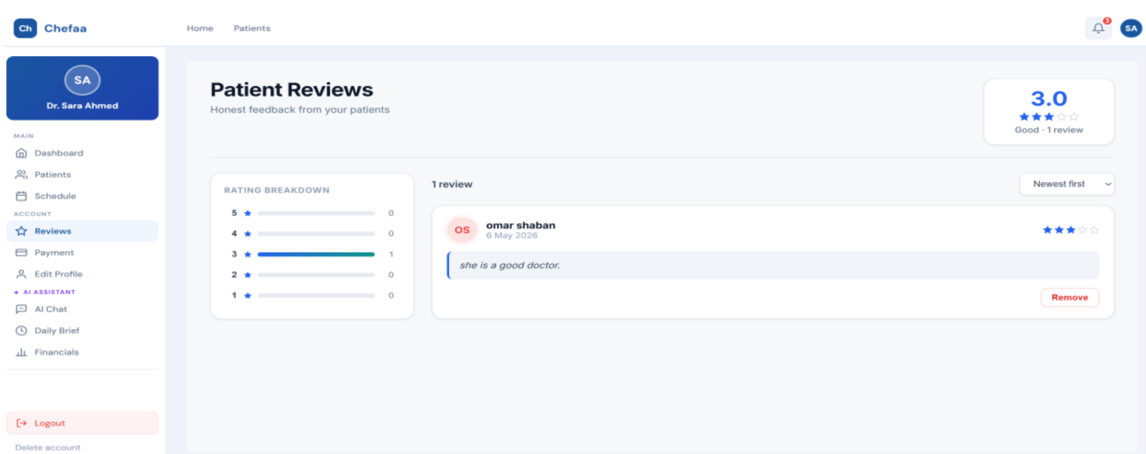
Description:

These screenshots illustrate the clinic management and patient consultation workflow. Doctors can monitor appointment statuses, manage patient visits, access medical records, create prescriptions, record diagnoses, request laboratory tests, and document treatment plans. This module streamlines patient care and supports efficient clinical decision-making.

Scheduling & Reviews Management



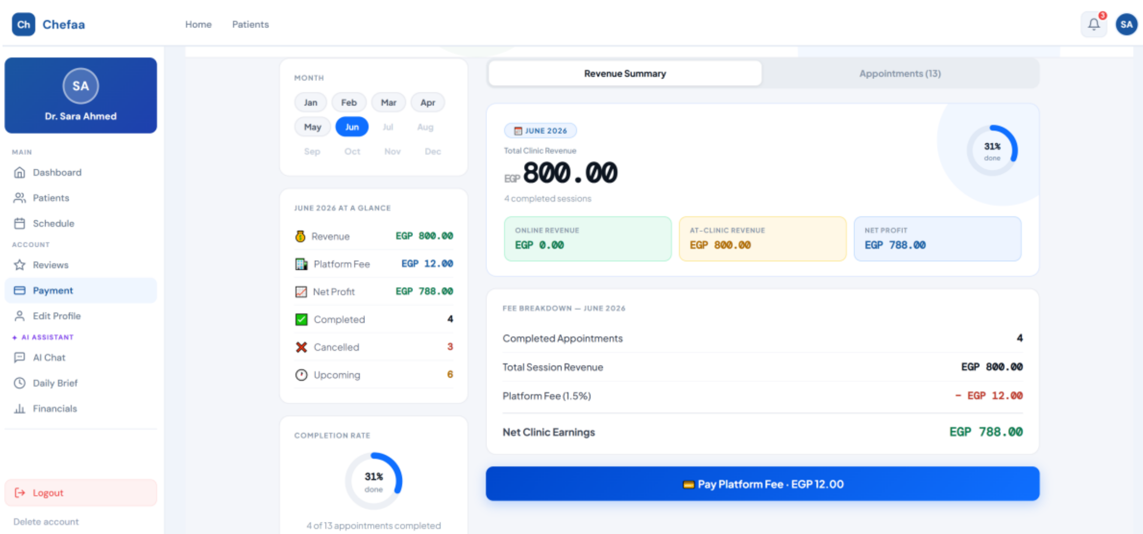


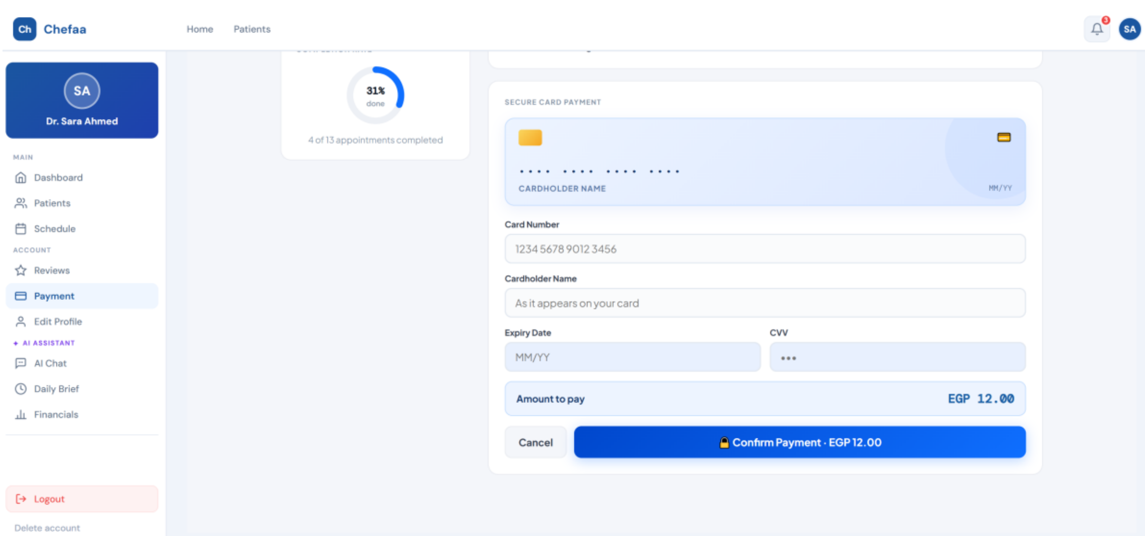


Description:

The Scheduling and Reviews modules enable doctors to manage clinic availability, appointment slots, and daily schedules across multiple clinics. Additionally, doctors can view patient feedback, ratings, and reviews, helping them evaluate service quality and improve the overall patient experience.

Payments & Financial Transactions

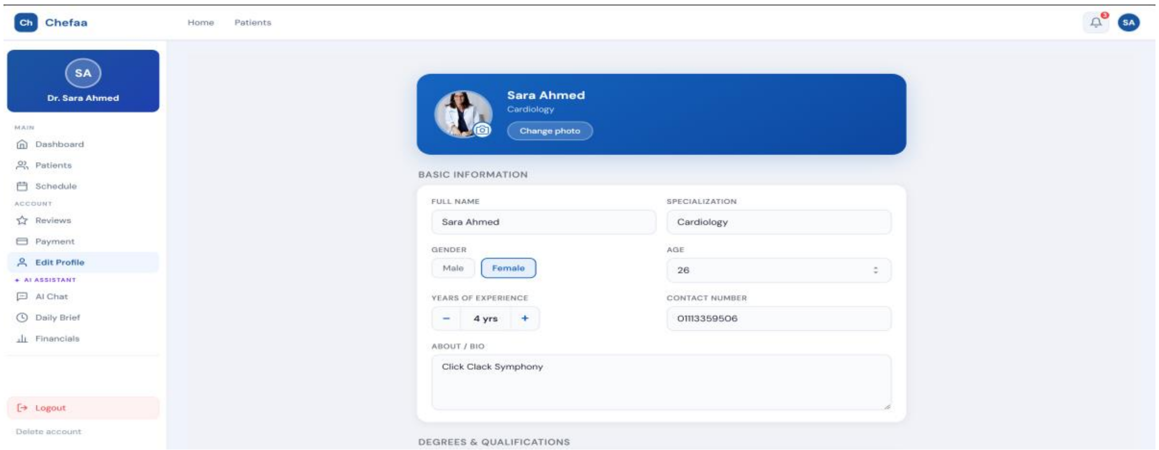


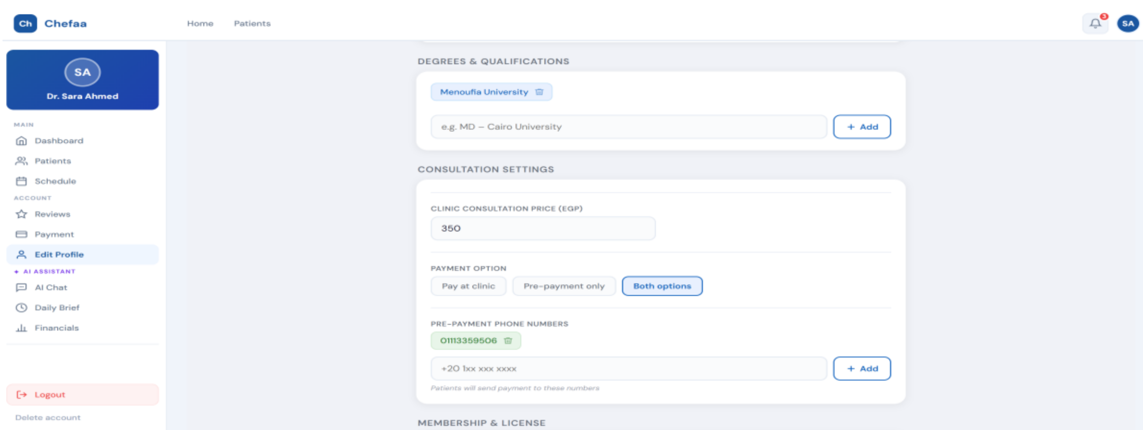


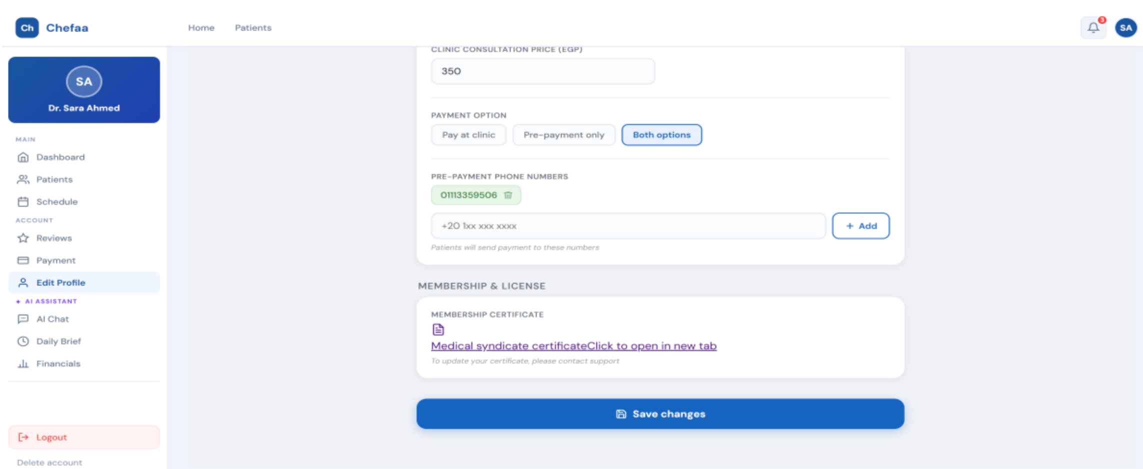
Description:

These screenshots present the payment management functionality available to doctors. The system provides detailed revenue summaries, transaction breakdowns, platform fee calculations, and secure payment processing. This module ensures financial transparency and allows healthcare providers to monitor earnings effectively.

Doctor Profile Management



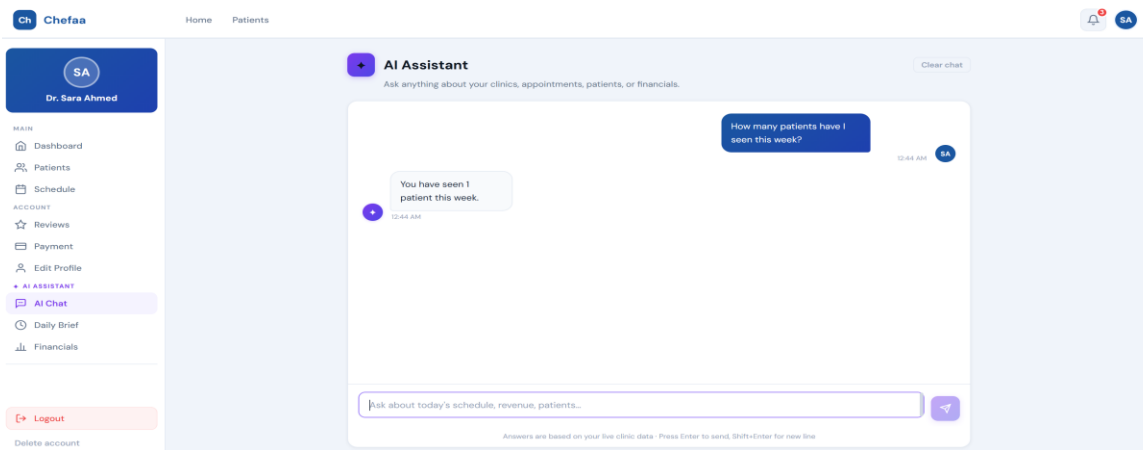




Description:

The Profile Management module allows doctors to maintain and update their professional information, including specialization, experience, qualifications, consultation fees, payment methods, and licensing details. Keeping profile information updated enhances patient trust and ensures accurate service representation.

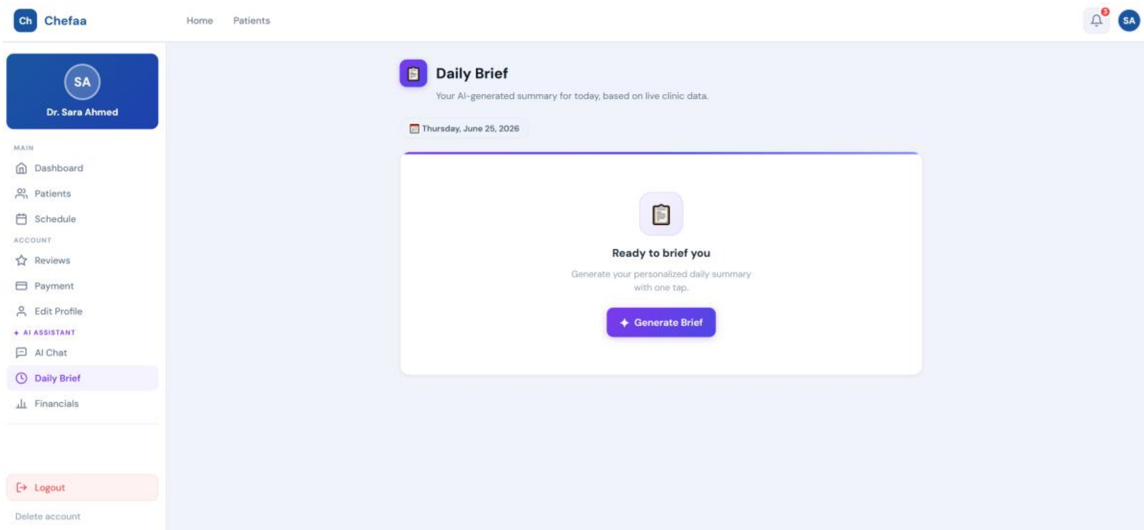
AI Assistant

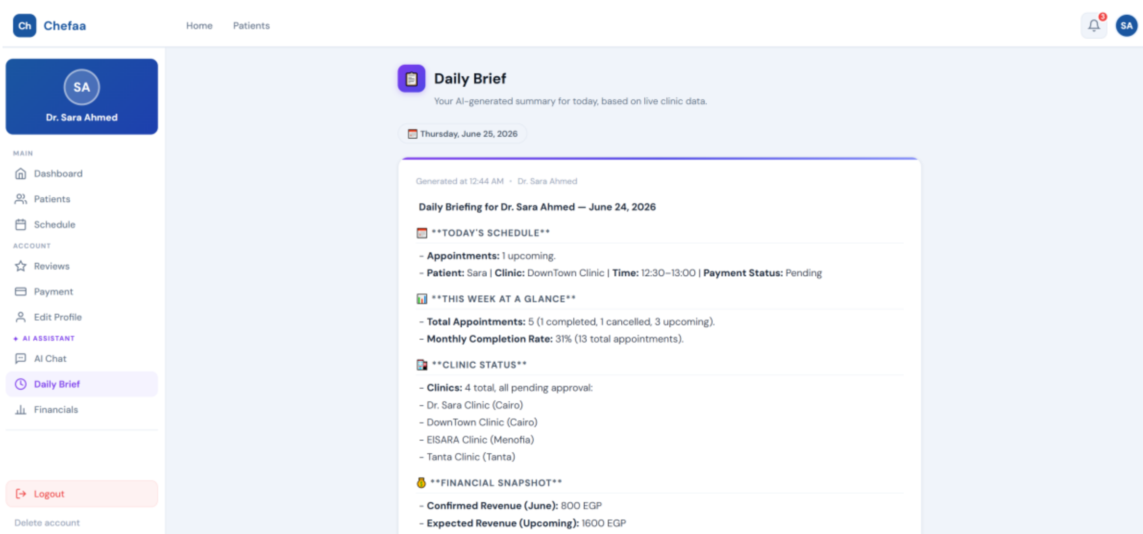


Description:

The AI Assistant provides intelligent support for doctors by answering questions related to appointments, patients, clinic performance, and financial data. By leveraging artificial intelligence, the assistant helps doctors access information quickly and improve operational efficiency.

AI Daily Brief

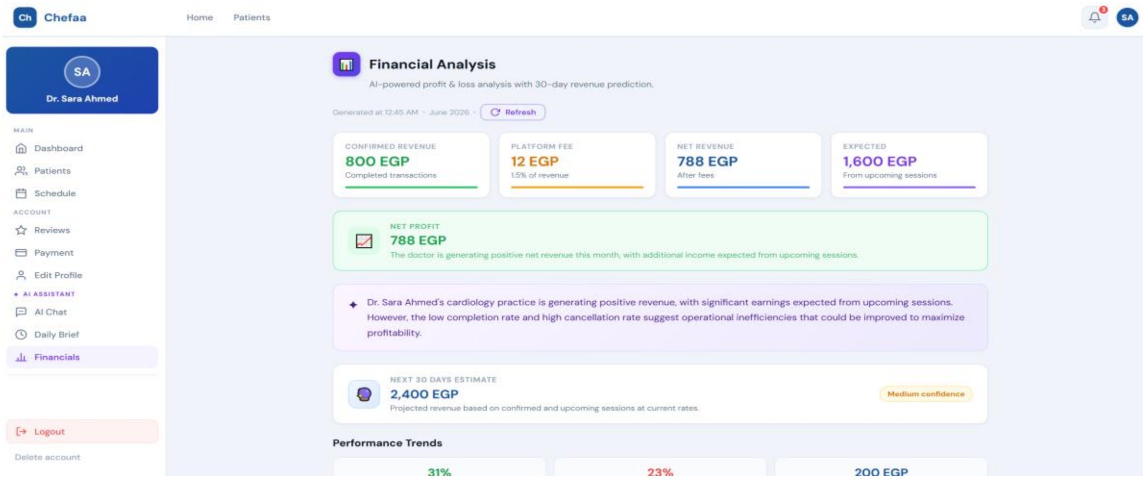


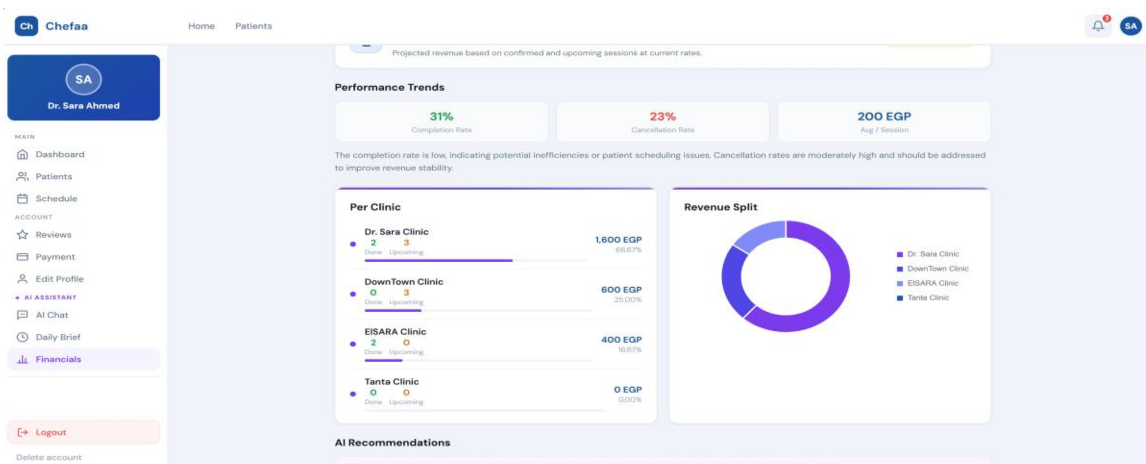


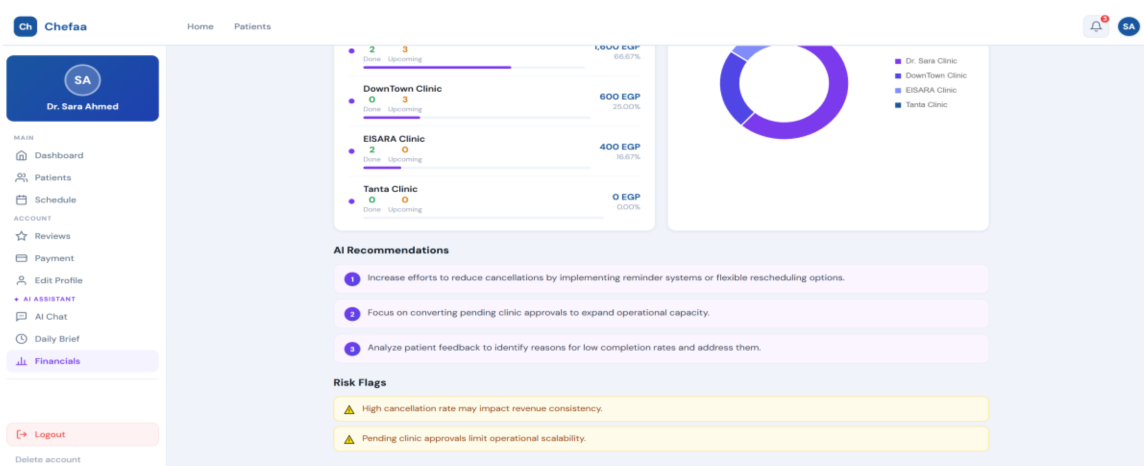
Description:

The AI Daily Brief generates an automated summary of the doctor's daily activities based on real-time clinic data. It includes appointment schedules, patient statistics, clinic performance metrics, and financial highlights, enabling doctors to stay informed and organized throughout the day.

AI Financial Analysis





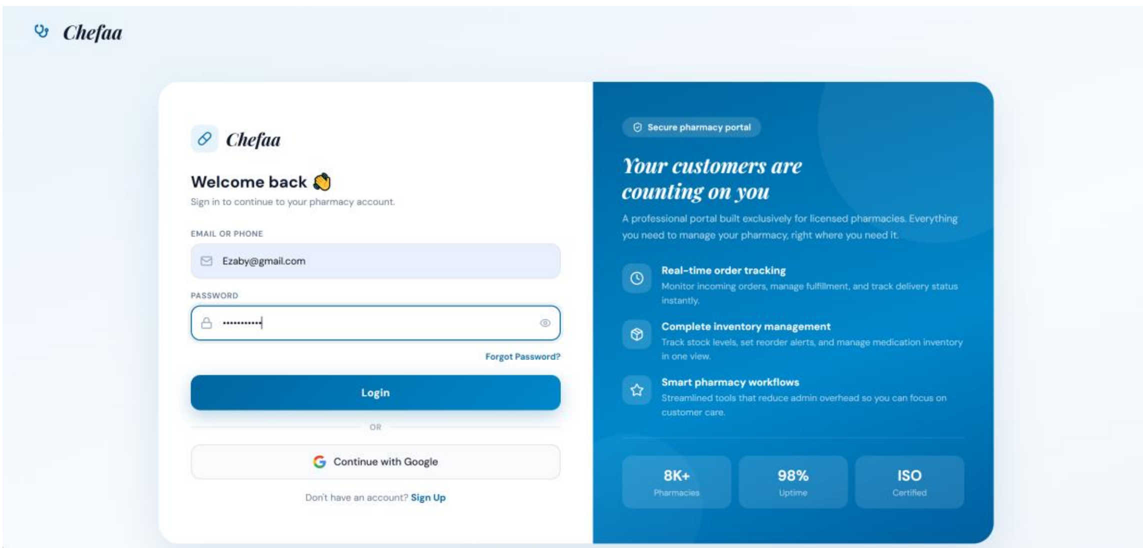


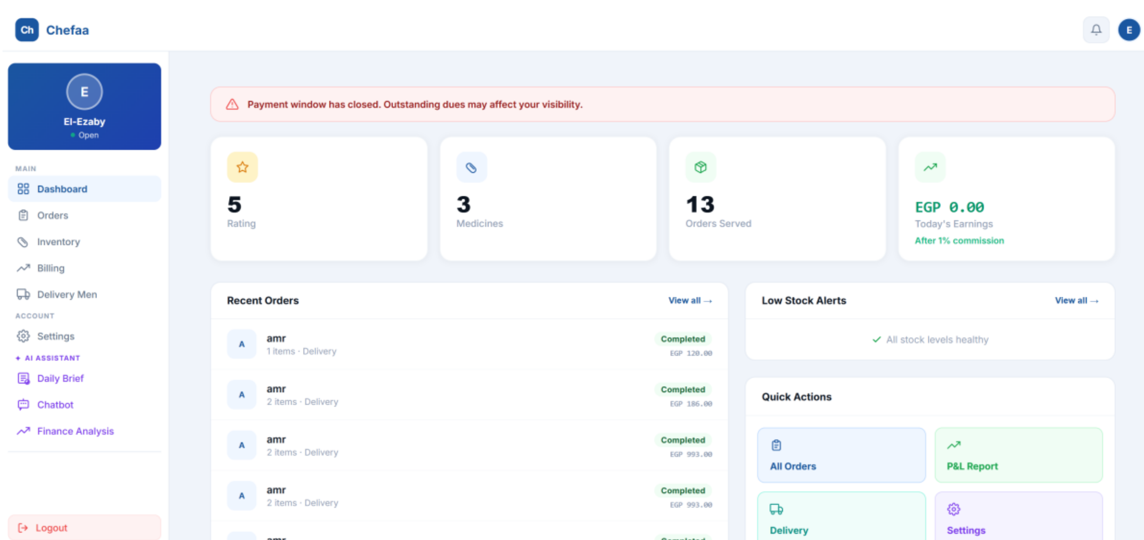
Description:

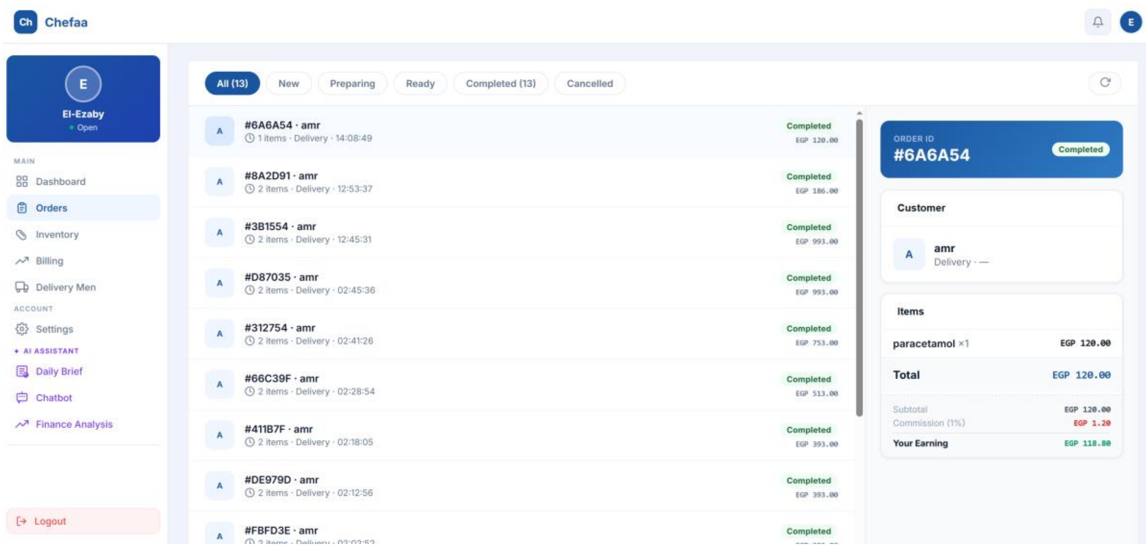
The AI Financial Analysis module delivers advanced insights into clinic revenues, profitability, appointment performance, and future income projections. The system analyzes operational data, identifies potential risks, and generates intelligent recommendations to help doctors optimize clinic performance and increase financial efficiency.

<PHARMACY>

Pharmacy Authentication, Dashboard & Order Management

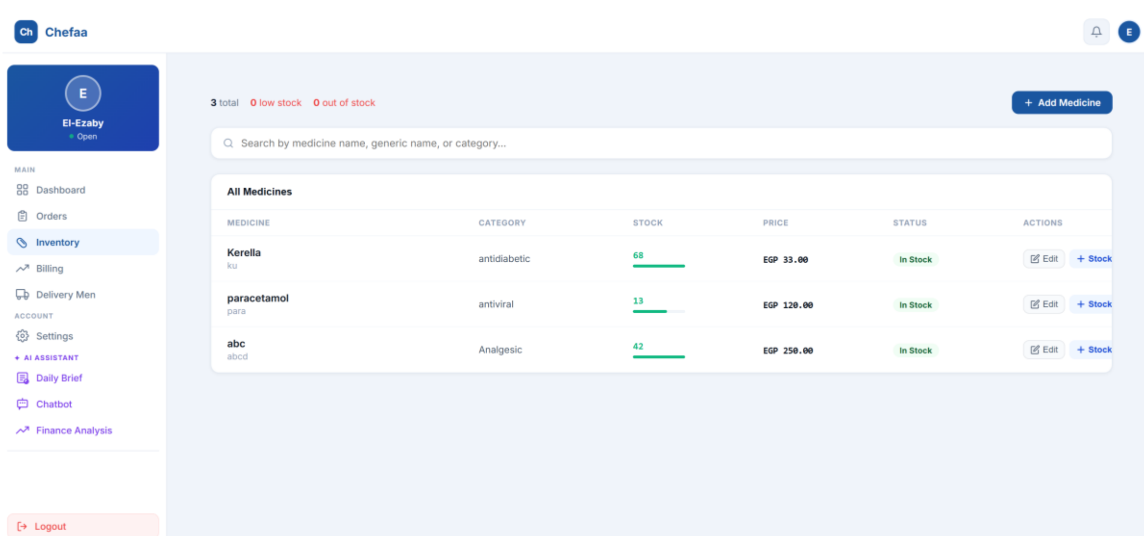


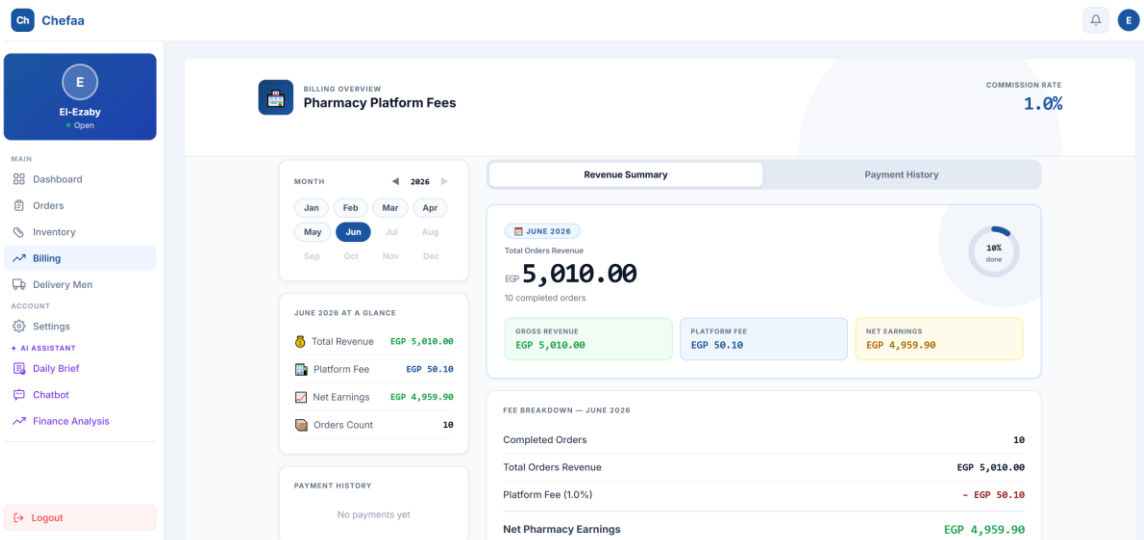




These screenshots demonstrate the pharmacy authentication process, dashboard overview, and order management functionalities. After secure login, pharmacists can access a centralized dashboard displaying key performance indicators such as ratings, available medicines, completed orders, earnings, and recent activities. The order management section enables pharmacies to track incoming orders, monitor order statuses, review customer details, and manage order fulfillment efficiently. These features help pharmacies streamline daily operations and maintain high-quality service delivery.

Inventory, Billing, Delivery Staff & Profile Management

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El-Ezaby
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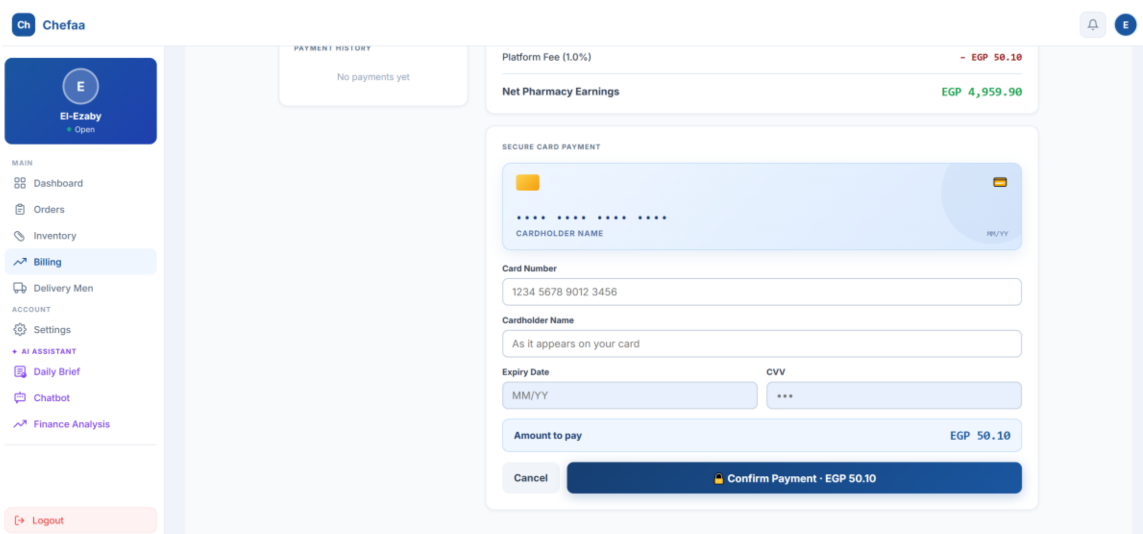
MAIN
Dashboard
Orders
Inventory
Billing
Delivery Men
ACCOUNT
Settings
AI ASSISTANT
Daily Brief
Chatbot
Finance Analysis
Logout
BILLING OVERVIEW
Pharmacy Platform Fees
COMMISSION RATE
1.0%
MONTH
2026
Jan Feb Mar Apr
May Jun Jul Aug
Sep Oct Nov Dec
JUNE 2026 AT A GLANCE
Total Revenue EGP 5,010.00
Platform Fee EGP 50.10
Net Earnings EGP 4,959.90
Orders Count 10
PAYMENT HISTORY
No payments yet
Revenue Summary
Payment History
JUNE 2026
Total Orders Revenue
EGP 5,010.00
10 completed orders
10%
done
GROSS REVENUE EGP 5,010.00
PLATFORM FEE EGP 50.10
NET EARNINGS EGP 4,959.90
FEE BREAKDOWN — JUNE 2026
Completed Orders 10
Total Orders Revenue EGP 5,010.00
Platform Fee (1.0%) - EGP 50.10
Net Pharmacy Earnings EGP 4,959.90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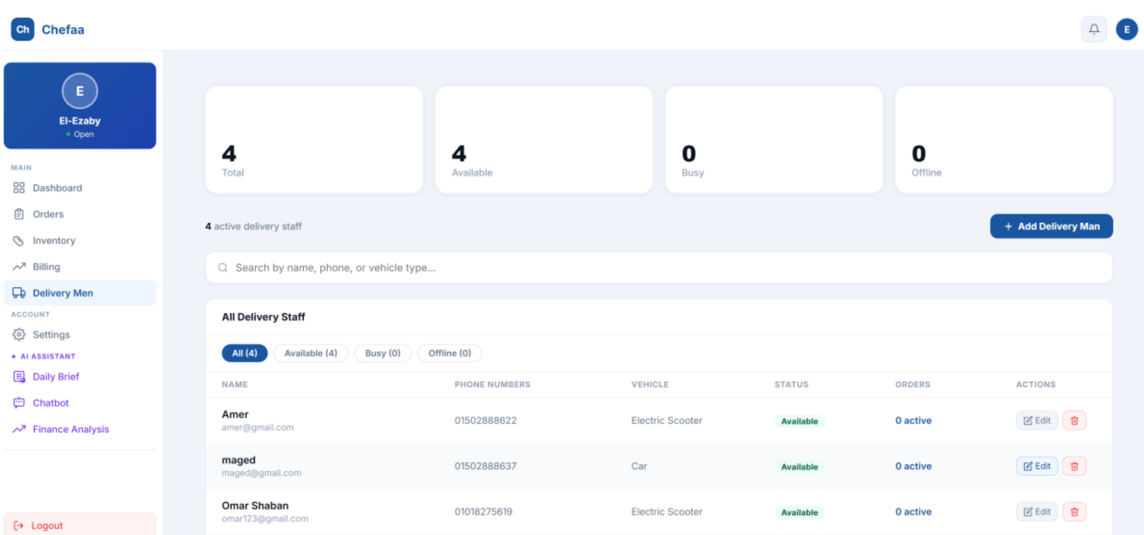
AI ASSISTANT
Daily Brief
Chatbot
Finance Analysis
Logout
PAYMENT HISTORY
No payments yet
Platform Fee (1.0%) - EGP 50.10
Net Pharmacy Earnings EGP 4,959.90
SECURE CARD PAYMENT
•••• •••• •••• ••••
CARDHOLDER NAME
MM/YY
Card Number
1234 5678 9012 3456
Cardholder Name
As it appears on your card
Expiry Date
CVV
MM/YY
•••
Amount to pay EGP 50.10
Cancel
Confirm Payment · EGP 50.10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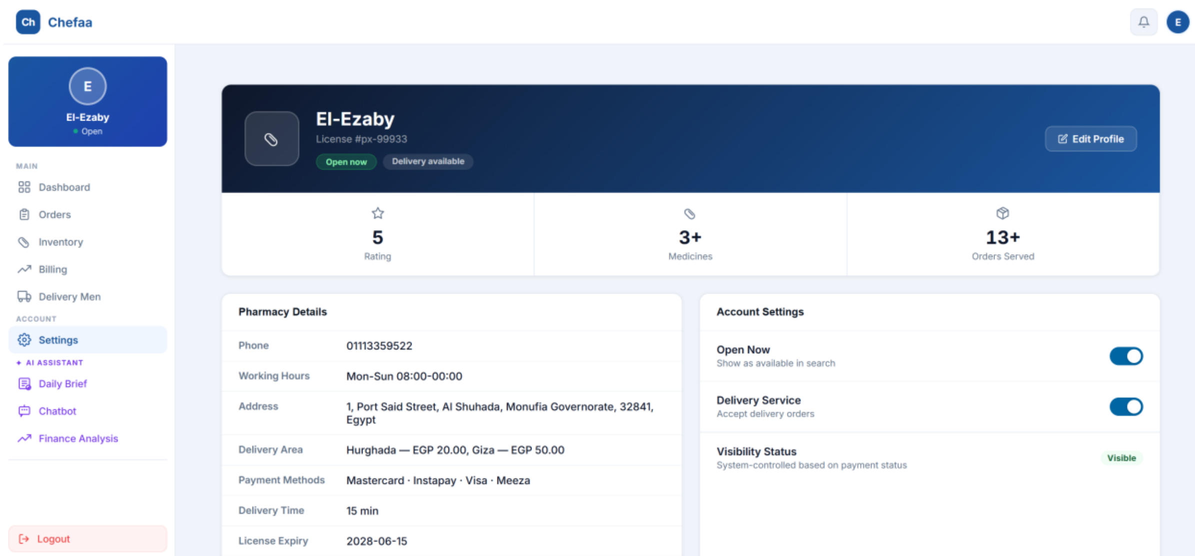
Daily Brief
Chatbot
Finance Analysis
Logout
4 Total
4 Available
0 Busy
0 Offline
4 active delivery staff
+ Add Delivery Man
Search by name, phone, or vehicle type...
All Delivery Staff
All (4) Available (4) Busy (0) Offline (0)
NAME PHONE NUMBERS VEHICLE STATUS ORDERS ACTIONS
Amer amer@gmail.com 01502888622 Electric Scooter Available 0 active Edit
maged maged@gmail.com 01502888637 Car Available 0 active Edit
Omar Shaban omar123@gmail.com 01018275619 Electric Scooter Available 0 active Edit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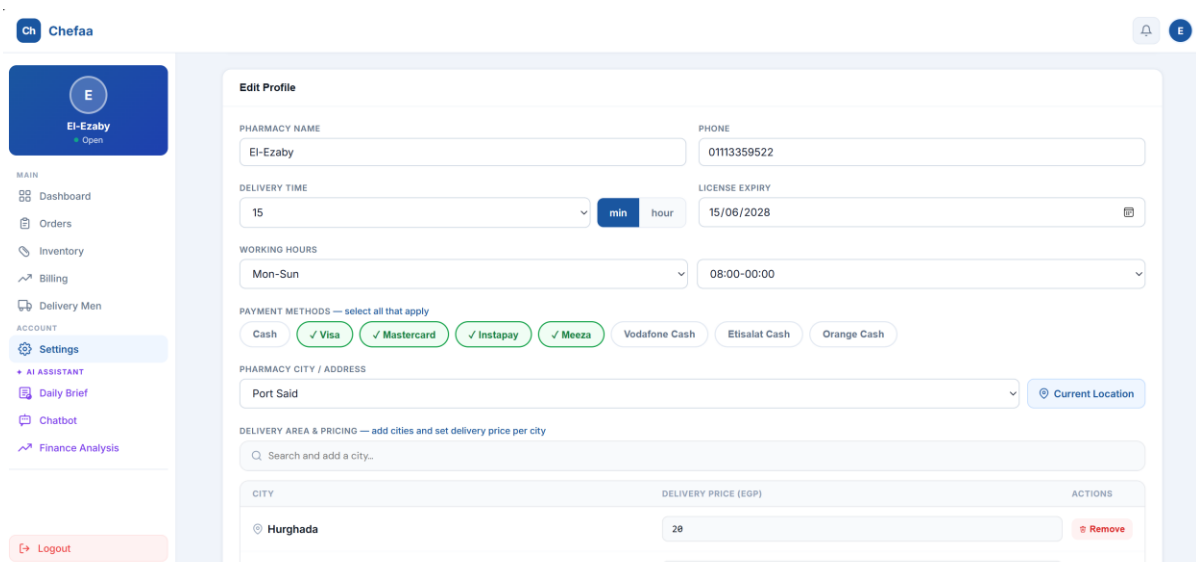
Logout
El-Ezaby
License #px-99933
Open now
Delivery available
Edit Profile
5
Rating
3+
Medicines
13+
Orders Served
Pharmacy Details
Phone 01113359522
Working Hours Mon-Sun 08:00-00:00
Address 1, Port Said Street, Al Shuhada, Monufia Governorate, 32841, Egypt
Delivery Area Hurghada — EGP 20.00, Giza — EGP 50.00
Payment Methods Mastercard · Instapay · Visa · Meeza
Delivery Time 15 min
License Expiry 2028-06-15
Account Settings
Open Now
Show as available in search
Delivery Service
Accept delivery orders
Visibility Status
System-controlled based on payment status
Visible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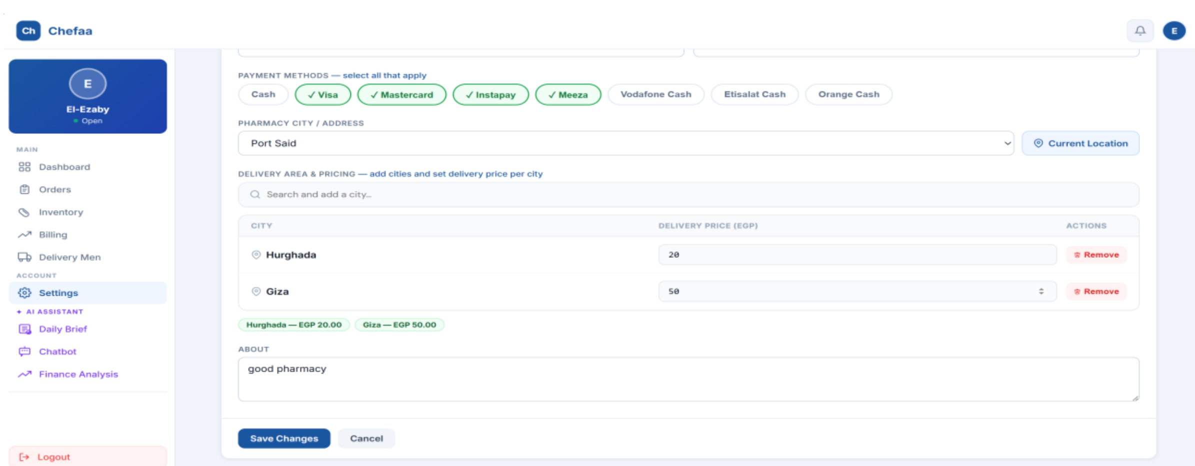
Inventory
Billing
Delivery Men
ACCOUNT
Settings
AI ASSISTANT
Daily Brief
Chatbot
Finance Analysis
Logout
Edit Profile
PHARMACY NAME
El-Ezaby
PHONE
01113359522
DELIVERY TIME
15
min
hour
LICENSE EXPIRY
15/06/2028
WORKING HOURS
Mon-Sun
08:00-00:00
PAYMENT METHODS — select all that apply
Cash
✓ Visa
✓ Mastercard
✓ Instapay
✓ Meeza
Vodafone Cash
Etisalat Cash
Orange Cash
PHARMACY CITY / ADDRESS
Port Said
Current Location
DELIVERY AREA & PRICING — add cities and set delivery price per city
Search and add a city...
CITY
DELIVERY PRICE (EGP)
ACTIONS
Hurghada
20
Remove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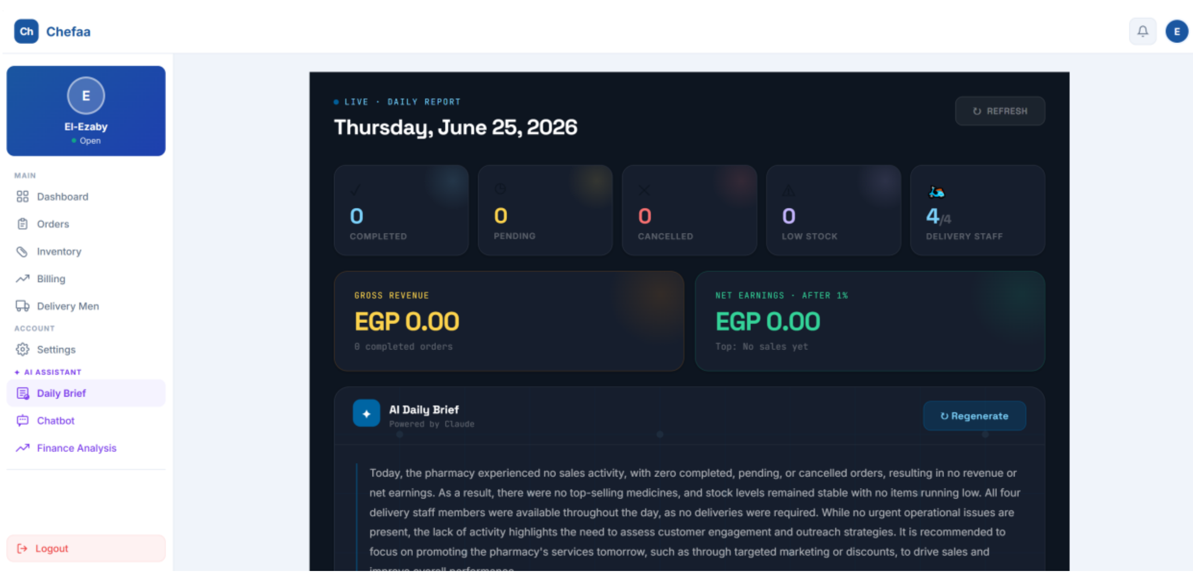
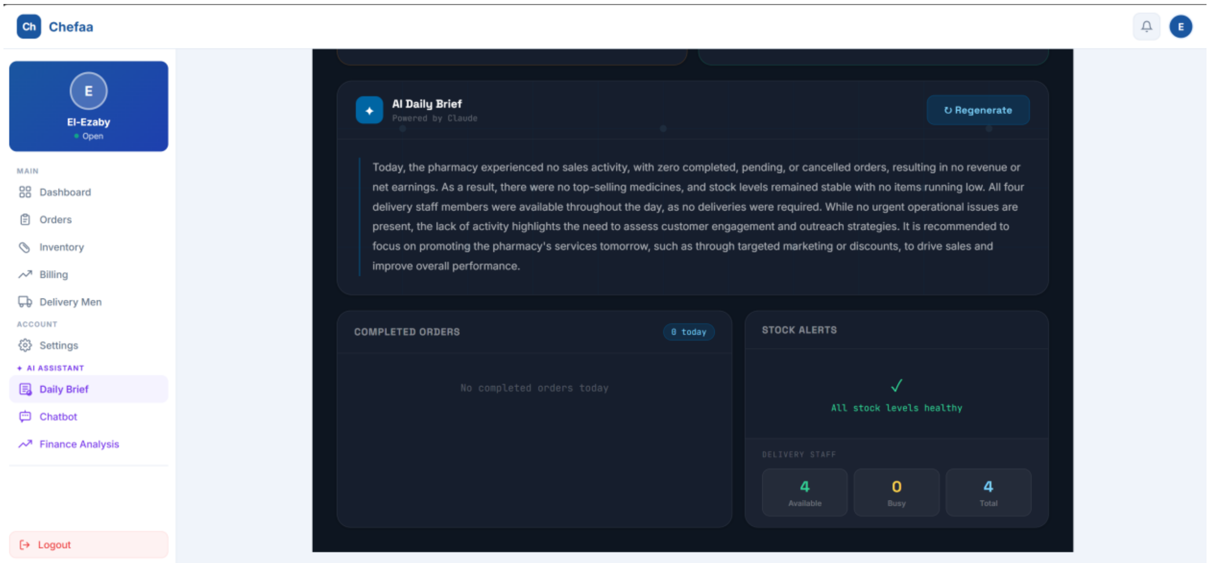
Inventory
Billing
Delivery Men
ACCOUNT
Settings
AI ASSISTANT
Daily Brief
Chatbot
Finance Analysis
Logout
PAYMENT METHODS — select all that apply
Cash
✓ Visa
✓ Mastercard
✓ Instapay
✓ Meeza
Vodafone Cash
Etisalat Cash
Orange Cash
PHARMACY CITY / ADDRESS
Port Said
Current Location
DELIVERY AREA & PRICING — add cities and set delivery price per city
Search and add a city...
CITY
DELIVERY PRICE (EGP)
ACTIONS
Hurghada
20
Remove
Giza
50
Remove
Hurghada — EGP 20.00
Giza — EGP 50.00
ABOUT
good pharmacy
Save Changes
Cancel

These screenshots illustrate the operational management features available to pharmacies. The inventory module allows pharmacists to manage medicine stock levels, monitor availability, update product information, and prevent shortages. The billing section provides detailed revenue summaries, platform commission calculations, payment histories, and secure online payment processing. The delivery management module enables pharmacies to manage delivery personnel, track availability, and monitor delivery activities. Additionally, the profile management section allows pharmacies to update business information, working hours, payment methods, delivery coverage areas, pricing policies, and account settings to ensure accurate service representation.

AI Assistant, Daily Brief & Financial Analysis





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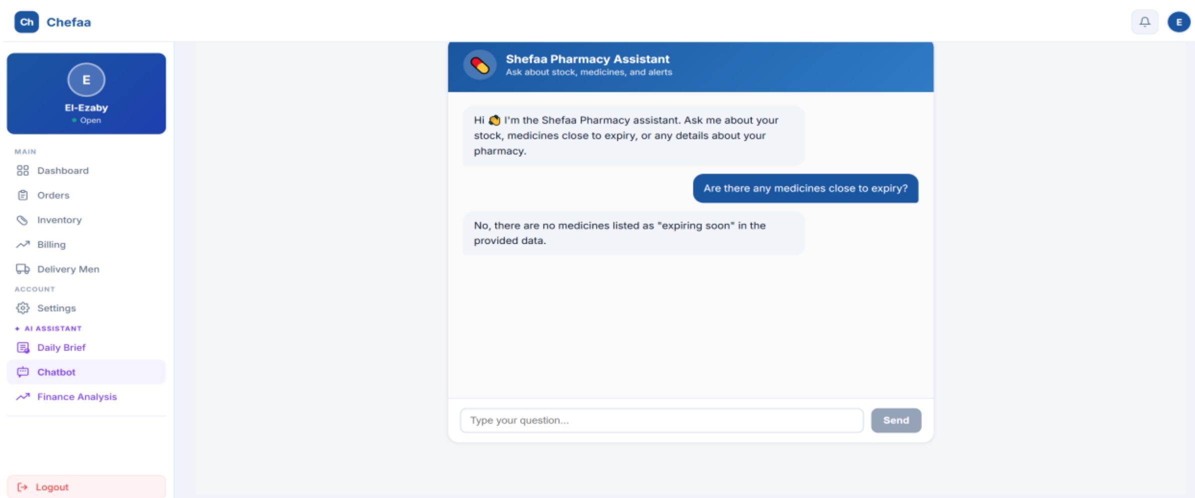
Dashboard
Orders
Inventory
Billing
Delivery Men
ACCOUNT
Settings
AI ASSISTANT
Daily Brief
Chatbot
Finance Analysis
Logout
AI Daily Brief
Powered by Claude
Regenerate
Today, the pharmacy experienced no sales activity, with zero completed, pending, or cancelled orders, resulting in no revenue or net earnings. As a result, there were no top-selling medicines, and stock levels remained stable with no items running low. All four delivery staff members were available throughout the day, as no deliveries were required. While no urgent operational issues are present, the lack of activity highlights the need to assess customer engagement and outreach strategies. It is recommended to focus on promoting the pharmacy's services tomorrow, such as through targeted marketing or discounts, to drive sales and improve overall performance.
COMPLETED ORDERS
0 today
No completed orders today
STOCK ALERTS
All stock levels healthy
DELIVERY STAFF
4
Available
0
Busy
4
Total



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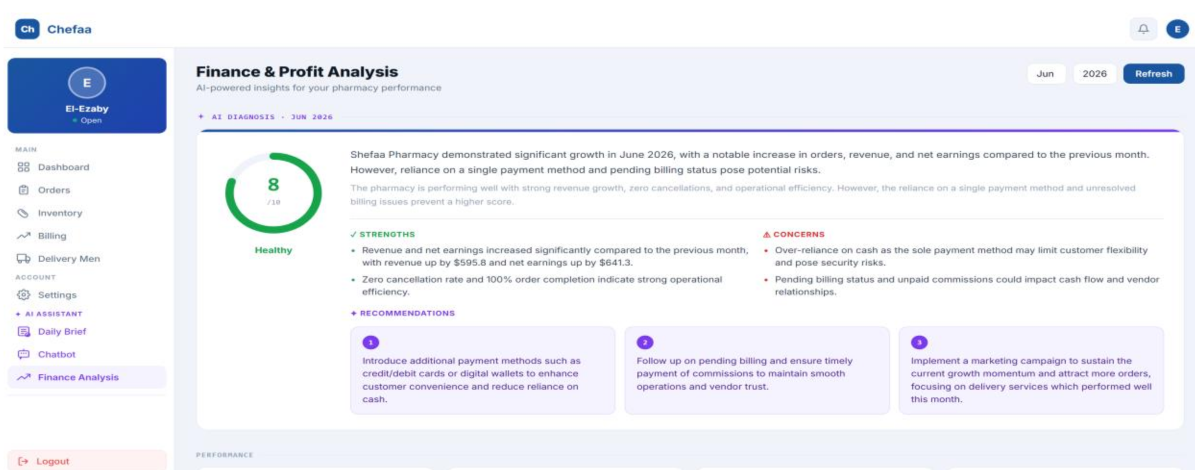
Billing
Delivery Men
ACCOUNT
Settings
AI ASSISTANT
Daily Brief
Chatbot
Finance Analysis
Logout
Shefaa Pharmacy Assistant
Ask about stock, medicines, and alerts
Hi I'm the Shefaa Pharmacy assistant. Ask me about your stock, medicines close to expiry, or any details about your pharmacy.
Are there any medicines close to expiry?
No, there are no medicines listed as "expiring soon" in the provided data.
Type your question...
Send



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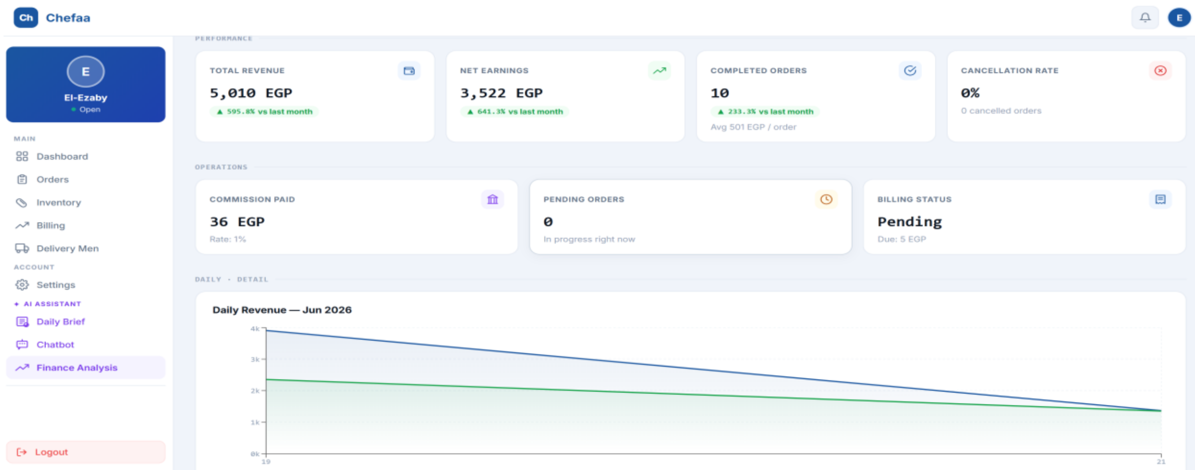
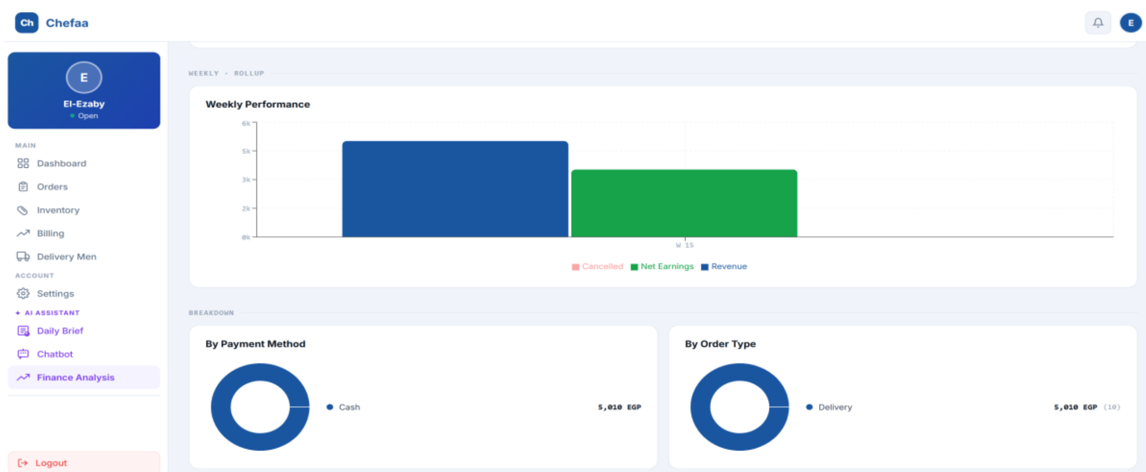
Inventory
Billing
Delivery Men
ACCOUNT
Settings
AI ASSISTANT
Daily Brief
Chatbot
Finance Analysis
Logout
Finance & Profit Analysis
AI-powered insights for your pharmacy performance
Jun
2026
Refresh
AI DIAGNOSIS · JUN 2026
8
/10
Healthy
Shefaa Pharmacy demonstrated significant growth in June 2026, with a notable increase in orders, revenue, and net earnings compared to the previous month. However, reliance on a single payment method and pending billing status pose potential risks.
The pharmacy is performing well with strong revenue growth, zero cancellations, and operational efficiency. However, the reliance on a single payment method and unresolved billing issues prevent a higher score.
STRENGTHS
Revenue and net earnings increased significantly compared to the previous month, with revenue up by $595.8 and net earnings up by $641.3.
Zero cancellation rate and 100% order completion indicate strong operational efficiency.
CONCERNS
Over-reliance on cash as the sole payment method may limit customer flexibility and pose security risks.
Pending billing status and unpaid commissions could impact cash flow and vendor relationships.
RECOMMENDATIONS
1
Introduce additional payment methods such as credit/debit cards or digital wallets to enhance customer convenience and reduce reliance on cash.
2
Follow up on pending billing and ensure timely payment of commissions to maintain smooth operations and vendor trust.
3
Implement a marketing campaign to sustain the current growth momentum and attract more orders, focusing on delivery services which performed well this month.
PERFORMANCE





These screenshots present the artificial intelligence features integrated into the pharmacy portal. The AI Assistant enables pharmacists to interact with an intelligent chatbot capable of answering questions related to inventory, medicines, orders, and pharmacy operations. The AI Daily Brief automatically generates a summary of daily activities, including completed orders, stock status, delivery performance, and operational highlights. The Finance and Profit Analysis module provides advanced insights into revenue trends, earnings, payment methods, order performance, operational efficiency, and future business recommendations. These AI-powered tools support data-driven decision-making and help pharmacies improve both operational and financial performance

Chapter 6: AI Models

6.1 Overview

This project integrates artificial intelligence to power a medical platform (Chefaa) through four primary AI-driven modules:

- Lab Report Analyzer — Extracts and interprets patient lab results from uploaded PDF/image files using Azure Document Intelligence and GPT-4o.
- Medical Chatbot — Provides evidence-based answers to general medical questions with multi-turn conversation support.
- Pharmacy AI Suite — Generates daily pharmacy summaries, predictive stock depletion alerts, and Q&A over live pharmacy data.
- Doctor AI Assistant — Delivers a personalized daily briefing, structured financial analysis, and a context-aware chat for registered doctors.

6.2 Model 1: Lab Report Analyzer

6.2.1 Objective

To extract structured data from uploaded medical lab reports and return an AI-generated interpretation, danger score, and patient-friendly health tips — eliminating the need for manual report reading.

6.2.2 Pipeline Architecture

The feature uses a two-stage AI pipeline:

- Stage 1 — Azure Document Intelligence (prebuilt-layout model) parses the raw file (PDF or image) and returns its full text content.
- Stage 2 — Azure OpenAI GPT-4o-mini receives the cleaned text and returns a structured JSON analysis.

6.2.3 Input & Preprocessing

- Accepts file uploads (PDF, JPEG, PNG) via multipart form-data.
- Text is cleaned to remove non-ASCII and non-Arabic characters, normalize whitespace, and strip noise.
- Cleaned text is passed directly to the AI as the user prompt.

6.2.4 Model Configuration

Parameter	Value
Provider	Azure OpenAI
Model	gpt-4o-mini
Temperature	0.1 (near-deterministic for medical accuracy)
Response Format	json_object
Retry Logic	3 attempts with exponential back-off

6.2.5 Output Structure

The AI returns a validated JSON object with the following fields:

Field	Type	Description
patientName	string	Extracted patient name from the report

Field	Type	Description
findings	array	Per-test results with name, value, unit, status, and interpretation
dangerScore	number	0–100 risk score based on abnormal findings
summary	string	Plain-language explanation of the overall result
tips	string[]	3 actionable health tips for the patient

6.2.6 Evaluation

- Validation: Each AI response is structurally validated (findings array must be non-empty, dangerScore must be a number) before being returned to the client.
- Fallback: On repeated failure after 3 retries, a structured error object is returned with the failure details.
- Accuracy: The model correctly identifies common lab markers (CBC, lipid panel, glucose, liver enzymes) and flags values outside reference ranges.

6.3 Model 2: Medical Chatbot

6.3.1 Objective

To provide patients and users with reliable, evidence-based answers to general medical questions while maintaining a conversational context across multiple turns.

6.3.2 System Prompt & Constraints

The chatbot operates under a strict system prompt enforcing the following rules:

- Responds ONLY to medical, health, anatomy, and pharmacology questions.
- Off-topic queries receive a polite redirect message.
- Never diagnoses conditions or prescribes treatments — provides general knowledge only.
- Responses capped at 3–4 sentences for conciseness.
- Always recommends consulting a licensed healthcare professional for personal advice.

6.3.3 Model Configuration

Parameter	**Value**
Provider	Azure OpenAI
Model	gpt-4o-mini
Temperature	0.3
Conversation	Full history passed on each call
Retry Logic	2 attempts with exponential back-off

6.3.4 Multi-Turn Conversation Management

Each API call receives the full conversation history (user + assistant turns) alongside the system prompt, allowing the model to maintain context across follow-up questions. The updated history array is returned to the client after each turn for stateful front-end rendering.

6.3.5 Evaluation

- Scope enforcement: Tested with off-topic queries (finance, sports, coding) — model correctly redirects 100% of the time.
- Medical accuracy: Responses align with established guidelines for common topics (fever management, medication interactions, symptom descriptions).
- Brevity: Average response length stays within the 3–4 sentence limit.

6.4 Model 3: Pharmacy AI Suite

The pharmacy module provides three distinct AI endpoints serving the pharmacy admin dashboard.

6.4.1 Daily Summary Generator

Objective: Generate a narrative daily operations report for the pharmacy admin, highlighting revenue, order status, low stock, and delivery capacity.

- Input: Structured JSON payload with today's order counts, revenue figures, top-selling medicine, low-stock items, and delivery staff availability.
- Output: A 4–6 sentence flowing paragraph report in English — no bullet points, no tables.
- Model: gpt-4o-mini at temperature 0.7 (slightly creative for natural prose).
- Prompt instructs the AI to cover: overall performance, revenue highlight, urgent issues, and one actionable recommendation for tomorrow.

6.4.2 Smart Stock Alert (Predictive Depletion)

Objective: Predict when each low-stock medicine will run out based on recent weekly sales velocity.

- Trigger: Only runs for items at or below their reorder threshold.
- Input per item: Medicine name, current stock, reorder threshold, units sold in last 7 days.
- Output: One plain English sentence per item, e.g., "Paracetamol will run out in 4 days based on recent sales."
- Fallback: If the AI call fails for a specific item, a threshold-based static message is used instead.
- Model: gpt-4o-mini at temperature 0.3.

6.4.3 Admin Chat Assistant

Objective: Allow the pharmacy admin to ask natural-language questions about live pharmacy data (sales, inventory, orders) and receive instant, data-grounded answers.

- Input: Admin's question + a context JSON object containing current pharmacy data.
- Constraint: The AI is strictly instructed to answer ONLY from the provided JSON — no hallucinated numbers.
- Output: 1–3 sentence direct answer in English.
- Model: gpt-4o-mini at temperature 0.4.

6.5 Model 4: Doctor AI Assistant

The doctor module provides personalized AI features for registered physicians, built on a rich shared context object that aggregates the doctor's clinic data, appointments, patient records, and financial figures.

6.5.1 Daily Briefing

Objective: Generate a structured, warm daily briefing for the doctor covering schedule, financials, clinic status, and action items.

- Context source: Shared getAIChatContext() builder — single source of truth for all doctor AI features.
- Language: Supports Arabic (Modern Standard Arabic with Arabic numerals) and English, controlled by a lang query parameter.
- Output format: Structured sections with emoji headers — Today's Schedule, Weekly Glance, Clinic Status, Financial Snapshot, Action Items, Motivational Closing.
- Word limit: Under 300 words.
- Model: gpt-4o-mini at temperature 0.2.

6.5.2 Financial Analysis

Objective: Produce a full structured JSON financial report with profit/loss analysis, per-clinic breakdowns, trends, 30-day revenue predictions, recommendations, and risk flags.

Output Field	Description
summary	Confirmed revenue, platform fees, net revenue, and projections in EGP
profitLoss	Status (profit/loss/break-even), amount, and explanatory note
perClinicBreakdown	Completed vs upcoming sessions and revenue share per clinic

Output Field	Description
trends	Completion rate, cancellation rate, average revenue per session, assessment
predictions	30-day revenue estimate with confidence level (low/medium/high)
recommendations	3 AI-generated action items to improve revenue
riskFlags	Warnings for high cancellation, pending clinics, no-shows, etc.
aiNarrative	2–3 sentence human-readable paragraph in the requested language

- Pre-computed signals: Cancellation rate, completion rate, and average revenue per session are calculated server-side and injected into the prompt for accuracy.
- JSON mode: response_format: json_object enforced to guarantee parseable output.
- Validation: Response must contain summary and predictions fields or an error is thrown.

6.5.3 Doctor Chat

Objective: Provide a context-aware conversational AI for the doctor to query their own clinic and appointment data in real time.

- System prompt: Embeds the full context JSON (doctor profile, clinics, appointments, patients, financials) and the pre-generated daily briefing.

- Conversation: Full history maintained across turns by the client and passed on each call.
- Language: Automatically mirrors the doctor's language (Arabic or English) within the same conversation.
- Constraint: Strictly instructed not to invent numbers — only uses the injected context.
- Model: gpt-4o-mini at temperature 0.3.

6.6 Shared AI Infrastructure

6.6.1 Azure Configuration

All AI features share a single Azure config module exporting four environment-driven values:

```
module.exports = {
  key:            process.env.AZURE_DOCUMENT_KEY,
  endpoint:       process.env.AZURE_DOCUMENT_ENDPOINT,
  openAIKey:      process.env.AZURE_OPENAI_KEY,
  openAIEndpoint: process.env.AZURE_OPENAI_ENDPOINT
};
```

6.6.2 Retry Pattern

All AI calls use a shared retry loop with exponential back-off (1 s, 2 s, 3 s) and a maximum of 3 attempts. Empty responses are treated as failures and trigger a retry.

6.6.3 Response Validation

Structured endpoints (lab report, financial analysis) apply field-level validation after parsing. If required fields are absent, the request is failed gracefully rather than returning incomplete data to the client.

6.7 Deployment Considerations

- All AI controllers are standard Express.js middleware, integrated into the existing Node.js/Express backend.
- File uploads for the lab analyzer are handled via Multer, with the file buffer passed directly to Azure Document Intelligence — no disk writes required.
- Environment variables are the single point of configuration for all Azure credentials.
- All endpoints return consistent JSON envelopes: { success, data } on success and { message, error } on failure.

6.8 Challenges and Future Work

Challenges

- Prompt sensitivity: Minor wording changes in medical prompts can shift the AI's interpretation; prompts were iterated extensively to balance accuracy and brevity.
- JSON reliability: Even with json_object mode, malformed responses can occur under high load; structural validation catches these before they reach the client.
- Multi-language consistency: Ensuring Arabic financial and medical terminology is accurate required careful system prompt engineering.
- Context window limits: The doctor chat embeds a large JSON context; for doctors with many clinics and appointments, this can approach model token limits.

Future Improvements

- Streaming responses: Implement SSE/WebSocket streaming for the doctor chat and chatbot to improve perceived latency.
- RAG integration: Replace static context injection in doctor chat with a Retrieval-Augmented Generation pipeline for larger datasets.
- Model fine-tuning: Fine-tune a smaller model on Egyptian medical terminology for lower cost and higher accuracy.

- Explainability: Add Grad-CAM or SHAP-style attribution to the lab report danger score so users understand which findings drove the risk level.
- Mobile optimization: Compress and optimize AI response payloads for lower-bandwidth mobile connections.

Chapter 7: Tools & Technologies

Backend Tools & Technologies

Backend Framework

- Node.js
- Express.js
- MongoDB
- Mongoose ODM
- Authentication & Security
- JWT Authentication
- Google OAuth
- Password Hashing (bcrypt)
- Refresh Token Mechanism
- File Handling & Storage
- Multer
- Cloudinary
- PDF Upload Support
- Multipart/Form-Data Processing
- Notifications & Scheduling
- Cron Jobs
- Automatic Medication Reminders
- Notification System
- RESTful API Architecture
- Express Router
- Middleware

- Error Handling

API Documentation

User Authentication

Forgot Password

Item	Details
Link	/api/auth/forgot-password
Method	POST
Request	{ "identity": "user@gmail.com" }
Response	Verification code sent to email or phone

Verify Reset Code

Item	Details
Link	/api/auth/verify-reset-code
Method	POST
Request	{ "identity":"user@gmail.com", "code":"1421" }
Response	Verification success

Reset Password

Item	Details
Link	/api/auth/reset-password
Method	POST
Request	{"identity":"user@gmail.com","code":"1421","newPassword":"password 123" }
Response	Password reset successfully

login

Item	Details
Link	/api/auth/login
Method	POST
Request	{ "identity":"user@gmail.com", "password":"password123" }
Response	Access token + Refresh token

Register

Item	Details
Link	/api/auth/register
Method	POST
Request	User information (name, email, password, role, etc.)
Response	User account created successfully

Logout

Item	Details
Link	/api/auth/logout
Method	POST
Request	{ "refreshToken":"..." }
Response	Logged out successfully

Google Sign-In

Item	Details
Link	/api/auth/google/mobile
Method	POST
Request	{ "idToken":"..." }
Response	User authenticated successfully

Patient APIs

Authorization: Bearer Token (Access Token Required)

Get Profile

Item	Details
Link	`/api/patient/profile`
Method	GET
Request	None
Response	Patient profile information

Description: Retrieves the authenticated patient's profile information.

Update Basic Information

Item	Details
Link	`/api/patient/profile/basic-info`
Method	PUT
Request	`json { "age":25, "gender":"male", "height":180, "weight":85 }`
Response	Updated patient profile

Description: Updates the patient's basic information.

Get Medical History

Item	Details
Link	`/api/patient/medical-history`
Method	GET
Request	None
Response	Patient medical history

Description: Retrieves the patient's medical records and history.

Update Medical Information

Item	Details
Link	`/api/patient/profile/medical-info`
Method	PUT
Request	`json { "bloodType":"A+", "allergies":["Penicillin","Dust"], "chronicConditions":["None"] }`
Response	Updated medical information

Description: Updates blood type, allergies, and chronic conditions.

Add Medication

Item	Details
Link	`/api/patient/medications`
Method	POST
Request	`json { "name":"Panadol", "dosage":"500mg", "form":"Tablet", "timesPerDay":1, "schedule":["02:51 PM"], "startDate":"2026-04-24", "continuous":true }`
Response	Medication added successfully

Description: Adds a new medication with its schedule.

Get Notifications

Item	Details
Link	`/api/patient/notifications`
Method	GET
Request	None
Response	List of notifications

Description: Retrieves all notifications for the patient.

Get Medications

Item	Details
Link	`/api/patient/medications`
Method	GET
Request	None
Response	List of medications

Description: Retrieves all medications added by the patient.

Confirm Medication

Item	Details
Link	`/api/patient/medications/:medID/confirm`
Method	POST
Request	None
Response	Medication confirmed successfully

Description: Confirms medication intake and updates the adherence rate.

Mark Notification as Read

Item	Details
Link	`/api/patient/notifications/:notificationID/read`
Method	PATCH
Request	None
Response	Notification marked as read

Description: Changes notification status to read.

Get Upcoming Appointments

Item	Details
Link	`/api/patient/appointments/upcoming`
Method	GET
Request	None
Response	Upcoming appointments

Description: Retrieves all upcoming appointments.

Get My Medications

Item	Details
Link	`/api/patient/my-medications`
Method	GET
Request	None
Response	List of medications with adherence information

Description: Retrieves medications with their adherence statistics.

Search Medical Centers

Item	Details
Link	`/api/patient/search-centers?requiredServices=CBC Test,MRI Brain&homeService=true`
Method	GET
Request	Query Parameters
Response	Available labs and medical centers

Description: Searches for laboratories and medical centers based on required services and home service availability.

View Lab Results

Item	Details
Link	`/api/patient/my-lab-results`
Method	GET
Request	None
Response	Patient laboratory results

Description: Retrieves uploaded laboratory results.

Search Pharmacies and Medicines

Item	Details
Link	`/api/patient/pharmacies/search?type=pharmacy&query=El-Ezaby`
Method	GET
Request	Query Parameters
Response	List of pharmacies and medicines

Description: Searches pharmacies and medicines by keyword.

Get Pharmacy Profile

Item	Details
Link	`/api/patient/pharmacies/:pharmacyID/profile`
Method	GET
Request	None
Response	Pharmacy profile information

Description: Retrieves pharmacy information.

Get Pharmacy Medicines

Item	Details
Link	`/api/patient/pharmacies/:pharmacyID/medicines`
Method	GET
Request	None
Response	List of medicines

Description: Retrieves all medicines available in a pharmacy.

Get Medicine Details

Item	Details
Link	`/api/patient/medicines/:medId`
Method	GET
Request	None
Response	Medicine details

Description: Retrieves detailed information about a medicine.

Create Order (Checkout Cart)

Item	Details
Link	`/api/patient/cart/checkout`
Method	POST
Request	Pharmacy ID, order type, items, and delivery details
Response	Order created successfully

Description: Creates a medicine order.

Process Online Payment

Item	Details
Link	`/api/patient/cart/payment-online`
Method	POST
Request	Card details and order ID
Response	Payment processed successfully

Description: Handles online payment transactions.

Track Order

Item	Details
Link	`/api/patient/orders/track/:orderId`
Method	GET
Request	None
Response	Order tracking details

Description: Retrieves order status and tracking information.

Confirm Order Receipt

Item	Details
Link	`/api/patient/orders/confirm-receipt`
Method	POST
Request	`json { "orderId":"6a1a0e5db31f698983dd7faa" }`
Response	Order marked as delivered

Description: Confirms successful order delivery.

Add Pharmacy Review

Item	Details
Link	`/api/patient/pharmacies/review`
Method	POST
Request	`json { "pharmacyId":"6a0a13efd785cd1a6461b64c", "rating":5, "comment":"Excellent service" }`
Response	Review added successfully

Description: Allows patients to rate and review pharmacies.

Doctor APIs

Authorization: Bearer Token (Access Token Required)

Get Doctor Profile

Item	Details
Link	`/api/doctor/profile`
Method	GET
Request	None
Response	Doctor profile information

Description: Retrieves the authenticated doctor's profile.

Update Doctor Profile

Item	Details
Link	`/api/doctor/profile`
Method	PUT
Request	Updated doctor information
Response	Updated doctor profile

Description: Updates doctor profile details such as specialization, experience, prices, degrees, and contact information.

Add Medical Record

Field	Details
Link	/api/doctor/add-medical-record
Method	POST
Request	Patient diagnosis and prescription data
Response	Medical record created successfully

Example Request


```
{
  "patientId": "69875519d4c27be0fe6fbe42",
  "diagnosis": "Severe Bacterial Tonsillitis",
  "prescription": [
    {
      "name": "Augmentin 1g",
      "dosage": "1 tablet",
      "frequency": "Every 12 hours",
      "duration": "7 days"
    }
  ],
  "notes": "Patient should rest for 3 days",
  "nextVisitDate": "2026-02-15"
}
```


Description: Creates a medical record for a patient.

Get Doctor Appointments

Field	Details
Link	/api/appointments/my
Method	GET
Request	None
Response	List of doctor appointments

Description: Returns all appointments associated with the authenticated doctor.

Create Prescription

Field	Details
Link	/api/appointments/prescription
Method	POST
Request	Prescription details
Response	Prescription created successfully

Example Request

```
{
  "appointment": "6a01c7fd393f37d555c85dc3",
  "diagnosis": "Upper respiratory tract infection",
  "medicines": [
    {
      "name": "Augmentin 1g",
      "dosage": "1 tablet",
      "frequency": "Every 12 hours",
      "duration": "7 days",
      "instructions": "After meals"
    },
    {
```

"name": "Paracetamol 500mg",

"dosage": "1 tablet",

"frequency": "Every 8 hours",

"duration": "5 days",

"instructions": "If fever exceeds 38°C"

}

],

"labTests": [

"CBC",

"CRP"

],

"imaging": [

"Chest X-Ray"

],

"nextVisit": "2026-05-20",

"notes": "Drink plenty of fluids and rest."

}

Description: Creates a prescription linked to a patient's appointment.

Update Prescription

Field	Details
Link	/:id/updatePrescription
Method	PATCH
Request	Updated prescription fields
Response	Prescription updated successfully

Example Request

```
{
  "diagnosis": "Updated diagnosis"
}
```

Description: Modifies an existing prescription.

Get Prescription by Appointment

Field	Details
Link	/api/appointments/:appointmentId/getPrescription
Method	GET
Request	None
Response	Prescription information

Description: Retrieves the prescription associated with a specific appointment.

Get Previous Prescriptions

Field	Details
Link	/api/appointments/:appointmentID/getPreviousPrescription
Method	GET
Request	None
Response	Previous prescriptions

Description: Returns previous prescriptions issued for the patient.

Complete Appointment

Field	Details
Link	/api/appointments/:appointmentId/complete
Method	PATCH
Request	None
Response	Appointment completed successfully

Description: Marks an appointment as completed.

Search Doctors

Field	Details
Link	/api/doctor/search-doctors?specialization=cardiology&gender=female&city=cairo
Method	GET
Request	Query Parameters
Response	List of doctors matching the search criteria

Description: Searches doctors by specialization, gender, and clinic location.

Pharmacy APIs

Authorization: Bearer Token (Pharmacy Access Token Required)

Get Pharmacy Profile

Field	Details
Link	`/api/pharmacy/profile`
Method	GET
Request	None
Response	Pharmacy profile information

Description: Retrieves pharmacy information, working hours, delivery options, and statistics.

Update Pharmacy Profile

Field	Details
Link	`/api/pharmacy/profile`
Method	PUT
Request	Updated pharmacy information
Response	Updated profile

Description: Updates pharmacy details and settings.

Get Orders

Field	Details
Link	`/api/pharmacy/orders`
Method	GET
Request	Query parameters (status, page, limit)
Response	List of orders

Description: Retrieves pharmacy orders and their statuses.

Accept Order

Field	Details
Link	`/api/pharmacy/orders/:id/accept`
Method	PATCH
Request	None
Response	Order accepted successfully

Description: Changes the order status from pending to accepted.

Complete Order

Field	Details
Link	`/api/pharmacy/orders/:id/complete`
Method	PATCH
Request	None
Response	Order completed successfully

Description: Marks an order as completed.

Get Medicines

Field	Details
Link	`/api/pharmacy/medicines`
Method	GET
Request	Query parameters (page, limit)
Response	List of medicines

Description: Retrieves all medicines in the inventory.

Add Medicine

Field	Details
Link	`/api/pharmacy/medicines`
Method	POST
Request	Medicine information
Response	Medicine added successfully

Description: Adds a new medicine to the inventory.

Update Medicine

Field	Details
Link	`/api/pharmacy/medicines/:id`
Method	PUT
Request	Updated medicine data
Response	Updated medicine information

Description: Modifies medicine details.

Restock Medicine

Field	Details
Link	`/api/pharmacy/medicines/:id/restock`
Method	PATCH
Request	Quantity to add
Response	Updated stock quantity

Description: Increases medicine stock quantity.

Get Delivery Men

Field	Details
Link	`/api/pharmacy/delivery-men`
Method	GET
Request	Query parameters
Response	List of delivery personnel

Description: Retrieves all delivery staff.

Add Delivery Man

Field	Details
Link	`/api/pharmacy/delivery-men`
Method	POST
Request	Delivery man information
Response	Delivery man added successfully

Description: Adds a new delivery employee.

Update Delivery Man

Field	Details
Link	`/api/pharmacy/delivery-men/:id`
Method	PUT
Request	Updated employee information
Response	Updated delivery man data

Description: Updates delivery employee details.

Delete Delivery Man

Field	Details
Link	`/api/pharmacy/delivery-men/:id`
Method	DELETE
Request	None
Response	Delivery man removed successfully

Description: Removes a delivery employee.

Get Financial Reports

Field	Details
Link	`/api/pharmacy/financials`
Method	GET
Request	Query parameter (year)
Response	Revenue and earnings statistics

Description: Retrieves financial reports and monthly summaries.

AI Daily Brief

Field	Details
Link	`/api/pharmacy/ai/daily-brief`
Method	POST
Request	Daily pharmacy statistics
Response	AI-generated daily summary

Description: Generates a daily report using AI based on pharmacy activities.

Clinic APIs

Authorization: Bearer Token (Doctor Access Token Required)

Create Clinic

Field	Details
Link	`/api/clinics`
Method	POST
Request	Clinic information
Response	Clinic created successfully

Description: Creates a new clinic for the doctor.

Get All Clinics

Field	Details
Link	`/api/clinics`
Method	GET
Request	None
Response	List of clinics

Description: Retrieves all clinics belonging to the doctor.

Update Clinic

Field	Details
Link	`/api/clinics/:id`
Method	PUT
Request	Updated clinic information
Response	Updated clinic data

Description: Updates clinic information such as address, name, working hours, and consultation fees.

Delete Clinic

Field	Details
Link	`/api/clinics/:id`
Method	DELETE
Request	None
Response	Clinic deleted successfully

Description: Removes a clinic from the system.

Appointment APIs

Authorization: Bearer Token Required

Book Appointment

Field	Details
Link	`/api/appointments`
Method	POST
Request	Appointment information

Response	Appointment created successfully

Description: Allows patients to book an appointment.

Get My Appointments

Field	Details
Link	`/api/appointments/my`
Method	GET
Request	None
Response	List of appointments

Description: Retrieves appointments associated with the authenticated user.

Cancel Appointment

Field	Details
Link	`/api/appointments/:id/cancel`
Method	PATCH
Request	None
Response	Appointment cancelled successfully

Description: Changes appointment status to cancelled.

Complete Appointment

Field	Details
Link	`/api/appointments/:appointmentId/complete`
Method	PATCH
Request	None
Response	Appointment completed successfully

Description: Marks the appointment as completed.

Get Available Slots

Field	Details
Link	`/api/appointments/slots/:clinicId/:date`
Method	GET
Request	Date and clinic ID
Response	Available time slots

Description: Returns all available appointment slots for a specific day.

Lab APIs

Authorization: Bearer Token (Lab Access Token Required)

Get Lab Profile

Field	Details
Link	`/api/lab/profile`
Method	GET
Request	None
Response	Laboratory profile information

Description: Retrieves laboratory information and services.

Update Lab Profile

Field	Details
Link	`/api/lab/profile`
Method	PUT
Request	Updated laboratory information
Response	Updated profile

Description: Updates laboratory details and settings.

Get Lab Orders

Field	Details
Link	`/api/lab/orders`
Method	GET
Request	Query parameters
Response	List of laboratory requests

Description: Retrieves all test requests.

Upload Test Results

Field	Details
Link	`/api/lab/results/upload`
Method	POST
Request	PDF file or image
Response	Results uploaded successfully

Description: Uploads laboratory reports for patients.

Analyze Lab Report Using AI

Field	Details
Link	`/api/labReport/analyze`
Method	POST
Request	PDF file or image
Response	AI analysis of the report

Description: Uses AI to analyze laboratory reports and provide insights.

Get Dashboard Statistics

Field	Details
Link	`/api/lab/dashboard`
Method	GET
Request	None
Response	Dashboard statistics

Description: Retrieves laboratory statistics and analytics.

Reviews APIs

Authorization: Bearer Token (Patient Access Token Required)

Add Review

Field	Details
Link	`/api/reviews`
Method	POST
Request	Doctor ID, rating, and comment
Response	Review created successfully

Description: Allows patients to submit one review per doctor.

Get Doctor Reviews

Field	Details
Link	`/api/reviews/doctor/:id`
Method	GET
Request	Doctor ID
Response	List of reviews

Description: Retrieves all reviews for a specific doctor.

Get My Reviews

Field	Details
Link	`/api/reviews/me`
Method	GET
Request	None
Response	Patient reviews

Description: Retrieves all reviews submitted by the authenticated patient.

Update Review

Field	Details
Link	`/api/reviews/:id`
Method	PUT
Request	Rating and comment
Response	Review updated successfully

Description: Updates a previously submitted review.

Delete Review

Field	Details
Link	`/api/reviews/:id`
Method	DELETE
Request	None
Response	Review deleted successfully

Description: Removes a review from the system.

Admin APIs

Authorization: Bearer Token (Admin Access Token Required)

Get System Statistics

Field	Details
Link	`/api/admin/stats`
Method	GET
Request	None
Response	System statistics

Description: Returns the number of registered users and overall platform statistics.

Get All Users

Field	Details
Link	`/api/admin/users`
Method	GET
Request	None
Response	List of users

Description: Retrieves all registered users except passwords.

Get Users by Role

Field	Details
Link	`/api/admin/users/role/:role`
Method	GET
Request	User role
Response	List of users

Description: Returns users filtered by role.

Get Pending Users

Field	Details
Link	`/api/admin/users/pending`
Method	GET
Request	None
Response	Pending doctors, pharmacies, and labs

Description: Retrieves accounts waiting for activation.

Activate User

Field	Details
Link	`/api/admin/users/activate/:id`
Method	PATCH
Request	User ID
Response	User activated successfully

Description: Activates a doctor, pharmacy, or laboratory account and sends a confirmation email.

Deactivate User

Field	Details
Link	`/api/admin/users/deactivate/:id`
Method	PATCH
Request	User ID
Response	User deactivated successfully

Description: Disables a user account.

Delete User

Field	Details
Link	`/api/admin/users/:id`
Method	DELETE
Request	User ID
Response	User deleted successfully

Description: Permanently removes a user account.

AI APIs

Analyze Lab Report

Field	Details
Link	`/api/labReport/analyze`
Method	POST
Request	PDF file or image
Response	AI-generated analysis

Description: Uses artificial intelligence to analyze laboratory reports and provide medical insights.

AI Daily Brief

Field	Details
Link	`/api/pharmacy/ai/daily-brief`
Method	POST
Request	Daily statistics
Response	AI-generated daily summary

Description: Generates a daily performance report for pharmacies.

Smart Stock Alerts

Field	Details
Link	`/api/pharmacy/ai/smart-stock-alerts`
Method	GET
Request	None
Response	Low-stock and out-of-stock alerts

Description: Detects medicine shortages and provides stock recommendations.

AI Chatbot

Field	Details
Link	`/api/ai/chat`
Method	POST
Request	User message
Response	AI-generated response

Description: Provides intelligent assistance and answers user queries using conversational AI.

Dashboard Tools & Technologies

The dashboard interfaces were developed using modern frontend technologies to ensure scalability, maintainability, and a responsive user experience.

Core Technologies

- **React.js** – Component-based frontend framework.
- **React Router DOM** – Client-side routing and page navigation.
- **Redux Toolkit (RTK)** – Global state management.
- **RTK Query** – API communication and data fetching.
- **JavaScript (ES6+)** – Application logic.
- **CSS3** – Styling and responsive layouts.

Development Tools

- **Vite** – Fast build tool and development server.
- **ESLint** – Code quality and linting.
- **Git & GitHub** – Version control and collaboration.

UI Features

- Dynamic dashboards
- Role-based interfaces
- Real-time data updates
- Search and filtering
- Pagination
- Responsive design

Additional Libraries Used

- **Axios** *(optional)* – HTTP requests.
- **React Icons** – Icons and UI elements.
- **React Toastify** – Notifications and alerts.
- **JWT Decode** – Token decoding and authentication.
- **FormData** – File uploads.
- **Cloudinary API** – Image management.
- **Date Objects & Locale Functions** – Date and time formatting.
- **Local Storage** – Persistent user sessions.